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Introduction to Geography

There are certain things we need to keep in mind in preparing lessons, in this case, in preparing geography lessons. The purpose of the story "Creation of the Cosmos" is not to entertain the children, though they should enjoy it. The idea is to set up the whole framework for everything else we're going to do in the elementary class. This is why we do the story. It is a different approach than that taken in the Casa dei Bambini. This is because the child has changed from his years in the first plane. The introvert has become an extrovert. The child has opened up to the environment and to everything that is going on in the world.

The focus of his interest has shifted to the outside environment beyond the home and the classroom. Geography, history, physics, chemistry are all parts of the same earth, the same whole that we are introducing to the child. The earth is one part of the universe. Because the earth, when it was created, was shaped in a certain way, and behaved in a certain way, that gave rise to the seasons, to day and night. Some elements combined together to form others. Other elements did not. All went through the three states of matter and all reacted to the law gravity.

When the earth was created, air and water, agents of the earth, also behaved in a certain way. Winds, oceanic currents, rivers, glaciers, the spread of vegetation – all of this constitutes an on-going story. All of these are taken up as follow-ups to the story. None of these are things to be memorized. But rather, they are things to become part of the child's world and the child's understanding. In this way, geography can become the true sense of the word: 'geo' means earth, graphy means to describe. The description of the earth. This is what geography means and this is how we will present geography to the children.

To enable the child to visualize, we have aids to the imagination. Sometimes they are illustrations on charts. Sometimes they are experiments or simple demonstrations. They're used to help the children to recognize a certain relationship in their own minds. They are not there to re-create a reality for the child. This is why they are called "aids to the imagination." We are not trying to give reality to

the child. But rather we are trying to give them the principles involved, so that they can work on imagining those principles. Aids to the imagination help the child to understand the physical world which is all around him.

We should remember our aim is not merely to make the child understand but to touch his imagination so that it enthuses him to innermost core. We must remember that in "To Educate the Human Potential" Maria Montessori said, "We consider it a crime to present such subjects as may be noble and creative aids to the imaginative faculty in such a manner as to deny its use, and on the other hand to require the child to memorize that which he has not been able to visualize. These subjects must be presented so as to touch the imagination of the child, and make him enthusiastic and then add fuel to the burning fire which has been lit."

This should remind us that our presentations have to be geared to something beyond a mere explanation. We don't just stand up and explain. It is your talk, your story that has to arouse the child's interest. It is not the demonstrations or charts that will do this. It is the story. Teachers must keep this in mind as one practices lesson-giving. What is the point of the lesson? What is the key which we hope to excite the child about and how can we get it across clearly and in a way that stirs the child up and gets his enthusiasm rolling.

Teachers must remember that his role is to form the link between the child and the environment. This is done by providing the child with the keys that he needs to begin exploring the universe. So, now the focus needs to be on how. We need to make sure that our presentations are not cluttered with too many details at any one time, so that the how or the why of that story comes across clearly. With too many details, the child will not know what to focus on.

At the elementary level, we don't have to count our words as we do in the first plane. The teacher is able to talk more because the child will allow it by virtue of a greater capacity for concentration. But we don't want to give too many different impressions, too much information and too many aspects at one time. This will make the picture confusing. Then, the presentation not only needs to be clear but needs to be enthusiastically delivered.

We also cannot know ahead of time exactly how far we will be able to go with a lesson. This is because we cannot gear our presentations to how much we think the children are ready for, or how much we

think they need to learn about something. Many times, as long as the interest is there, the teacher can keep going. Other times, the teacher will not be able to get to even the first potential stopping point. They must try again another day.

When a teacher-trainer presents the children's presentation, he will tell us where logical stopping points are. But in practice, if the interest is there in the children, it doesn't mean the teacher must stop at the first logical point. Teachers should keep going in this case. And this is learned with experience – when to stop and when to continue. In general, it is better to prepare to go further than one might imagine, so that if interest is present, the lesson can continue.

Teachers discover that different groups of children are interested in different things. The teacher has to get to know the children to perceive this. He discovers that sometimes one group will show little interest the first time something is presented, but more interest the second or third time it is presented. This is why we must practice lessons with the children in order to have these experiences.

Teachers must ask themselves, how much of an album's presentations is for the 6 year olds and how much is for the 9 year olds. In general, it is all for the 6 to 9 year olds, since the 9 to 12 year olds often work from these presentations as jumping off points for deeper study.

All the broad ideas must be presented to the children as early as possible because this is what forms a general picture of the universe for them. This is what sets up the whole framework for cosmic education, in a general impressionistic way. This does not mean teachers will go into every detail of every topic introduced. The 9-12 year olds will be ready for that additional study and additional details and they can do this if they get this general introduction when they are younger.

In order to give structure to geography albums, it has been divided into six chapters. It is an arbitrary division that accords with the teacher's adult mind. The chapter headings help the teacher look at geography differently than does the traditional educator. The headings are:

The first chapter: "The Creation of the Cosmos."

The second chapter is "The Nature of the Elements".

The third chapter is "The Sun and the Earth."

The fourth chapter is "The Work of Air."

The fifth is "The Work of Water"

The sixth is "Interdependencies and Economic Geography."

Again, this is a format to help the adult. It does not mean the child's work should be organized around these chapters. These chapters are organization for the adults. In practice, no chapter is completely independent of others. The children can move back and forth from one chapter to another – in terms of which presentations are given. They don't need to follow the sequence these chapters imply.

Because we are following the interest of the child, we might begin with a presentation in one chapter and realize in fact, they are more interested in another chapter, another area or topic of work. So the next presentation should, in that case, switch gears in order to follow their interest.

In conclusion, teachers should look at geography not as a subject to be taught but rather as a help to orientating the child to his surroundings. Teachers are going to be orientating the child to the world which contains certain physical phenomena. This world which is part of the universe, follows a set of laws that are innate in its being, laws that operate and dictate the behavior of all of the elements – both animate and inanimate. These elements, as they live and act, create and maintain the environment as we find it today - an environment in which human beings are acting out their own drama. Teachers must keep this in mind as they build up cosmic education in the classroom.

Date: September 27, 2004

Notes on the Charts

The charts are meant to be aids to the imagination. This should tell us that they are to help something. It is to help the story. The charts should help the children visualize what your story is talking about.

Sometimes we also refer to the charts as impressionistic. This implies they are meant to give an impression rather than an exact representation of fact. For example, the first chart 1a showing flames off the sun. We would say it is impressionistic because it does not give an exact picture of the size difference between the sun and the earth. It is just giving an impression.

Some of the charts we use verge on being more on what we might call fanciful. Botany chart j is an example, where little people are tugging on roots below a plant to hold it strongly into the ground. There aren't people there so we might say it is fanciful where we are only making use of imagery to make an impression, to help them remember what you are talking about.

The reason we use charts like this is because the more striking the impression, the better chance it will remain with the children. This is what they will remember. It appeals to their characteristics that we've talked about at this age. We make use of allegory, for example, so the children will like and respond to them.

There are certain other charts that we would consider more factual. For example, maps tend to be more factual than not. They give more an idea of this is how it really is, for example showing the hottest and coldest places on earth. Even though this chart is more factual, it is still a simplification. A map of this size cannot possibly give all the detail to show the reality and it is not meant to do so; it gives the general impression and then the children can look up in books, the detail which refine their mental image. Even more factual charts are simplifications.

The books we put into the environment and the trips outside the classroom are meant to provide the children with true facts and

current information that they are looking for. The charts normally just get the children started.

There are also some other charts, one of which we have seen in the Time Zone Chart, which are called “**work charts**.” They have been designed to give the children a manipulative activity, to give the children time to dwell on a certain factor and sometimes they provide opportunities to use their reasoning minds.

None of the charts are meant to be used as teaching material in the traditional sense. This means they are not meant to be copied by the children and those copies colored in (unless the children think of this on their own and suggest this). This often tends to be busy work without generating better ideas or inquisitive mental activity. Rather than copying, it makes better sense to draw your own chart which shows layers of the earth; in this case the child has to calculate the layer size and so on. In this way, we help the child develop their reasoning minds and other skills.

This is one of the reasons why the charts have been made as large as they are. They are made *not* to fit on a child's table along with materials for copying. Another reason they are so big is because we are generally speaking of big ideas. The size of the charts help to reinforce that bigness of concept.

In the classroom, after you give a presentation that uses charts, if you have room in the classroom, it is nice to **leave a couple of them on display for a couple of days**. In this way they serve to remind the children of the lesson. They give the children the chance to look at them more closely. Sometimes they notice new things on the charts that weren't pointed out in the presentation. Sometimes pondering the charts will stir up questions in the children that they will come and ask about. Or maybe they will figure something out on their own and come to tell you what they noticed. At that point, you can see what direction or detail to pursue in the next presentation.

Most of the charts have a number (some have a letter). At one point the charts were lined up more or less sequentially and a filmstrip was made of the charts. It was at that point that the number was assigned to each chart to identify its frame in the filmstrip. The numbers stuck and are still used today. From the child's standpoint, the numbers have no relevance. They are useful for us to organize ourselves. We can find the charts more easily this way.

In terms of coloring charts, when you color them keep in mind that they are aids to the imagination. The chart itself should not be what is most attracting the child. It should be the story that attracts him primarily. It also means that charts should be neatly done because a sloppy job will also attract them – but they also should not be done as a work of art. This again detracts from the artist. Focus must always remain on the story.

We also don't want to add, to the chart, things that are not there because this could become a hindrance. We should not point out every detail of every chart as we present them. Some of the details will come up in practicals. Keep as close as possible to how the charts are colored. There is often a color significance to things that may not appear significant during the presentation; so don't veer away too much.

When working with charts here or wherever you practice teach, or in your own classroom, be careful with how you handle them, how you **show the children to handle them and where you place them**. We need the children to take care of them so we must model how they should be handled. Corners are crushed and so on very easily. If care is taken, they should last years and years.

There is nothing wrong with getting help coloring the charts as long as you carefully supervise. Earning your diploma depends on coloring them correctly. Albums need to be done by oneself; charts are not this way. They need to be colored and they are not available colored. You can have a family coloring party and so on. Some people do all the coloring themselves. They sometimes review lessons as they color.

Don't color any chart until you've seen them presented. We will see almost all of them by winter break. You have the rest of the year to complete the coloring in. We can use the copies from Judy but it is up to us to check the color beforehand to ensure they match what is in our environment. Thematically related charts must use same hues; otherwise hues can vary a bit. We will see charts thematically related in the same presentation. Avoid smears by placing paper under your hand on wet sections.

September 17, 2004

Practical Considerations for the Experiments

These are guidelines. First of all, we consider purchasing the substances we use. Many of the things are available at a grocery. Salt, ammonia, sugar, to name a few. Things like acids and chemicals, one needs to get from a scientific supply store or catalog. NASCO, in Janesville WI, and Fischer Supply, on the east coast are two such places.

There are state regulations that control what a school can buy versus what an individual can buy. Ammonium dichromate can only be ordered on school letterhead or maybe a school purchase order. Plan ahead accordingly. Very often the minimum you can buy is too much for the classroom. So these are things that a school should order so each teacher can take from a common supply. Otherwise, the leftover may go bad before you can use it.

The containers come with information on them, how to store them, the date that you purchased it and any antidotes for anything poisonous. Any information like that on a big container needs to be placed on a small container. **This information must be handy in the classroom in case of accident, along with whatever antidote may be needed.**

Make sure that if the directions say to only store in a glass bottle, only store it in a glass bottle. Really follow the guidelines because it makes a difference in safety and usability.

Let us consider how we dispose of the chemicals and substances we use. Most things can be cleaned up with lots of running cold water and then the glassware is washed in very hot water. The piece of coal that forms in this presentation can be dumped into the garbage. The two experiments that form a **hazardous waste** are **Chemical Reaction with Copper Sulfate** and **Chemical Reaction with Potassium Dichromate**. These two chemical reactions form a hazardous waste so we will dispose them as hazardous waste. Both can be dumped into our hazardous waste container. These are the only two we do that create hazardous waste and they can be mixed together as waste. Sometimes a high school or university nearby may allow you to dispose of this through their lab. You need to work out how you dispose of this hazardous waste.

How do we know when something is a hazardous waste? It is something not environmentally friendly such that it has to be disposed in a special way. Find out about the substances you plan to use beforehand from an authority.

Some left over substances can be reused but often this is not possible. Copper sulfate is a dry substance that normally can be returned to the copper sulfate container. With acids, use only what you need to use; if there is any left, you cannot reuse it since it is no longer as pure. Also for safety - one never wants to pour sulfuric acid into hydrochloric acid and they are easily confused. If any acid is left over, dilute it with cold running water and pour it down the drain.

Cleaning up of the equipment, i.e., glassware, etc., needs to be taken care of diligently. This needs to become part of your lab habit. Everything needs to be washed with hot soapy water. One problem is it take so long to get hot water from a tap, boil it if necessary; test tubes need to be scrubbed to get all the residue out. This is only cleaned using a test tube brush and hot soapy water.

If it is not removed soon, the tubes need a mild acid bath just like chemists use. The test tube racks have pegs on them that are used to hold tubes after they've been washed so they can dry. So, after tubes are washed, put them onto the pegs to dry. You can see after they dry if all the oil is taken out; if not wash again before it becomes impossible to remove without the acid bath. **Wear rubber gloves if handling acid**; pick up beakers with acid at sides well away from the spout.

In the elementary class, ideally there would be a sink to use for cleanup. But the sink and cleaning the sink is the last step in the cleanup process, teaching the children also this is the last thing to be cleaned. The rule to establish in the classroom is that nothing is put back until its clean and dry, ready for the next use.

Storing the chemicals and supplies in the classroom: anything that is poisonous or at all dangerous is stored under lock and key. In the training environment, there's a black cabinet that is always locked containing these substances. The teacher is the only one with a key. Anytime anyone wants anything there, they have to come and ask the teacher. The one supervising practicals generally has the key.

Things that are not poisonous are kept outside the cabinet. Some classrooms have a two-sided cabinet where one side locks and the

other does not. All containers are labeled so everyone knows their content. As an additional safety measure, containers for dangerous substances have red marks or tape around each container. The antidote is listed on each of those containers. Make sure when we fill containers that they are filled from the central supply *at a time when no children are around*. They should not know of the central supply which remains locked away. Make sure not to put too much in any container; if it's all used up in one day, children will have to wait until another day. This also limits danger for accident or spill.

As part of a daily routine, check the chemical supplies and material supplies to see that all is prepared for use and in order.

The experiments using the acids are not ever for the 6 to 9 year olds on their own. Even 9 to 12 year olds have to be normalized to allow them to access these things.

In setting up classrooms, keeping in mind ideally there is a sink, as much as possible; set up the area for experiments as close to the sink as possible. Water supply is close and spills are less likely while being carried. Never carry test tubes in bare hands; it's too easy to drop them. Test tube racks carried with two hands are the ideal way. If there is no rack, put the tube into a glass that can be held with two hands.

As well as water, you should have sand readily available in case of accident or spill. Keep a bucket of sand and pitcher of water handy. Everything needs to have its own place. Smaller things should be in front of taller things so things are easily seen. Everyday, we need to check that everything is put away.

Sometimes different things will work for an experiment. Sometimes a glass or beaker are interchangeable. Using acid, this is not possible because the beaker has a spout that directs the liquid.

Take some basic precautions for the actual presentation. Try out the experiments yourself before presenting them to the children. Even if this was done months ago, refresh your skill and practice beforehand. Master any special techniques involved. Anytime you use a heat-source, turn it off and unplugging it as soon as you are done. That is important. Anytime we use anything dangerous, we bring it out last – just before presentation beings.

Likewise, you can never leave in the middle of a presentation, when there are dangerous materials or chemicals present. Prepare ahead of

time for the case of a fire-drill. Don't try to do the experiments on a day when you are not feeling up for it – mentally or physically. If you are not ready, wait a day until the time is right. We are more accident prone if psychologically or physiologically distracted.

Date: September 23, 2004

Notes on Experiments

We've seen the experiments we do with the stories such as with Creation of the Cosmos. Experiments are done this way to aid the child's imagination. We've also seen experiments geared to arouse the child's interest. Different ways of combining come to mind. The children enjoy the experiments and want to do them themselves. We need to keep in mind why we give the experiments. Unless we constantly keep this idea in mind, it's tempting to add other experiments in. But we need to remember that we give the key lessons.

The experiments we give must be regarded as key lessons. They hold the keys to some of the basic laws of nature. They can be used to explain a lot of phenomena that happen on the earth. Because they are so basic, we have to make sure that they come across clearly to the children. This is why we wouldn't want to have experiment after experiment – showing the same idea but in different ways. This would clutter the children's minds. The key ideas would become obscured if we did too many.

If we limit ourselves to the keys, we stand a better chance of the children understanding those ideas, and then being able to use those ideas themselves, to figure out things they see happening around them. A piece of chocolate left in the sun; what happens and why? This is the type of question the child confronts with his reasoning mind.

So the limitation in the number of experiments that we give always needs to be given some consideration. This does not mean that if the children find some other experiments that they want to do that we deny them. That's precisely the difference between us presenting something – versus allowing the children to explore and follow up. Sometimes the children will receive a science kit or chemistry set as a gift and want to bring that into the classroom to share. We would, of course, allow this, though it doesn't mean it will become a permanent part of the classroom.

Sometimes these other experiments the children find and do themselves will contain a key idea. Perhaps you can stop by, have a little discussion with the children, ask a couple pertinent questions – to help them realize what the key idea might be. Make sure the kit or set is visited long enough and then goes home. This idea applies to all materials in the classroom. There is a distinction between what we present, related to the keys, and what the children bring in and want to do as follow up.

The direct purpose of the demos we've been doing and we'll continue to do is to aid the children's understanding, to help them visualize physical phenomena taking place around them. This is what we saw with the initial creation of the universe story, for example. This is why they are called aids to the imagination.

We want to help the children become aware that each little item that is part of our earth is behaving in a particular fashion, and the way in which it's behaving is directed by its own inner laws. This is what will eventually help the children understand how the world functions as a whole unit in which each element makes its own little contribution. That's really the direct aim of all the experiments we do.

Indirect aims are that we're laying the foundation for chemistry and physics. The children are handling test tubes, stirrers, heat sources and so forth. They are being introduced to terminology such as solution, mixture, matter in suspension, chemical reaction – all of which is providing them with a foundation for science. This work is teaching them how to observe and what to look for – all parts of lab work. Because the experiments call for careful handling and for following a sequence, the children are helped by these activities in the same way they are helped by the practical life exercises in primary classroom.

Another activity with the experiments involves reading and writing. That activity helps the children develop their thought processes, as any language composition does.

The work with experiments has three stages to it. The **first stage** is what you see when I present the experiments for the very first time. We present the experiments either with a story or a little talk, making use of allegory and personification, sometimes a little magic. The teacher is doing the experiments without giving the ingredients to the children or without explaining the techniques.

The **second stage** comes when you've already developed some interest in the children. The first presentation has hopefully aroused their curiosity. Then here, at the second stage, you might *repeat some of the experiments in a more factual way, laying the emphasis more on the experiments themselves* rather than on a story or topic. Teacher begins to explain to the children what he uses, what the ingredients are. He explains how the experiments are done. He explains anything that needs to be taken care of. It is at this *second stage that one would introduce goggles, wearing an apron - those aspects of lab work*. If we really understand the purpose of the experiments, we understand we don't rush into the second stage. There must be unrest built up in the child's mind. The child thinks, becomes impatient, because they want to know more. This gets them stirred up and anxious to know more.

Often, the children will ask to see some of the experiments again. At this time, the teacher says, "sure, let's set a time and this time would you like to begin to help me. I'll show you where we keep the materials that we use, so you can help me gather up the materials that we need." Teachers can also begin to show children how to clean up. Very gradually, teacher begins to help the children carry out some of the experiments themselves. This leads to the **third stage: The Command Cards**.

Date: September 23, 2004

The Command Cards

The third stage, using the command cards, is where the children become able to do the experiments completely on their own, without needing your help at all. If we realize the main purpose of the experiments and the importance of them, we won't get to the command cards too soon. The first stage is most important and the second is of the longest duration.

The command cards are made up of three sets of cards. In training material, we have them in separate racks. The **first set contains a title or a subject, a list of materials needed and what is called the command** (an indication of what to do). The text of the command ends with a question. For example "What do you observe immediately and what do you observe later on?" is a question one might see in the first set.

The **second set of cards are the statement cards**. Each of these contains the statement which will answer the question or questions posed in the command cards. "The sugar dissolved in the water and remained dissolved later" is an example of a statement that answers the question posed in the first set.

The **third set is the control set, question and statement together** along with all other details. Third set is the control of error. This is like the nomenclature cards in the primary.

The command cards are not numbered on the back. Numbering implies order or sequence. They are meant to be done in any order. Also, if they were numbered, some children would just look for the corresponding answer just to get the answer. There is, however help for organization. There are letters on the back. We've separated the command cards into groupings and we've assigned each group a letter. So if a child happens to mix the cards we are able to sort them out.

Each grouping should have at least five or six cards to the group. In our materials, we have ten groups. Headings include:

- A. Different ways of combining
- B. Precipitation and crystallization
- C. Matter in suspension
- D. Three states of matter
- E. Heat and matter
- F. The sun and the earth
- G. Currents (e.g. Ocean)
- H. Further states of matter
- I. Erosion
- J. Changes in the earth's crust

Teachers can reorganize these groups but what is important is that there is enough cards in each group (6 or so). The groups with dangerous substances have cards that are bordered at bottom in red. Any card with a red line on them like this can never be done on their own unsupervised. The teacher's presence is always needed and the children must be told this as well. Often in a classroom this means that they need to make an appointment with you to do the experiment. This is another way of building their sense of responsibility; for some things one does have to plan ahead and make arrangements. We can't always do what we want to do at a given moment, and this they will learn here.

The dangerous experiments are not for the 6 to 9 age group of children. In a 6 to 9 age class, teachers should not have the red-slash cards present at all. Take them out of the environment completely. The teacher has this control. In a 9 to 12 age class, if there are some children who are ready, then the teacher can add some of these experiments.

What is done with the command cards? There are three levels of work with the command cards. The **first level** is when teacher introduces them to the children for the very first time. This is perhaps one or two cards at first, that is introduced to a small group of children. These should be experiments that the teacher has already done at stage one, and that teacher has worked with the children at stage two. Now there would be the one or two cards; read through the card together with the children. They read it and teacher makes sure they understand what it says. Supervise them in gathering the materials and in carrying out the command. Remember the question or questions at the end. Help them summarize their answer(s) to whatever question(s) are asked. This is the first level with the command card; teacher is right there with the children supervising.

The **second level** with the command cards is when the children are able to carry out the commands on their own but the teacher is still keeping an eye on what's going on. If necessary, the teacher is ready to step in. The teacher might be involved in giving another lesson or presentation but would ask the children working at the second level, to work near to where he is working to enable observation of their activity. It is at this second level, when the children have worked through some of the command cards, that the teacher can begin to introduce the statement cards to them.

The statement cards can be used in two different ways. The first thing one might do is ask the children to write down their answer to the question. Prior to this we've just been verbalizing. Then show them the second set of command cards, the statement cards, and show them that they want to look through that group of cards and find the one that best answers their question. Compare what they have written as their answer to what's on the statement card. It becomes a small language lesson in composition. Make whatever comparisons are valid. Perhaps statement card is worded more concisely, for example. Perhaps they forgot punctuation or forgot to break ideas into sentences. We can show them how the statement card is broken into sentences. Perhaps they forgot some important detail that we can compare and bring out to them.

The second way statement cards can be used, especially for those who have difficulty writing, is that, after carrying out their command, they simply look for the statement card that best answers their question. Then the teacher should also show them the third set of cards as their control of error. This is how you make them independent so they don't keep coming to you and begin to work on their own. This would be the second level of work.

The **third level** in the work with the command cards is that the children are left completely on their own without supervision, except for the one's slashed in red. (They never arrive at level three for the dangerous cards.) This would be the third level.

It is a natural outcome of this work with experiments and work with the command cards, that the children often want to make up their own command cards and experiments. Now they would have to write their own command, determine what it is they want to do, what materials they would need to use to do it. Think of what a language lesson this is.

This is work particularly suited to the 9 to 12 age group. They have to think through what their purpose is, write it down, give it someone else to read to make sure it makes sense to somebody else (e.g., does it say what they wanted to?). This is why we have the non-dangerous chemicals out and available to the children. These are the things they can use in making up their own experiments. Or sometimes they can use other things in the environment as well.

As a last step before they actually carry out their experiment, they need to bring it to you. Because once again, the teacher is responsible for everyone's safety in the classroom. If the teacher does not know exactly what will happen as a result of this experiment, he should try it first. If not sure what each substance, when put together, might do – go to an expert and ask. In fact, the teacher will show the children how to do this as well. The teacher must become at least phone-friends with someone who is an expert. The teacher has to make sure all are safe; so he must check it out with an expert.

If there is scientific terminology the children need to learn during their years with you, especially coming from the public curriculum, the teacher can make some card material for those terms and definitions. This is like the card material in primary where the term is missing but the definition is present or vice versa – along with a control set. You can have cut-up definitions of terms, as in primary as well. This holds true for any definition and term needed by the child.

Date: September 15, 2004

Eric Ferris
December, 2004
Adapted from
"God With No
Hands", a work by
Mario Montessori

The Creation of the Cosmos

From the earliest times, people noticed a great order in the universe. The sun rises and sets each day. A caterpillar always becomes a butterfly. A seed of corn will always grow corn and not pumpkins. Planets circle the sun the same way year after year. The seasons repeat themselves every year. This is what they noticed. And we see this today, even after thousands of years later, and we admire the beauty of it all. People wished to know the source of this order and were in awe and wonder at this mystery and the possibility of some great creator or protector as its cause. Among humankind, it became a great quest to find this source or creator. People around the world, speaking different languages, living at different times and different places were curious about this.

The order we see in the universe is natural, meaning people did not make it. We could not have created something so vast and seemingly perfect. And it has balance and pattern that we can observe day after day. Each part of the universe seems to cooperate with all other parts, not always willingly, but nearly always to the benefit of all others. Each type of creation has received a unique nature, that means its own set of rules and wishes. Many people believe that this natural order came about by a great string of events over many many many years, that led us to the universe as it is today; the sun and earth, the plants, the animals, the mountains, rivers – everything we see today.

So, how were these things created the very first time? Where did they come from? If there was a great creator, where is he or she now? Why were all these things created? Why was I created? If everything seems to play an important part in the universe, what is my part? These are questions human beings have wondered about since anyone can remember.

People also follow the rules of this natural order, just like everything else does. This is why we find we cannot create these natural things for ourselves, like a new star, and cannot control what is not in our nature to control, like creating a pumpkin from a corn seed. Everything in the universe has a set of rules that it absolutely must follow, no matter what, because it is not their nature to do otherwise. For example, have you met a man who could flap his arms and fly? Have you met a rabbit that could talk? Does the sun take an

afternoon off every so often, and set a little early? How did all this order in the universe come about? This is the story of how people think all these things came to be.

Billions of years ago, there was darkness and cold - and not much else. The natural order we see today did not exist. No one could survive this cold and dark place that was like a blank canvas on which nature and the natural order was to be painted.

Our earth was part of a great cosmic storm that swirled around this cold dark place. Due to the great swirling, things mixed together and combined to form heat in the storm and this caused other things to form; such as light, air and water. All the things we think of as hard, such as steel, were only dust at that time. Can you imagine steel before it became steel and was only dust?

Out of this vast cosmic storm came the sun, the earth, the stars and immeasurable others like them, that we never think of. It makes us imagine, maybe for the first time, how immense the universe must be and wonder what else exists that we don't know of and never think about. Is it possible for us today to get some idea of what we don't yet know?

Knowing how these things were created, by this tremendous fiery cosmic storm in the past, today we can look at a flower differently than before. We have learned we are part of the same creation as the flowers we see. It has a nature. And I have a different nature. And yet, it appears we are destined to work together. Why do we say this?

Again, because this is the pattern humankind sees in the natural order around us, in the nature of the universe, which is beyond our control and came to the earth before we did. And nature has done quite well for billions of years before people arrived - even seeming to have been intended to work closely together all along. We see how plants create what we need to live and people create what plants need to live; this is an example of that awesome order that began even before human beings were found in the world.

Even though people are unique in this created world, meaning there are special rules for us that are very different from rules of other animals, and even though each person has a unique personality among people due to the choices he makes, we see that all humanity and all of nature plays an important part in this natural order.

Science has shown us we are part of a great web of life, along with the rest of nature. It is part of our nature that we are connected to the rest of creation. This is what people have noticed from the beginning. So we search for the facts about this mystery to know the truth for ourselves. And we'll talk today about things we feel we can prove.

Now, as the huge cosmic storm chaotically traveled across the

universe, dropping planet here and stars there, it traveled great distances so that things that were dropped from the storm, were far apart from one another. They are so far away from each other that we know today, light from some stars takes millions of years to reach us on earth. Millions of years. Is anyone here a million years old? Maybe you think that light must travel very very slowly.

Do you know how fast light travels? (accept various answers.) It travels 186,000 miles per ... second - not per hour - per second. Imagine how fast that must be. If you traveled that fast and held a mirror in front of your face, the mirror would be blank because the light of your reflection would never catch up with the mirror. So, that's pretty fast! It's hard to imagine so let's put it another way.

How long does it take light to travel around the world? (Snap fingers). Want to know how long it takes light to go around the world two times? (Snap fingers). In fact, light travels around the whole world seven times in the time it takes to (snap fingers). Seven times, it just circled the world. Maybe you think the world must be small then.

How far is it around the whole world? If there were a highway stretching around the world, we could leave today after school, taking the highway and when do you think we would arrive here again - going around the world and arriving from the other direction? (accept various answers.) Better bring a toothbrush and 16 days of food and clothing - and that's if you don't sleep - at all. That would be one trip around the world for us. Yet light travels it 7 times in a mere second. And it takes light millions of years to reach us from distant stars. So they must be very very very far away.

At the same time, there are so many stars that scientists have calculated that if each star were a grain of sand, all the stars together would cover the whole state of Alabama - up to a height of a sixty-story building. So now, one can begin to see how huge the universe must be to have all those stars, each one far far apart. One of these stars among those thousands of billions of grains of sand - is our sun and one millionth of one of those grains would be the size of our earth. Is it even possible to imagine how immense our universe must be?

And yet surprisingly, each person and each animal, plant and thing in the world seems to have a usefulness within the natural order that resulted from this huge cosmic storm billions of years ago, even though the earth is only small compared to the sun and tiny compared to the universe. So we are very fortunate to have such an important role in the universe! So small and yet very very important.

Now, the earth may seem larger than the sun, but actually, the sun is a million times larger than the earth. The great sun is so far away that it looks small to us, just as huge buildings look like miniature toys

when we look down from an airplane. In fact, the sun is so far away that eight minutes pass before light from the sun reaches the earth. If we were to travel 100 miles per hour it would take 106 years to reach the sun. Anyone here 106 years old? A single flame from the surface of the sun could contain 22 earths.



Chart 1a - The sun

Science has also shown our earth is made of many tiny particles. They are so tiny we can't even see them. And these particles follow the rules of natural order, just like us, since they are also part of the universe.

Sometimes particles come together with other ones, just as we people sometimes like to come together with other people. Sometimes, particles want to stay apart from other particles, also like we do. It is the amount of heat in the particle that gets to decide this. That is the nature of particles. They follow different rules depending on if they are hot or cold or just in between - and this was also true when the huge cosmic storm began the work of forming our earth and sun while traveling through space.

So, we are surrounded by particles everyday, in things like ice, water and steam. The difference among these three things is their temperature - the amount of heat in the particles of the ice, water or steam. The temperature controls how tightly the particles hold together and this decides whether something is a solid like ice, a liquid like water or a gas like air - or the steam we can see after someone takes a very hot bath.

Experiment 1: Three States

When the tiny particles are cold they want to come together with each other. A group of particles that are cold will shrink in size because of this. These particles form solids. But when there is a lot of heat, the particles will want to separate from one another. A group of particles with a lot of heat will grow in size since they spread apart from each other. These particles form gases.

In between the colder temperatures of solids and warmer temperatures of gases is a medium temperature that forms liquids. This is part of the nature of particles and the states they form - solid, liquid or gas.

Let's see if we can detect particles of gas as they move apart from each other.

Experiment 2: Attraction and Repulsion

In a solid state, particles cling so strongly together that they are almost impossible to separate. Solid matter cannot be separated without applying force. Even if a piece is broken off, the particles still try to cling together. So for example, if you chip off a piece of stone, the stone and its chips break off in chunks, thanks to the particles holding the chunks together. This is the nature of solid particles.

Liquids have particles that cling together, but not tightly. These particles roll over each other and so they cannot form a single shape like a solid. This causes a liquid to spread over a surface when it spills, filling every dip and groove in its path. If we accidentally spill milk on a table we see how the liquid spreads across the table and falls to the floor. It is because the particles are loosely held together that we can put our hand into water. We cannot do this with a solid because a solid's particles cling together so strongly. So the nature of liquid particles is to hold together only loosely and to roll over one another as the liquid spreads out.

Experiment 3: Liquid Motion

The nature of gas particles is to move away from each other freely in all directions, not just down and sideways. We say that particles expand as they move away from each other, spreading like liquid but in all directions, because they barely cling together at all. These particles practically float away because they are hot; hotter than liquids and much hotter than solids.

This is why smoke from a smoke stack seems to rise and disappear. It is because the hot particles have moved away from each other, that

we no longer see their collection together as a cloud of smoke. The particles still exist but they have moved away from where we first saw the cloud of smoke – moving up, down and sideways.

We can see that we behave in a similar way. When I am hot, I don't want to hug another hot person. I want to be further away from that person and if I hold onto him, I only hold his hand instead of putting my arm around him. It is more comfortable this way. This is like the gases which are rather more hot.

But if I am cold, and my friend is also cold, we want to get close to each other to warm ourselves and we hold each other tightly, not just with hands but we hold our bodies close and this way we take up less space also, since we are close together. This is like the solids.

When I am neither hot nor cold, I am not too close, nor too far from my friend and if we hold onto each other, it is neither loosely nor tightly, but in between. This is like liquids that are neither the hottest nor the coldest.

So we said that things in the world are made of particles and each thing has its own nature; each thing becomes a solid, liquid or gas according to its own rules. This means one temperature creates solid ice for example, while another hotter temperature creates water as a liquid we can drink, and a third even hotter temperature creates a gas, like steam we see in the air after a hot bath. But water is just one example.

The temperature needed to make a liquid water from solid ice is different than the temperature needed to make liquid chocolate from solid chocolate. This is because ice and chocolate have their own different natures. And though they can both become solid, they become solid at different temperatures, according to each of their natures; the nature of water and the nature of chocolate. This is why on a hot day, an ice cube will begin to melt in your hand faster than a chocolate bar. Have you ever had a chocolate bar melt in your hand? This is because a solid is turning into a liquid because of heat.

Experiment 4: Temperature and States

There is another rule particles must follow. If I am cold and you are cold and you come to stand by me to become warmer, what happens if I am standing on an old floor that has become weak? Now we have two people where before there was only one, right? The floor will sink a bit because there is more weight on the weak floor, right? We might even fall through the floor if it's weak enough.

Particles have this same nature. When the colder particles come together to form a solid, together, they become heavier and so they

fall downward with more weight, just as two people might fall through the weak floor, where one person alone did not.

On the other hand, when you become warm and move away from me, you step away and off of the weak floor. What happens? The floor rises back to where it was. It expands upward because there is less weight on it, and I rise a bit too since I am still standing on the floor. This is like the gases that are very warm. They rise into the air as they separate from each other and there is less weight from the particles.

But liquid particles, since they are neither very hot nor very cold, behave a little like both solids and gases. They push downward from their weight but also want to push sideways. So another part of the nature of particles, besides the 3 temperatures that tell us when they will become a solid, liquid or gas, is their weight. So just as water and chocolate were different, if we have three different liquids, we can find out which is heavy, which is lighter and which is the lightest. Which of these do you think will be the lightest?

Experiment 5: Separation by Weight

So, following these laws, the particles, in this giant cosmic storm billions of years ago, began to form what we know today as the earth. In the middle of this storm was this great intense heat. Particles close to it were very very hot and barely held onto each other; therefore they became lighter and rose away from it, to the edge of the storm. When they again became cooler and held onto each other more tightly, they became heavier and fell back toward the center of the storm again, falling back toward the heat in the center. This went on and on for a very long time – millions and millions of years. Heat rising and coldness falling.



Chart 3a - Dance of the elements

Like little workers, the hot particles carried their heat toward the outer edge while the icy cold particles carried their coldness toward the center and gradually the center of the great cosmic storm became a little cooler and the outer edge of the storm, closer to the coldness of space, became a little warmer.

As they passed by each other, many different kinds of particles met each other and formed new substances. Again, this continued for millions of years until the heaviest substances, like stone, had collected toward the center of this huge storm, while the lightest materials, the gases like air, had collected further toward the edge of it. And gradually, over eons of time, it began to look more and more like the earth we see today.

But the heat of this great storm had been so great and the chaos had been so extraordinary that, as cooler substances fell toward the center, hotter substances trying to rise toward the outer edge of this great storm were trapped underneath the cooler ones. And being unable to do anything but exactly what their nature required them to do, the hot particles tried to rise from underneath the surface created by rocks and other cooler and heavier substances. There were great battles on the surface of the earth as the hotter things, below the surface, pushed upward to escape, expanding from their heat, while those cooler things above them pushed downward heavily to stop their escape. And this, also, went on for a very long time.

Finally the forces under the surface began to break through. The water that had formed on the surface immediately vaporized, turning into gas once again. There were hot ashes thrown far away from the

surface and an enormous cloud formed around the earth from all the smoke and ash that escaped from below the surface of the earth.



Chart 4a - Expanding elements

Eventually, in time, the great battles at the surface of the earth were settled and the earth became more orderly. As everything cooled, according to the law of particles, some gases became liquid and some liquids became solids. What had been part of this vast storm, shrank and became wrinkled like a dried up apple. The wrinkles created the mountains and valleys of earth.

It rained and rained and rained as the gases collected into liquid and fell to the earth. The lowest spaces of the earth, where the water poured into every hole and crevice, became the oceans. In the sky, there was mostly just one special gas left, the air we breathe. Most of the other gases had become liquids or solids as they cooled. The veil of smoke and ash was gone and the sun shone on the earth, providing the environment needed for life, for my life and your life.

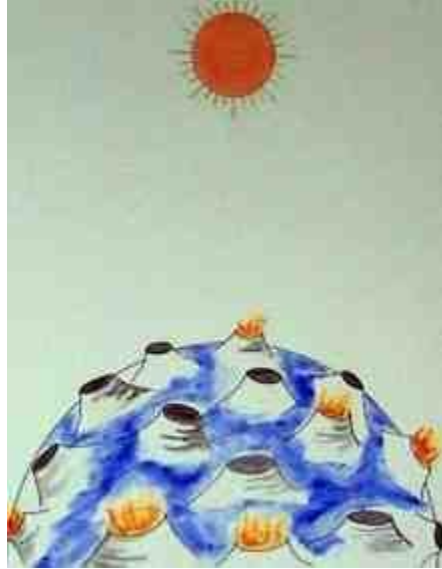


Chart 5a - After expansion

From that time until today, the earth has continued to spin once a day and has continued to go around the sun once per year. Corn seeds have only grown corn and caterpillars have only become butterflies. The elements of the earth have continued to follow their nature just as all things do, including human beings. Solids, liquids and gases have followed their nature. The great chaos at the beginning became great order and it continues to this day.

Demos for The Creation of the Cosmos

Charts

Charts in order (from Geography chart set 1a-28a):

1. 1a - The sun
2. 3a - Dance of the elements
3. 4a - Expanding elements
4. 5a - After expansion

Experiments

1. Three States

- 1.1. Materials: 3 identical glasses on a tray containing ice, water, and air; use thermos to keep ice cold.
- 1.2. Procedure: **Display each glass, holding glass at the top so contents are in view of all.**
- 1.3. Purpose: Demonstrates three states of matter

2. Attraction and Repulsion

- 2.1. Materials: a clear glass bowl about 10 in. diameter about 2/3 full of water. Small pieces of paper (about size of hole punch) in a small container. Both are on a tray.
- 2.2. Procedure: **drop pieces of paper, one at a time, into the water, toward the outer edge and slowly work your way into the center a little bit gradually. When you see some coming together, stop**
- 2.3. Purpose: Demonstrates particles coming together (or opposing one another).

3. Liquid Motion

- 3.1. Materials: small clear glass on a tray, 1/4th full of BBs (iron shot, ball bearings)
- 3.2. Procedure: **pick up glass and rock it side to side so BBs roll over each other**
- 3.3. Purpose: illustrates how particles of a liquid roll over each other

4. Temperature and States

- 4.1. Materials: three identical dishes, either metallic or heat-proof, each containing a solid substance (1. candle wax or paraffin 2. string solder 3. piece of nondescript iron) all placed on a tray, heat source like a hot plate, e.g., Bunsen burner on a heat-proof mat, a heat-proof stirring stick, a hot-pad for emergency removal, test-tube holder to pick up the dishes with.

4.2.Procedure: **turn on heat, place the dishes on the hot-plate to apply heat evenly to all 3. (May need to turn hot-plate down gradually to avoid the paraffin from flaming)**

4.3.Purpose: Demonstrate substances change their state according to their temperature

5.Separation by Weight

5.1.Materials: three test tubes in a test-tube rack on a tray. First tube is filled about 1 inch of water. Second tube has 1 inch of molasses. Third tube has about 1 inch of olive oil.

5.2.Procedure: **Pick up tubes with water and that with molasses (holding at the top). Put some molasses into the water by pouring it right down the middle so as not to color the sides of the tubes. Then the third tube with oil is poured into the water in the same manner.**

5.3.Purpose: Demonstrate separation of substances due to heavy and lighter characteristics (Molasses at bottom, water in middle and oil at the top).

6.Heat Expansion

6.1.Materials: Volcano made of paper-mache built up around chicken wire (hollowed inside but formed into the shape of volcano). It needs a heat-proof mouth that is 1-1 and ½ inches in diameter and no more than 2 inches inches deep (not too large a chamber so as not to use much chemicals). Ammonium dichromate (ignites and forms ashes) and a tiny pinch of sulfur (provides the smell). Long (fireplace) matches. Fill mouth ½ to 2/3rd full with ammonium dichromate, then put sulfur on top. Then place another small layer of ammonium dichromate (which will light up from the match). Clean up the ashes wearing rubber gloves that are kept handy for general use in scientific experimentation. Put ashes into small plastic bag; dispose of through hazardous waste.

6.2.Procedure: **Light the match and light the chemicals in the top of the volcano.**

6.3.Purpose: Demonstrates hot elements expanding, breaking out, from under surface of the earth.

NOTE: This demo is not to be repeated by the children; but normally they can perform the demo on their own. So, for their follow up repetition, use a vinegar (added last) and baking soda version (mouth of the volcano).

Date: September 16, 2004

Notes: The Creation of the Cosmos

The story "Creation of the Cosmos" is not meant to contain any particular theory about how creation of the earth took place. It contains some basic facts given in a story format. At the same time it's not just facts strung together. The story is meant to appeal to the imagination.

First of all, the teacher should consider the language to be used. It is not the same language we would use everyday in talking to each other. "There appeared a vast fiery cloud, a raging fiery cloud of nothingness..." for example, provides color and drama. The language is something that achieves this and must be considered.

The whole tone of the story is to arouse much more than just interest in a subject. We recall Maria Montessori said, "If the idea of the universe is presented to the child in the right way, it will create his admiration and wonder." The language of the story provides this. Each teacher has his own story in his own words, though he can use Mario Montessori's story from 1958. The teacher's story has to feel comfortable and he should feel conviction in that story. He must practice telling this story over and over to refine its delivery.

There are some essential facts that do need to appear in the story, some key ideas. If the teacher rewrites the story, he must keep these keys therein. Besides the tone of the language, the story should include four important aspects. They are: 1) the idea of the immensity of the universe in which the earth is set. This is because, to the young child, he is often at the center of his own world (as a natural function per the first plane), but at the second plane, the child is eager to use his imagination. We can aid this and help him become aware of the reality that exists outside himself. This is a by-product of the story but an important one. The teacher himself may need to reflect on this as well. The light of the nearest star takes 4 light years to reach us. In the galaxy in which we are situated, it takes 100 million years for the light from the edge of the galaxy to reach us. "Big" simply does not convey this. The teacher has to make this hugeness something distinct in the mind of the child, has to create some new reality for him; 2) in this vastness, everything that we know of was already there in its elementary form as tiny particles. We don't say atoms or molecules

because they have specific scientific connotations. We aren't trying to give a scientific explanation as yet. Later on, the teacher can go into these but initially we are drawing an impression which contains only large strokes of fact. The 9 to 12 year olds may want to use 'atom', but this is not initially desirable; 3) that each little particle was given a set of laws to follow and these laws are universal. Every particle has to follow these laws. Likes and dislikes for other particles are noted. Reactions to temperature are noted. Heat creates motion and expansion, for example. The earth we have was created because of these two basic laws. As the earth cooled, particles got together according to the likes and dislikes and formed different substances. Some assumed a solid state, some assumed the liquid state and some assumed a gaseous state; 4) that even if we don't want to use the word God, some representation of this idea needs to be part of the story. Everything that was created has its own nature. The nature of a substance is inherent in its being. Water will always boil at a certain temperature at a certain altitude no matter what. Gold will melt at a certain temperature but at a different temperature than ice will. Plants grow toward light; it is in their nature to do so. The basis of all of this is laid in the order of creation. So, however this order is expressed, it needs to be part of the story. It is because all particles responded to their nature in an individual way that we have all the diversity we find in the universe, the different rocks, plants and so on. The key to this idea must be in our story. These ideas set this creation story apart from other creation stories. Another by-product of this idea of order is that both children and adults tend to condition and to be conditioned by their environment, but without those laws, one can see there would not be order. We need to help the child understand a new basis for these rules and for order generally.

The story should be given very early in the life of the class. The children don't need to be able to read and write. They don't need math facts. All they need is to be able to listen. If a teacher is starting a brand new class, he may want to wait until the class is settled down and he knows the children a bit. If the class is already established, the teacher can present the story the second or third day of school.

The teacher begins by inviting children who are new to the class as their obligation but then also invites everyone else as they please. The older children who have heard the story already generally will hear new things when they hear it again. We need to announce the story ahead of time to the children. One needs to do the entire story along with the charts and demonstrations. Some will say the story should be told from memory in one's own words. Some will say it can be read. But either is workable and must be used as needed in order to assure the story IS given.

The most important factor is HOW the story is delivered. The teacher must transmit that sense of wonder and sense of awe, a sense of adventure. The teacher should not try to tell it on a day when he is not feeling well. It is better to wait a day or so than to give a presentation that is lacking in enthusiasm or energy. Teachers themselves, need to take time to experience and appreciate the wonders of nature; watch a sunrise or sunset, go out at night and look up at the stars and planets. They are always there for our inspiration and wonder. We then see how early humankind must have felt the same wonder and awe. The teacher must feel these intense emotions before the child can feel them. "A Sense of Wonder", by Rachel Carson, is a good book that help a teacher prepare for this. If we can get the children aroused, then true inspired learning can unfold.

Date: September 16, 2004

Follow Up to the Story

Follow up is generally done in smaller groups than what had attended the creation story initially. Because “Creation of the Cosmos” is to arouse interest, the teacher definitely does not want to ask the child to rewrite or summarize the story. The child should not be asked to copy the charts or perform the demonstrations. (If they want to do those things of their own volition, that is different and is encouraged.) Often the children want to hear the story again. So the story may simply be repeated on another day and maybe fill in details a bit differently or express things with different colored effects. The teacher might do part of it in a more developed way or the whole story can be repeated. Subsequent repetitions are thought of as a review for increased understanding in order to allow the child's engrams to organize and to amplify a sense of awe and hopefully inspiration.

If the child shows no follow-up interest, the teacher must take the initiative. The teacher should make sure the ideas offered are at the level of capability of the children. They will go deeply into any aspect capturing their imagination. There should be good books available; on the meaning of cosmos, on volcanoes, planets, the solar system and so on.

For pupils of ages 9 to 12, it would be appropriate to go into other creation stories such as those of American Indians and others in our midst. A good book for this is “In the Beginning: Creation Stories from Around the World” by Virginia Hamilton.

Some ideas for follow up by the teacher may take the following forms:

1. Bring up the story yourself in some other context. Offer more about the story, a few more facts and more clarification about ideas. For example, the teacher could pick up on the stars, giving more facts about the stars (i.e., how numerous are they, how big are they, how far away are they ... all are areas that can be used).
2. Emphasize the extremes of cold and hot in the creation story. A simple experiment can be used to show something colder than ice: two containers filled with crushed ice. To one add some salt and then take the temperature of each of them. The one with salt will be colder. Older children can look into the chemistry of this.
3. The teacher can develop the idea of the relative size of the sun and

earth using models. What does it mean to be one million times the size. What is more accurately true? The sun chart shows the earth in the flame of the sun but it is not to scale. This can be used to call attention to scale. We could tape four charts together for a sun (to approximate the scale) and then add the single dot within the flame to represent the earth. Older children can help doing math calculations.

4. The child can be drawn into looking into the solar system (using chart 2a of the solar system) as a result of hearing the creation story. Wonder aloud "I wonder how far away our earth really is from the sun." Point out that the chart does not present proportional areas and sizes of the solar system and planets. "My First Book of Space" is another great book for children interested in space. It introduces the solar system and each of the planets. The child may want to explore this book. The child may want to make his own drawing of the solar system or a model of it in proportion.
5. Since our story mentioned the earth traveling around the sun, the children can look into how all of the planets we know of orbit the sun. Show how it looks using a (plastic) nut tied to a string when planets closer to the sun travel around it faster (by swinging the nut in a circle around and above your head.) The fact that all planets move around the sun and never bump into each other can be used to capture the imagination. Why don't the planets fall into the sun? If we drop something, this trajectory occurs; it drops to the floor. Why does this not happen to the planets? This is another idea that may intrigue a child ... the fact that the planets keep moving and avoid this falling trajectory. Planetary and lunar motion is a combination of the facts of their bodies' constant motion and their gravitational pull, mutually applied to the sun.
6. When we swing the nut attached to a string around the head, we feel a pulling force; this is what is called the centripetal force that keeps planets in their orbit. The child feels this force just as we do. This is partly why planets don't fall into the sun and this can be investigated as force of physics.

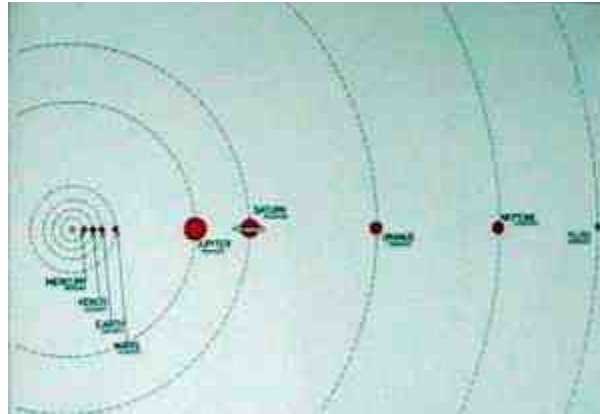


Chart 2a - The solar system

Date: September 16, 2004

Composition of the Earth

Prologue

The following presentation is given to small groups, as **the only required follow-up to the creation story**, given after the child has processed the story for an appropriate period of time and after other follow up has begun.

Begin

Remember our lesson about the creation of the cosmos? We talked about how, in the beginning, everything was very hot. Because everything was so hot, it was first in a gaseous state, and then some things became liquids.

As they cooled down and became liquids, they started to settle according to their weight. Remember we saw with the test tubes that heavier things go to the bottom and lighter things stay at the top.

The heaviest substance always goes to the bottom nearest to the center, or the heart of the earth. This is how things settle, with the heaviest at the bottom, closer to the center of the earth, and with the lighter things on top.



Chart 6a - Barysphere

The earth's center region is called the barysphere. Barus is a Greek word meaning heavy. This what Barus means – heavy. On top of the barysphere, settled the lighter part, this part we see right around here...

Experiment 2: Lithosphere

It is made up of all the rocks and land we see around us. It is called the lithosphere. Lithos is a Greek word which means stone.

But then on top of the lithosphere is settled a still lighter part. This part here...

Experiment 3: Hydrosphere

This is called the hydrosphere. It is all of the water that we see around us in the oceans and lakes. Hydro come from the Greek work hydor – meaning water.

And on top of the hydrosphere settled, a still lighter part, the atmosphere. All of the air that is around us is part of the atmosphere.

Experiment 4: Atmosphere

It provides all the air that we breathe. Atmos is a Greek word meaning vapor.



We live on the lithosphere. And it is really quite thin. You can see here how thin it is compared to the barysphere.

Experiment 5: Width of Earth's Layers

To help us visualize this, think of the earth, which has a diameter of 8000 miles straight across the sphere. The lithosphere is only about 23 miles thick. That's pretty thin compared to 8000 miles, isn't it. The hydrosphere is even thinner; it's only about 2 ½ miles thick.

Experiment 6: Relative Width of Layers

Let's imagine this yardstick is the diameter of our earth - 8000 miles across.

The lithosphere would only be this thick - $2/16^{\text{th}}$ of an inch.

The hydrosphere would be practically invisible on our yardstick.

The atmosphere would be only $1/16^{\text{th}}$ of an inch.

And the part of the atmosphere we can breathe is only about $1/4^{\text{th}}$ of that $1/16^{\text{th}}$.



Why don't we fall off of the lithosphere?

Experiment 7: Gravity

It is because the barysphere pulls everything toward itself. It pulls the lithosphere, it pulls the hydrosphere and it pulls the atmosphere. It's sort of like what you see here...

Experiment 8: Gravitational Pull

So everything is pulled and held onto the earth's surface by the barysphere. It pulls onto everything with a force. This force has a special name. It is called the force of gravity. This is why something sent flying eventually lands.

Experiment 9: Everyday Experience of the Barysphere

It goes down to the ground rather than continuing in the air. It is the force of gravity pulling it downward. That's why we also don't go flying off into space, why we stay on the lithosphere.

End

Demos for Composition of the Earth

Charts

Charts (from Geography chart set 1a-28a):

1. 6a – barysphere

Experiments

1. Barysphere

1.1. Materials: Chart 6a

1.2. Procedure: **Show chart of earth with barysphere**

1.3. Purpose: Show location and relative position of barysphere in the earth

2. Lithosphere

2.1. Materials: Chart 6a

2.2. Procedure: **Point to the earth's edge - purple on the chart**

2.3. Purpose: Show the location and relative position of lithosphere

3. Hydrosphere

3.1. Materials: Chart 6a

3.2. Procedure: **Point out the thin blue line**

3.3. Purpose: Show the location and relative position of hydrosphere

4. Atmosphere

4.1. Materials: Chart 6a

4.2. Procedure: **Point to atmosphere**

4.3. Purpose: Show the location and relative position of atmosphere

5. Width of Earth's Layers

5.1. Materials: Chart 6a

5.2. Procedure: **Point to relative sphere sizes**

5.3. Purpose: Inform children about each of the earth's layers

6. Relative Width of Layers

6.1. Materials: Chart 6a and a yardstick

6.2. Procedure: **Point to 1/8th inch mark (litho) and 1/16th inch mark (atmos)**

6.3. Purpose: Show relative thinness of lithosphere, hydrosphere and atmosphere.

7. Gravity

7.1. Materials: Chart 6a

7.2. Procedure: **Point to barysphere and other layers**

7.3. Purpose: Show how the barysphere is situated to act on other spheres.

8. Gravitational Pull

8.1. Materials: An index card, magnet and paper clip on a tray

8.2.Procedure: **Move paper clip across card using magnet on opposite side**

8.3.Purpose: Show how barysphere effects its force on other layers

9. Everyday Experience of the Barysphere

9.1.Materials: Wad of paper

9.2.Procedure: **Throw the wad of paper and watch it fall**

9.3.Purpose: Demonstrates the gravitational pull on other objects

January 13, 2004

Further Details of the Composition of the Earth

Prologue

We began looking at composition of the earth, talking about things settling according to their weight. Heaviest things went to the bottom or center; lighter things formed outer spheres of the earth. Now we get more specific. This is a small group presentation.

Prerequisites

Make sure these things are covered: Further States of Matter (degrees of solidity), The River (sediment therein).

Age

About 9 years old.

Material

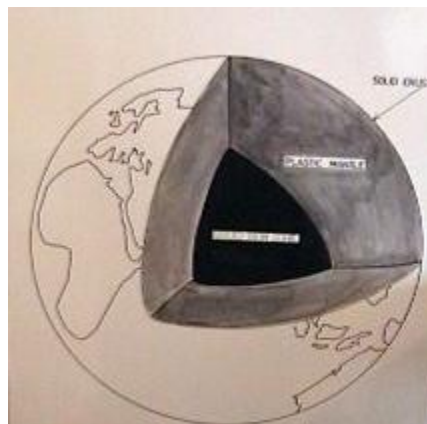
- Glass pan of water $\frac{3}{4}$ full, yellow prism representing granite that floats on the water (representing basalt). The yellow prism is part of solid geometry material set of 5.
- Towel (to dry off the wooden prism right away after taking it from the glass pan)
- Charts: u1 and u2 (traditionally called unnumbered charts, titled core-mantle chart and granite-basalt chart), Chart showing Isostatic balance,
- Plastic bag sealed and filled with air and a heavy weight from the drawers
- Two simple pieces of cardboard, and piece of bubble-wrap, representing plates.
- Steam pressure demo: flask partly filled with water and with a rubber stopper top with a hole at top, heat source, rubber tubing, gray clay mountain with a funnel set-in upright (connected to rubber tubing) filled with dry coffee grounds on the top trapping pressure under the funnel.

Begin

Remember we talked about how the earth formed and everything settled down according to its weight. Heavier things settled toward the bottom of materials and lighter things settled on top.

This is how the earth formed – heaviest things in the center. We called this the Barysphere. Lighter parts settled around the Barysphere: the lithosphere – the part we live on.

Inside the Barysphere, things also settled according to weight. (hold up chart - globe with a cutout of layers). The heaviest things went to the very center (point) of the earth. We call this the **core** of the earth.



On top of that settled a lighter part called the **mantle**. And on top of that settled a still lighter part called the **crust**. This is the part we live on. The crust of the earth is also called the lithosphere.

We know the crust is solid since its the ground underneath us. Our chart says “solid crust.” But underneath the crust is the mantle; its not quite as solid as the crust.

The **mantle is in a plastic state** – that's why it says “plastic mantle.” Underneath that plastic mantle is the core of the earth.

Scientists don't all agree about what the core is like - no one's ever been there to see it! We don't know for sure what it's like.

Some scientists feel the **core is in a liquid state** and always has been. Others think it's in a semi-liquid state. Others believe it is in a solid state. The most common and current thought is that the **core is a solid surrounded by a liquid**. Our chart says "Liquid Iron Core."

Scientists do agree on this though: the further down one goes into the earth, the hotter it gets. That's why they assumed the core was a liquid - since it is the hottest part.

Let's talk about the crust a bit more. Do you know that just within the crust, things settled according to their weight.

(bring glass baking dish of water to table along with a yellow thin prism)

The crust is made up of two types of rock: one is granite. Granite is composed of tiny little crystals and it's the lighter of the two rocks. The heavier part is called basalt.

Now if you could go back in your minds to when the earth was just forming - everything was in a liquid state and was just coming together. The granite and the basalt were liquids.

The lighter rock, the granite (hold up yellow prism) floated on the heavier rock - the basalt - just like you see here (place yellow prism onto the water - it floats).



If you get low and look at the side of the glass pan, we see most of the granite is under the water. (yellow prism floats but most of it is submerged)

This is how the **granite was floating on the basalt as things started to cool and the granite became trapped in the basalt** – just like a ship trapped in a frozen sea.

(hold up chart u1)



It's like we can see here, where we have the granite sticking up out of the basalt (lighter gray is where granite is sticking out and darker gray is the basalt)

Wherever granite is sticking up out of the basalt is where one finds the **continents** of the earth. In between the granite, in the hollows, is where we find the **oceans** (there is a thin layer of blue on outer edge). This gives us the oceans on earth.

Epilogue

According to current science, the temperature increases 135 degrees F per mile as one descends toward the center of the earth. At a later time, we could go into the thickness of these layers: mantle is about 1800 miles thick, core is about 3000 miles from edge to center – 6000 miles in diameter. The crust is very thin – 22-23 miles thick.



Begin

Together, granite blocks and basalt blocks form the crust of the earth. However, the **crust is not one solid continuous shell** covering the earth. It's more like a cracked egg shell, broken into huge pieces.

These huge pieces are called **plates**. That's what makes up the crust of the earth. Now these **plates are floating on the plastic mantle**. Sometimes they drift, bumping into each other. Most of the time the plates move but we are not aware of it. Sometimes we do become aware of it.

But remember when we did the river model and we talked about how much sediment is washed down by a river? Year after year, the sediment is deposited into the ocean at the end of the river ... the Mississippi alone deposits 406 million tons of matter in one year. Can you imagine that?

If we took all of that sediment and formed it into a column with a base of 1 square mile, that column would be 241 feet high. Imagine that! That's only for one year! The river flows on day after day, year after year.

(take sealed plastic bag, filled with air onto table)

You can imagine how much pressure that sediment exerts on the

earth. As that sediment is dumped year after year, the crust starts to get pushed down

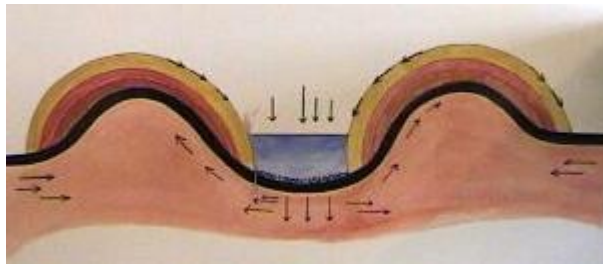
(set weight on plastic bag - it is suspended on the air).



Also like we can see here, when pressure forces the crust to go down in one place, it expands in other places. Of course, on earth this happens very very slowly. If we take the weight off, the crust comes back up. This also happens very slowly.

So in one place, the crust will go down but in another place it is forced up.

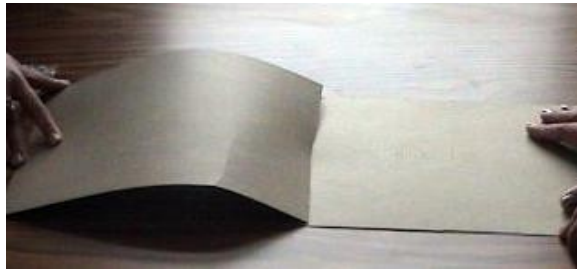
Like you see here. (hold up chart - water between two raised portions of crust)



Land is washed down and the land rises up, here on the edges. Here, where sediment is coming down and settling, the crust is forced down. This is possible because the mantle is plastic and can allow for this movement. And the crust can become plastic when pressure is applied to it - as we saw when speaking about states of matter.

Remember, we said the crust is broken into huge plates and those plates can move around. As these plates move about they rub against

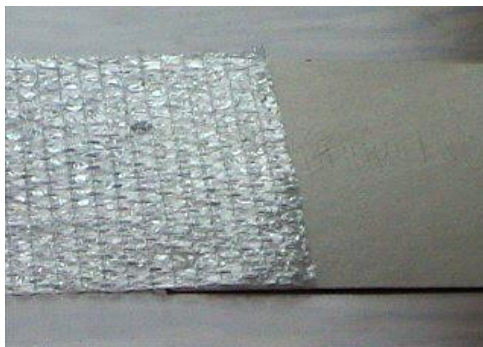
each other a bit (push cardboard pieces together to show their collision) Sometimes, one pushes against the other like this (push edges together until one bubbles upward).



Very slowly, one begins to rise as a result of the pressure applied at the edge; one of the plates develops a fold – just like we see here. This happens over many many years.

Sometimes, mountains are formed in this way. Sometimes, instead of rubbing together or folding, scientists have found there are ridges that form which mark off the edges of these plates. Along the ridges are cracks where molten rock gets pushed up – especially under the oceans. The molten rock is pushed up through these cracks and it forms new crust.

The crust starts to spread away from the ridges and the ocean basin starts to get wider. As this happens, sometimes the edge of one plate is forced under the edge of another plate.



(use piece of bubble-wrap and a piece of cardboard to demonstrate)

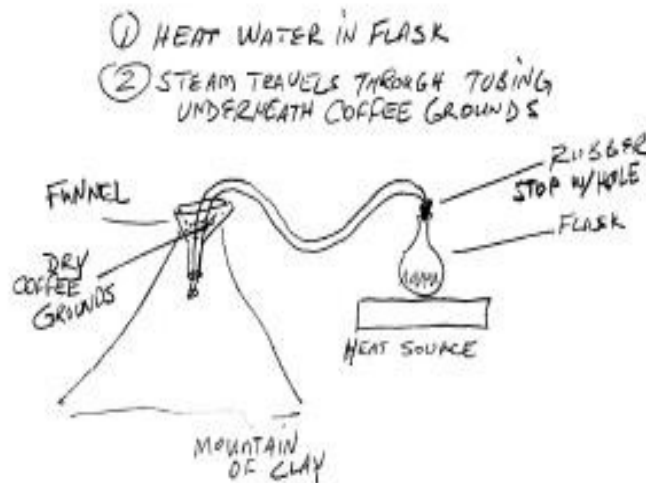
The sinking rock of one plate goes down – is crushed or melted from the heat under the other plate. This can be one cause of

earthquakes.

Occasionally, as the molten rock comes across water, it turns the water into steam – into gas. And this starts to build up under the crust of the earth. The crust is a solid and it can take a lot of pressure – but at a certain point, it can't take anymore. The steam, molten rock and other gases explodes out of the earth – this is a **volcano**.

Where do you think those points are where the crust of the earth gives way? Toward the edge of the plates – right? (point out on the earlier charts where volcanoes appear)

(do demo of steam pressure if appropriate to interest and readiness / preparation)



Epilogue

Isostatic balance or isostasy is the quality where the land rises in one place but must drop on another or vice versa. In the book "Planet Earth", there is very clear information about plate tektonics . We need good current books on this for follow-up.

Subduction area is when one plate goes under another. Go into as much detail about earthquakes as interest warrants. Living in California or area with many quakes, one might do much detail.

See illustration for demonstration of the force of steam. From here, the children may want to study volcanoes – all different kinds and how they got their names, parts of a volcano, etc. Granite and basalt are

igneous rock – the children may also want to study rocks after this lesson.

January 13, 2004

Formation of Mountains

Material

- Books with pictures of volcanic and fold mountains, etc.
- Charts: one has 3 illustration of mountain types, second has horst / rift crust movement.
- From solid volume geometry material (blue hexagonal prism)- on end we attach paper with colored strips to show layers of strata - rubber band to hold pieces together.
- From solid volume geometry material (green rectangular-rhomboidal prism)- on end we attach paper with colored strips to show layers of strata - rubber band to hold pieces together.
- From solid volume geometry material (2 gray rectangular prisms and a third yellow but otherwise prism divided into 5 equal rectangular prisms - on end we attach paper with colored strips to show layers of strata - rubber band to hold pieces together.

Begin

Remember, we talked about volcanoes. Some of the mountains on earth have been **created as a result of volcanoes**. They are called **dead volcanoes**.

The lava has come out and hardened as it cooled - and has left this raised mass of land - a volcanic mountain. These mountains are quite distinctive - because of their shape and because the land around them is typically flat, like we see here.

(hold up picture of flat land with on volcanic mountain)

We can also see that here on this top illustration. (point to new chart)



In fact there is a ring of mountains like these in the Western US, in Oregon and thereabouts.

But **most mountains are formed by pressure** on the crust of the earth and the movements that result from that pressure. A fold is created by crust pushing against other portions of crust.

It creates a fold like we see here (orange and other felt pieces folded upward).



These are called **fold mountains**. The Andes and Appalachians are fold mountains. That's what we see on the bottom two illustrations on this chart.

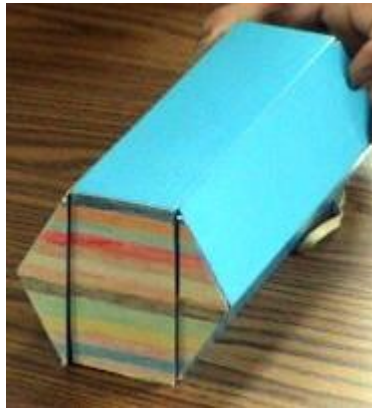
(hold up the chart just shown)

Here at the bottom, the layers are straight.

If there is pressure exerted, you end with something like **the middle illustration** - a fold develops in the layers.

Layer Movement

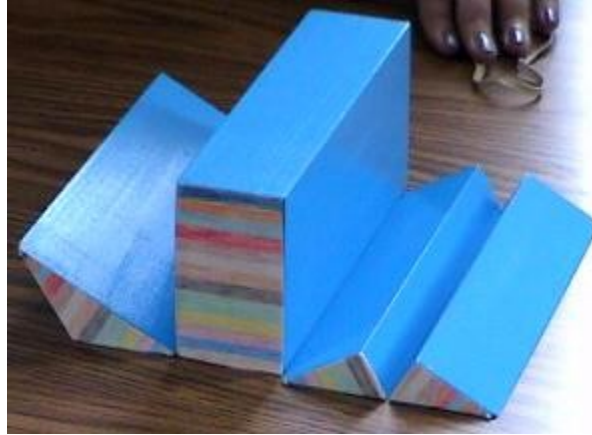
Sometimes, if there is a landslide or a cut has been made into a mountain, one can see how layers rearrange themselves.



Sometimes there is pressure on the crust and it causes the crust to crack. Then the layered pattern doesn't line up perfectly any longer. The layers of strata move around.

These cracks are called **faults**. As the pressure continues, the land might slide.

(allow solid to divide into pieces dramatically)



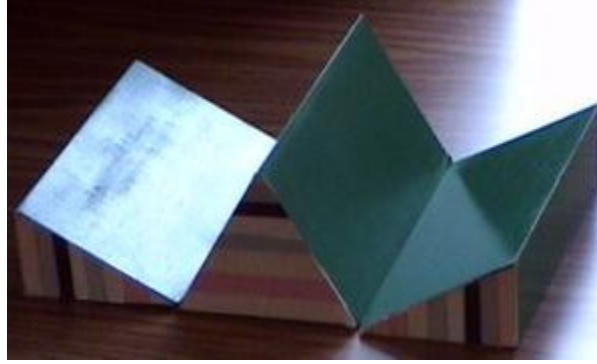
Do my layers line up now? They don't line up at all, do they? Look at how they've been rearranged. Here, the cracks are vertical – these are called **vertical faults**.

But faults are not always vertical. (use new figure to show in same manner)

Sometimes they are oblique – like this one.



The pressure on the crust continues to make those faults bigger and bigger and at some point – they slide and rearrange themselves.



(allow figure to come apart again in dramatic fashion)

Sometimes something else happens – cracks appear.



(three prisms side-by side simulating one piece of crust; gray ends and yellow in center)

The land between the cracks slides down and you end up with something like this.

(take one layer from yellow middle stack)



This is called a **rift valley**. It's caused by a rifting in the earth's crust.

Sometimes the opposite happens. The cracks appear but this time the land in-between the cracks raises up due to the pressure.



(place hand under yellow stack of prisms in center to move it above the other two parts)

This kind of mountain is called a **block mountain** or also called a **horst**.

(show new chart - horst / rift mountains)



We can see the same thing here. (shows horst and rift movement)

Here is where land has been pushed up forming a block mountain or horst - and this is where the earth's crust has been moved downward.

Often times you will find a rift valley right next to a horst and they are inter-related - one has caused the other.

You may have heard of the Great Rift Valley - primarily in Ethiopia but it spans several countries in Africa. It was caused by this type of movement in the earth's crust.

Epilogue

“The Wonders of Nature” is a great book for the classroom, showing fold mountains and other specimens of geographic formations.

Following this presentation, the children may want to look into the history of various mountain ranges on earth; which are old, which are new, how was each formed – and so on. Appalachians are very old. Himalayans are very new.

This makes sense to us if we know their sizes. This lesson is a good segway into relief maps or names of mountain ranges. We are showing results, not direct causes of these mountains.

January 13, 2004

Geography Nomenclature

Prologue

These ideas apply more broadly than to just geography. The first thing we remind ourselves of is that we have **second plane children**. What are their needs, sensitivities and characteristics? We know they have **changed from an introvert in primary to an extrovert** - not concentrated on themselves now; they have turned outward to a community - working in groups. They are eager to find out about society and explore the physical-cultural environments that surround their lives. They need to be able to explore things that are beyond the reach of their senses.

In the first plane, language serves the child as building blocks, creating order in his mind - serving to organize the impression he is taking in. When he learns a word such as table or cup, all of the impressions that have been taken in prior to that related to those kinds of objects - are stored in that mental grouping.

In the **second plane of development, the child now needs language mainly for his thought processes**. Now, that is how his mind develops. This is why the sensitive period for language and for vocabulary which was so characteristic of the first plane of development - now disappears. It is replaced by other sensitivities that are characteristic of the second plane.

It should therefore follow naturally that nomenclature work, which forms such a large part of what the young child does, cannot play the same kind of role in the elementary class. Because of the changed characteristics, **most elementary children will not sit with nomenclature for long periods** and repeat the activity over and over again - like primary children will do - though there are exceptions.

Introduce Required Terms Early

Yet, in the elementary class there will be new terms and nomenclature the children will be required to learn. One of the

first things teachers need to do is to become familiar with what that terminology is; what does the public system require – as one example. Teachers need to make sure these terms are incorporated into lessons and helping the children use them. Anything they learn over and above that is a bonus.

Now that we have **understood the terms they need, we now must see if any of the terms can be given very early** on in their elementary experience. If they are still close to the first plane developmental make-up, it might be easier for them to learn these new terms right from the start of their elementary class. **Sometimes, the terms can be given as an extension to the work they did in primary.** This work can then run parallel to other activities in the classroom.

Much of the geography, botany and zoology nomenclature can be given in this way. If we know the details of what they've already learned, this is **useful in segueing to the next level of knowledge** with nomenclature. If they've worked with **puzzle-maps**, there is much nomenclature that relates to geography which could be gone into in greater detail. Names of rivers and mountains – **any geographical aspect that can be related to those puzzle maps** can be picked up on in the elementary classroom.

Requirements for Learning

Interest

There are two things required for learning to take place: interest and repetition. They **need to care about learning something and then they need to see it more than once.** If we are able to create an interest, then generally, half the battle is won. The interest will often help the children engage in repetition on their own.

In the first plane of development, that interest is provided by the sensitive periods. In **elementary, one needs to make use of other characteristics of that plane, in order to stimulate interest** in them. Relating the nomenclature to the larger picture before them – of air and water, for example – is one thing that can ensure interest. If there are names the children need to know but don't know, the **terms can be related to that bigger picture.** For example, they often need to know the names of types of clouds – often part of the public school curriculum; this is related to the work of air and work of water. So, though we don't have those particular names related to clouds, we

can bring those names into the lessons we give on air and water.

After new terms are introduced, **help the children realize that they will not remember them in just one hearing.** Maybe they themselves can think of some of their own means to reinforce the new vocabulary through repetition. They are always more receptive and enthusiastic when ideas for their development come from them. Remember, these children can be reasoned with and we need to do this for their greater development – as this is how they become responsible for their own personal development in all areas.

Other ways to awaken interest would be to explain, in an initial introduction, some of the differences that characterize two or more similar geographical features. For example, the terms hill, mountain and plateau – are all raised pieces of land; what distinguishes one from the others. We could explain those distinguishing characteristics as we introduce these terms. Waterways can be introduced in the same manner. This is simply more grading and categorization.

Another way to awaken interest is to **explain how a particular geographical feature came to be formed.** We do that with “The River” as well as in our discussion of mountains. When we did “The River”, we also had the overlay map in which we saw various cities that developed along some of the waterways. This can be one way in which the children become interested in major cities and in learning their names and other information. Names of cities could also be brought in through history studies or demographical studies. Sometimes, the children will want to research the origins of names of cities, rivers or mountains; what does the name mean, who named it, when was it named and so on. Lake Disappointment in Australia is an example. How did it get *that* name?

We can imagine that **this relates to tales of adventure and this appeals to the child's characteristic of hero worship and admiration.** They find tales of adventure stimulating and fascinating – which can be made use of to help them learn those names required. They might be doing a history project about a group of people who lived in a certain place – and they might become involved in studying the geographical terrain of that place. Mountain passes have played an important role in history. Rivers and waterways are equally central to the movement of people and development of societies. The starting point for geography work may very well come out of history projects. Again, this helps us integrate the terms they will need to know.

Still **another way to awaken interest is through etymology.** We've used this in all different areas of study already. We want to help

the children become aware of how, knowing that a giving prefix means, also helps them understand other words beginning with that same prefix. The etymology of terms can make them curious and interested in some of these ideas. By the same token, we need to be careful not to force etymology on all of the children all of the time; it can then become a chore and a bore. Use it judiciously.

Another way of **awakening interest is through making models**. Sometimes the children get involved in making a model of their local terrain – sometimes even to scale – bringing in math. Maybe they will model their state's terrain. Again, we can expand their vocabulary in these contexts. They may model flowers, insects, railroads or whatever – we must bring ideas in that capitalize on this – while ensuring the work remains within their capacity to be successful. They need to finish and be proud of their work in all cases when possible.

This is not a definitive list but is a beginning in thinking of how we can stimulate interest.

Repetition

One of the ways in which we **ensure repetition is that we give our lessons to small groups** of children; so each child in the group gets a turn – but the others in the group watch and receive repetition passively.

We also provide **material for repetition**. When it comes to nomenclature, it means we provide nomenclature material. We have **nomenclature material for geography and botany**.

There is a booklet, pictures with no names, loose labels, pictures with names attached and definitions with the terms missing. These are used just like all other nomenclature material is worked with. The material is made incorporating the Montessori principal of isolation. For parts of the river, as one example, the first thing is a picture of all parts. Afterward, we see each part isolated in the context of that same whole picture.

In **geography, we have two racks - one that has to do with water, hence the blue covers - and the second having to do with land forms, using a brown covering for the booklets**. Besides parts of the river and parts of mountains, there are some similar features grouped together in each of the other booklets; for example, strait, channel and canal can be found together in one of the blue booklets.

The cards with loose labels and matching picture with the names on it – that set is **meant for children who aren't yet reading**. They are at the word-reading level. So this is what the labels are for. If you have children who are all readers, they aren't needed. For them, one goes right into definitions with the terms missing – to make use of their reading abilities. **Loose labels are for non-readers or beginning readers**. So make sure to use some color coding to help everyone distinguish booklets – land forms, water forms, etc.

Another way to ensure repetition would be to encourage them to write their own definitions or to make their own nomenclature material. They can **draw their own pictures, color in the isolated part, write their own definition** and so on.

Remember periodically to **stop and ask the child, “How would you describe a river?”** Why don't you write it down. That was a great definition!” Later on, we can encourage them to compare their definition with the one in the nomenclature material or in other sources. This helps them refine their ideas, while making it their own at the same time. Finding pictures of specific geographical features to make their own collections is another means of repetition. **Visiting places to experience geographical features** is another. These sorts of activities are what make the work come alive for the children. The children might want to make their own overlay maps – implying that we need to care for our environment rigorously to enable them to do such projects.

Sometimes, we'll have a group of children prolific in making their own nomenclature and maps. We need to guide them to take them home at a certain moment. This causes clutter but more importantly, if work from other children is there, we take away from the work and opportunities of children that follow who might want to do the same activities.

When we think of activities to suggest to the children, **make sure that these activities involve the mind rather than suggesting things that merely copy work**. We avoid copying in this second plane of development; they can recreate something but the original should be put away first – to allow the child to reconstruct, express and refine his own image. Remember that some of this work will be done connected to research projects. Make them aware that this information is required by the public system if appropriate.

Be careful to not make nomenclature the be-all and end-all of what goes on. It is not cosmic education but merely part of its foundation. We have all of our stories to tell and we can connect

nomenclature work to those stories. Avoid the danger of focusing on nice deliverables – pretty products pleasing to parents and others. It becomes an easy means of gratification but misses the mark of mental development. Others need to be aware of this also.

Suppose you want to introduce the parts of the mountain to the children. You could build it up as you describe it or prepare it ahead of time.

Example Lesson - Mountain Nomenclature

Material

Small mountain of clay, tickets, tooth picks and a pencil.

Begin

A mountain is a raised mass of land which is quite high.

It has steep sides here (show sides of clay mountain) which are called **slopes**.

The part down here where the mountain starts to grow from is called the **base** of the mountain (point to base)

The top of the mountain is called the **peak**. The very highest point is called the **summit**.

(second period)

Who would like to show us the slopes? **Who can point to the peak?**
Where is the base of the mountain?

(write 'peak')

I'm going to write something down.

(write one of the terms on a ticket and have the child place it using the tooth pick, as a flag into the mountain, at the appropriate location)

Here is the peak. **(continue for other terms)**

Epilogue

The **children can take up the same type of presentation for themselves**. If there is a writing problem in some manner, they can use prepared tickets. This could logically be followed up with the nomenclature material on the parts of the mountain. "I wonder how this booklet defines the peak"?

Anywhere there are **terms and definitions that the children will need to know, we handle them like the third rack in botany** – each term with a cut-up definition on strips of the same dimensions. These can be mixed up and then the child can reconstruct the definitions; this gives the children a manipulative activity but also makes them think and discuss related to those terms and definitions.

Names of countries, cities, states, waterways, mountains and so on can use **pin-maps**. There is the map itself along with a box of color-coded flags. For North America, red flags are the cities, green are the countries, blue are waterways and yellow flags are mountains. The map itself marks these points as well – for example, mountains have a small star located where the holes are placed that receive the flags. Encourage the child to place as many flags as he can based on what he knows; set aside the other for review using an atlas and other materials.

Another material is that of **models of continents and small models of mountains to scale**, for example. These maps are teacher-made of stiff paper. The models of the mountains, or whatever is modeled, is simply placed on the table. Again, the child should have an atlas nearby to find correct placement or to control-check his work.

The Nature of the Elements

Date: September 16, 2004

Three States of Matter

Materials

- Heat source with heat-proof mat, spoon, piece of wax, clear glass bowl with water in it, towel for drips; all on trays.
- Use solids the children won't have another association with if possible; something large like a brick or prism, variation of shape such as a steel rod, a piece of broken flower pot, an index card or piece of wax paper. Include a brick on the floor, along with a hammer used to break the terra cotta.
- Clear bowl half full of water on a tray with towel for drips.
- Clear glass containing iron shot (bb shot) on a tray.
- Clear vessels of various shape and size - not too large, 3 to 4 in all. Colored water in a pitcher. All on a tray.
- Pitcher with a few holes in bottom and sides toward the bottom, basin large enough to catch water spray, pitcher with no holes filled with water. Colored tape to cover holes.
- A large prism or brick.
- Some perfume in an atomizer on a tray.
- Clear small glass, basin, water in pitcher, two pieces of paper about 6 in sq., towel.

Prologue

This presentation goes a different direction after the creation story. One small group might hear composition of the earth while another group might receive this one. Important to remember is that there should be some time between these presentations to allow engrams to organize a bit. Begin by making a reference to the creation story in order to refresh their memory about what was said before.

Once again we are giving the child impressions, this time about solids, liquids and gases. Some of the characteristics of each and some of the results of those characteristics. The teacher needs to use personification and some magic. He needs to have fun with this because they appeals to the second plane characteristics. This is different from traditional education as well as from the primary level.

If we can make these impressions in a dramatic way, they are going to remember them. They will be able to refer to them and use them to develop their understanding about the earth. This is why we say the teacher has to be an actor or an actress. Stand on the chair, crawl on the floor ... whatever it takes to keep attention and capture imagination.

Begin

Remember when we talked about the creation of the earth? We said that each tiny particle was given a set of laws to follow. One of those laws was that with more heat, particles become looser and spread away from each other. With more cold, particles hold more tightly together and move more slowly. When something melts, the particles of that thing hold together very loosely and are spreading away from each other.

Experiment 1: Melting a Solid

With heat the particles of this solid become looser and spread out. It's turning into a liquid. You can see it as it changes.



continue heating, attention to the demo, pour liquid out into cold pool of water at the end

With cold temperature, it once again becomes a solid

(hold up solid, remove hot plate and other material for this demo from table.)

Once again, this is what happened when the earth formed. As it

became cooler, particles of the same kind started to cling closer and closer together, until solids were formed. The particles of solids cling together very tightly.

(take tray with solids and place it on the table)

Here I have some solids. Let's see if I can separate this solid.

(Struggle to bend or break the first solid)

Maybe this one can be broken apart.

(Try again)

The particles of a solid cling so tightly that the only way we can separate them is with force.

Experiment 2: Breaking a Solid

Even then the particles are still clinging together. *(hold up the broken pieces)*



We've looked at solids. What about liquids. Are they the same as solids? The particles of liquids cling together. Remember we talked about that. But they cling together more loosely than the particles of solids do.

Experiment 3: Liquid Particles are Loose

With liquid there's no problem passing my hand right through it like this. The solids are not so accommodating. I can't pass my hand right through one of the solids like that. There's no problem passing your hand right through the particles of the liquid because we recall that their particles move over each other.

Experiment 4: Liquid Particles Roll

This is why they are so much more accommodating. Their particles are so loosely held together. In fact, the particles of a liquid are so loose and so accommodating, that they take the shape of whatever

they are in.

Experiment 5: Liquid Has No Shape

They have no shape of their own but follow what they are in, as you can see here. One is tall and regal. Look how curvy this one is. This one is short and rectangular, isn't it? Those liquids are so accommodating, they take the shape of whatever they rest upon. The solids we have over here aren't like that are they?

(move back to staging table)

These each have a shape of their own. (picking up a solid from the table)

Experiment 6: Liquid Press Sideways and Down

As the particles of liquid are always rolling over each other, they push both sideways and downwards.

(Pause to watch water leak out of pitcher. Return pitcher to side table)

So what about solids? Do they press both downwards and sideways like the liquids? Let's see.

(hold a solid on top of the hand)

I can feel it pressing downwards.

Experiment 7: Solids Press Down

We can feel it pressing downwards can't we? But what about pressing sideways? Can I feel that?

(Try to hold the solid lightly on the side to see if there is sideways force. Allow children to try to feel it as was done just above.)

Would you like to feel it? So solids only press downwards. What about the gases? Their particles are very very loose. So loose that there is practically no resistance. I can move my hands this way, that way - they just let me pass right through. They say, "be my guest, I won't stop you." Their particles are so loose, that they push in all directions.

Experiment 8: Vapor Presses Every Direction

Has it gotten to you yet? Do you smell it towards the floor. Oh, Vicki smells it now. Maybe if you stand up on your chair, you can smell it up there. Has it gotten to that side of the room? Oh, it's gotten to Becky. Is it also under you by the floor?

(Place basin on table to catch the water)

I have one more magic trick to show you.

Experiment 9: Air Tight

Why doesn't the water fall out? Air pressure, the force of air is holding it in.

Epilogue

What did we show? For solids, we can conclude they have a shape of their own because the particles cling so tightly. They also push downwards. For liquids, we conclude there is not shape but liquid takes the shape of whatever it rests on or within. Liquids press downward and sideways. For gases, we conclude that they don't have a shape of their own and they tend to go all directions.

After we've done demos like this, make sure to connect this experience to their environment. For example, when eating lunch, ask the child if their chocolate milk is a solid, liquid or gas.

Demos for Three States of Matter

Experiments

1. Melting a Solid

1.1. materials: heat source, heat-proof mat, spoon, piece of wax, clear glass bowl with water in it, towel for drips. All on trays

1.2. procedure: place wax on the spoon, wait till it melts and dramatically dump it into the water. Pull the wax from the water as a solid again.

2. Breaking a Solid

2.1. materials: Use solids the children won't have another association with if possible; something large like a brick or prism, variation of shape such as a steel rod, a piece of broken flower pot, an index card or piece of wax paper, a brick on the floor, hammer.

2.2. Procedure: Place solids on table within the tray, a random assortment of solids. Try to dramatically and enthusiastically, try to separate each one. Take the hammer, place terra cotta flower pot on top of the brick and break it into pieces.

3. Liquid Particles are Loose

3.1. materials: clear bowl half full of water on a tray with towel for drips

3.2. procedure: pass hand through the water in a cutting motion.

4. Liquid Particles Roll

4.1. materials: clear glass containing iron shot (bb shot) on a tray

4.2. procedure: tip glass side to side to show how the bearings roll over each other

5. Liquid Has No Shape

5.1. materials: vessels of various shape and size not too large 3-4 in all. Colored water in a pitcher. All on a tray.

5.2. Procedure: fill each vessel with water, each time calling attention to the shape.

6. Liquid Presses Sideways and Down

6.1. materials: pitcher with a few holes in bottom and sides toward the bottom, basin large enough to catch water from, pitcher with no holes filled with water. Colored tape to cover holes.

6.2. procedure: pour water in and pull off the tape one by one. The tape saves the drama for the last moment. Let the water leak out into the basin. Use towel to cleanup drips.

7. Solids Press Down

7.1. materials: large prism or brick

7.2.procedure: lay one hand on table and place the brick on top of the hand. We demonstrate it this way in case the child repeats the exercise and the brick is too heavy. When we ask, "Does it press sideways?", place both hands lightly on the sides to sense any force or pressure from the sides. Let every child feel these things as well if it is a small group.

8.Vapor Presses Every Direction

8.1.materials: Some perfume in an atomizer on a tray

8.2.procedure: at one end or side of the room, spray it once. Then wait for it to get to all parts of the room and querying those in the group to see if they can smell it. Be sensitive about how much is sprayed and where due to allergic reactions.

9.Air Tight

9.1.materials: clear small glass, basin, water in pitcher, two pieces of paper about 6 in sq., towel

9.2.procedure: fill glass as full as possible with water so there is no air left in the glass. Put paper on top of the glass. Make sure there is a seal between paper and the glass rim. In one smooth movement, turn the glass over.

Date: September 16, 2004

Further States of Matter

Solids

Materials

1. Felt mat
2. Balsa wood strips
 - a) 3 of varying length between 1 and 3 feet long
 - b) 3 of varying thickness between $1/16^{\text{th}}$ and $1/4^{\text{th}}$ inches
3. Set of palm-size weights – 100, 200 and 500 grams
4. A candy-bar shaped clay figure
5. 4 bricks
6. Some solids on a tray, one of which cannot change shape, including:
 - a) a steel rod
 - b) An eraser as a elastikos solid
 - c) A piece of clay to illustrate plastikos
7. Three spheres, about the same size, including:
 - a) a Styrofoam ball
 - b) a rubber racquet ball
 - c) a sphere of clay

Prologue

This is presented a bit after and complementarily to Three States of Matter.

Begin - different type of solids

We're going to look into solids a little further. Since these three objects have a shape, they must be a solid, right? Let's find out.

(choose and try to bend the steel rod)

I can't do anything with this one.

(choose and try to bend the eraser, see shape change and restore itself)

Oh, look at that – this one bends a bit. (show the bend to each child)

What about this one?

(pick up and bend a small candy bar of clay. Take it from the container just before presenting, so the clay is still soft)

Look at that, when I apply pressure to change its shape it keeps its new shape. Let me try this one.

(choose and try to bend or crush the Styrofoam sphere)

This one doesn't bend at all does it? And how about this one?

(choose the racquet ball and show it will crush a bit)

It's taking a lot of force to bend... but it bounces right back!

Let's try this one.

(crush the clay sphere)

It keeps its new shape, doesn't it?

So, solids don't seem to all be the same. Some solids won't change shape. Some will change shape, and bounce back to its original shape, no matter how much force I apply. And some change their shape when

I apply force and stay in the changed shape.

The ones that keep their shape under force are called **rigid**, from the Latin word rigidus. They are stiff.

The ones that change their shape under pressure but go back to their original shape when the pressure is removed, are called **elastic**. That's a word coming from the Greek word elastikos, meaning to revive or recover.

And the third kind of solid that changes its shape under pressure and keeps its new shape is called **plastic**, from the Greek word, plastikos, meaning to mold or to form.

End



Begin - Rigid Solids Become Plastic or Elastic When Conditioned by Pressure

Remember (the other day) we saw rigid , elastic and plastic solids. Today we're going to look closer at rigid solids.



Place bricks and balsa wood as above. Begin with long balsa strip and with bricks toward the middle.

We're going to watch and see what happens.

(place the weight, which clearly stresses the wood only when bricks are at strip's ends, onto the long strip. place the bricks toward the center of the strips)

Did anything happen? Not really, right?

(separate the bricks further apart and replace the board)

Now what happens. It is bending isn't it. How interesting. Let's separate them a bit more.

(separate the bricks further until they are at the long strip ends)

Are you watching? Look at that.

(dramatically drop weight on the board and see it collapse to the table under the weight)

So a rigid solid can become plastic when a constant pressure is applied, depending on its length. Now we'll look at the same thing but in another ways. We're going to use these three strips this time.

(set shortest balsa strip on the bricks. place a weight, that bends the longest strip, on top of the shortest strip at strip ends)

Anything happen? Not really, huh? Let's try the next one.

(place medium length balsa on bricks, with bricks at strip ends. Carefully place the same weight on top)

Anything happen this time? Yes, just a little bit of bend this time. Let try the last one.

(Place longest balsa on the bricks, with bricks at strip ends,

and place same weight on top)

What happened this time? It bent a lot didn't it? **So we can see that depending on its length, a rigid solid can become elastic when pressure is applied.** When we take the pressure off what happens? It goes back to its original shape.

End



This time, I'm going to use these balsa strips of the same length but differing thickness.

(place thickest solid balsa strip on the bricks and place a weight, which will bend the thinnest strip, on top)

It moved just a little bit this time didn't it? Let's try the next one.

(place medium thickness balsa on bricks and the same weight on top)

Look at that, this time the wood bends more. Now let's try the last one.

(Place thinnest board on bricks and dramatically place the weight onto the board)

This time it bends much more.

So we see that the thickness of a rigid solid will determine how elastic it becomes when pressure is applied.

Now we'll take the middle thickness of these three strips and this time we'll use three different weights (500 gr., 200 gr. and 100 gr.).

This time we'll change the pressure or force we apply. Should we start with heaviest or lightest? Lightest, okay let see what happens.

(place 100 gram weight on medium balsa)

What do you see. Yes, it bends a little bit. Let's try another, one a bit heavier.

(place weight on board, using same strip with a 200 gram weight)

It bends a little more doesn't it? Let's try out heaviest weight and see what happens.

(place the weight on the board, using same strip with a 500 gram weight)

Now it bends much more. So, **we see that, how elastic a solid becomes, changes according to how much pressure is applied.**

(place 200 gram weight on the medium thick board, right in the middle of the strip)

What's happening here? It bends a little. But suppose we put the weight closer to the end.

(place weight on the edge close to the brick instead of the middle so the bend is gone)

So what happens? This is another fact of nature that if you ever walked across a rope or bamboo bridge, you've experienced. **The amount of bend or how elastic a rigid solid becomes *depends on where the pressure is applied*.** This is something that has to be taken into account when building a bridge, right?

End

Epilogue

Recap of realizations: First, a rigid body can become elastic when pressure is applied. We saw that in two ways (moving bricks apart and using different lengths of balsa strip). Second, we saw a rigid body can become elastic when pressure is applied depending on its thickness and length. Third, we saw a rigid body can become elastic depending the amount of pressure we apply. Fourth, we saw that how elastic a rigid solid becomes depends on where the pressure is applied.

Liquids

Materials

Three test tubes of water, oil and honey (or molasses). Small smooth glass flat plate, bowl into which the glass plate will terminate and catch liquids. Heat source, heat-proof pad, Pyrex of water into which test tube can be placed. Test tube holder.

Prologue

This lesson is given on another day, shortly after the lesson on solids.

Begin

What about liquids? Are they all the same? We saw that solids are not all the same.

I have three test tubes here and we are going to run a race. The thing is I can only pour one at a time, so you need to watch closely and see who moves the quickest. Can you do that?

(turn on the heat source to begin heating the water. pour water down an incline into a glass baking bowl)

Pretty fast, right? What about this one?

(pour oil down the incline)

Were you watching? This one seemed slower, didn't it.

Lets try this one.

(pour molasses down the incline)

This one's really slow isn't it?

The one that ran the fastest is called **fluid**. The ones that don't flow as fast are called **viscous**; that word comes from the Latin word viscum, meaning mistletoe, because it clings just like mistletoe clings to what it is growing on.

I wonder if it matter if we applied heat to one of our viscous liquids.

(submerge the test tube with molasses into water in the Pyrex container on the hot plate. When the molasses is heated, pour it down the sloped glass as before)

Wow, with heat, a viscous liquid becomes more fluid. But why does it slow down as it moves down the incline? What do you think? (Because it is cooling as it moves.)

End

September 17, 2004

Different Ways of Combining Matter

Materials

Solution Tray A - glasses / bowls

1. small clear glass with water, small clear bowl with salt or sugar, scoop, stirring rod
2. small clear glass with water, copper sulfate (dry blue) in small clear bowl, scoop, stirring rod

Solution Tray B - beaker / bowls

3. small beaker with a lip, containing a small amount of full-strength hydrochloric acid, a large glass bowl on a heat-proof mat containing a small marble chip in a smaller heat-proof Pyrex bowl

Suspension Tray - four test tubes in rack

4. test tube with 1 inch water and second test tube with small amount of fine yellow sand such as that used for sand art, stirrer and stopper
5. Test tube with 1 inch water, test tube with powdered chalk, stirrer and stopper

Mixture Tray - two flat dishes

6. Coarse iron filings and fine salt or sand, mixed together on a small saucer
7. Coarse iron filings still in container, fine salt or sand in a container, small dish, stirrer

Chemical Reaction Tray A - four test tubes in rack

8. test tube containing blue liquid (water plus copper sulfate, just blue enough so reaction can take place but you can see what is happening). Second tube containing a drop or two of ammonia, tube holder.
9. test tube containing orange solution (water and very little potassium dichromate so it's not too orange), second test tube containing several drops of a lead-nitrate solution (comes in dry form, so add water), tube holder

Chemical Reaction Tray B - cup, saucer, Pyrex dish and beaker

10. white cup containing a drop of hydrochloric acid, white saucer,

hidden cotton ball with ammonia on it.
11. larger beaker containing sulfuric acid barely covering the surface of the bottom of the beaker, Pyrex dish with 1/3 full of sugar on a heat-proof pad

Prologue

Some particles like each other and other do not. We are again going to do some experiments to provide some impressions. It is because of this law, how things come together that we have all of the diversity that we find all around us, the variety of minerals and rocks, the variety of plant life. What is important is to see how we will do these presentations with the children. This is on another day after we've done some work, following the creation story.

Begin

Remember we had the creation story. In that story we talked about all of the particles that make up our earth, each coming with their own particular laws that they had to obey. Each of them had to follow the laws that they were given. Some particles like each other, and other particles did not. We are going to look at this law today of likes and dislikes.

Demo Solution of Salt

(tray with glass of water and a bowl of sugar in the table)

Here we have two substances. When we put them together,

(mix water and substance)

they like each other so much that the one disappears into the other one.

(hold up glass to see. Switch trays, new is same setup with other substances. Mix them as before)

Here we have two other substances. Just like the first ones, they like each other so much that when we put them together, the one disappears right into the other. They become one.

Demo Solution of Copper Sulfate

(switch to new tray as before)

Demo Solution of marble and hydrochloric acid

Now I have two more. This time I have a little stone.

(place liquid onto stone in the bowl)

Let's see what happens when we put those together. You may come closer and take a peak but don't touch anything. Even the stone has somebody that likes him so much that when they are together, the stone melts away, and they become one. This one takes a little while so this is why we started with a small piece. We'll set this aside and check on it later.

(put tray away, set test tubes in holder on table)

Demo Suspension with Sand

Here I have two more and I'm going to put these two together.

(mix test tubes one into another and mix)

Hm, not much is happening here is it. Maybe if I put a stopper in and shake it. Let's try that.

(put stopper and shake)

They seem to like each a little but the moment I stop, what happens? Yes, they separate again. Let me shake it again. (do so) But again when I shake it they mix but then separate. Let try another two.

Demo Suspension with Chalk

(take two more test tubes and combine as before. Mix as before)

Oh, they seem to like each other, don't they?

They seem to like each other more, but after I stop stirring, after a time, we can see they are separating. At the top it is getting clear and at the bottom something is settling.

Let's shake it as before to see if shaking makes a difference.

(use stopper and shake as before)

They mix as before but again begin to separate don't they. They don't seem to like each other very much.

(check on the stone)

There's only a small piece of the rock left.



Here we have two more substances

Demo Mixture with Unmixed Substances

(mix two substances on a plate)

These substances neither like nor dislike each other. But when one is ready to go home he just picks up and leaves. (hold up plates to show, place plates on the table)

(switch to two other plates on a tray with substances on each)

Demo Mixture with Premixed Substances

Here we have two other substances on a plate and we can mix them up, and mix and stir and mix. But no matter how long we mix them, we find they neither like each other nor dislike each other.

They simply sit side by side and when it is time for one of them to leave, he just picks up and goes home.

(use handkerchief to pick up metal shavings by magnetism, tray goes to side. Take test tube rack with tubes and remove stoppers.)

Now we have two more substances. (hold up two of them)

Demo Chemical Reaction with Copper Sulfate

These two when put together (pour one into the other slowly) like

each other so much that something brand new is created. Do you see right at the top, something brand new has been created. Now its starting to happen. Pieces of something brand new are falling down. (show to class by walking around)

Demo Chemical Reaction with Potassium Dichromate

Here (hold up two tubes left) I have two more. Let's see what happens when I put these together. (pour one into the other – one is a shade of orange) These two like each other so much that again something new is created, a brand new substance that was created that wasn't there before. (something is settling at the bottom)

Now I have a magic trick to show you.

Demo Magic in a Cup

I have an empty cup and empty saucer. I'm going to put them together like this. (attempts to hold both by putting cup under the saucer, lifts slightly to release small cloud of smoke)

And we have one more.

Demo Chemical Reaction Making Coal

In my dish I have some sugar (same as beginning, bowl with sugar and glass with clear liquid. Pour liquid into bowl)

Let's see what happens when I put them together. (distributes liquid evenly on top of the sugar) What's happening to our sugar. Its turning brown, how strange. Now what color is it turning (stick probe into sugar)

Epilogue

The first three we put together, and the two substances that like each other so much that they become one, are giving impressions of a **solution**.

Four and five, the two which kind of looked like they might like each other when mixing or stirring, the substances separate. So four and five are going impressions of matter in **suspension**.

The next two (six and seven) that neither like nor dislike each other that sat together side by side are impressions of **mixtures**.

The last four which like each there so much that something brand new was created are giving impressions of **chemical reactions**.

Especially with the first we could see the precipitate falling.

We don't give these terms to the younger children yet. We are giving only impressions. Only with older children would you bring in terminology.

Each demonstration has its own stirring rod, its own scoop, its own stopper and so forth since this is good lab practice.

September 17, 2004

Demos for Different Ways of Combining

1.Solution of Salt

1.1.Materials: small clear glass with water, small clear bowl with salt or sugar. A Little scoop and a stirring rod, all on a tray

1.2.**procedure:** put some salt into the water and stir it.

2.Solution of Copper Sulfate

2.1.Materials: liquid in a glass as in one and a dry blue substance (copper sulfate) in the bowl, scoop, stirring rod, etc. as above

2.2.**procedure:** same as one

3.Solution of Marble and Hydrochloric Acid

3.1.Materials: smaller beaker with spout edge, bowl is on a heat-proof mat, marble chip in the heat-proof Pyrex, bowl, all on a tray (liquid is full-strength hydrochloric acid - only a small amount in the beaker).

3.2.**Procedure:** With acid, the liquid is poured over the dry substance to avoid a splash. Otherwise similar to one and Two. Hold acid against Pyrex dish until the last drop was in. Otherwise the tray will be marked from the acid. Exercise caution with acid. Add the acid just before you begin the experiment. If full reaction is not visible, set it aside and check on it later.

4.Suspension with Sand

4.1.Materials: 2 test tubes, one with 1 inch of water and a small amount of fine yellow sand such as that used for sand art.

4.2.**Procedure:** (first two tubes) pour sand into the water, use a stirring rod and try to mix them together. When you can't get it mixed, put a stopper and shake to get it to mix.

5.Suspension with Chalk

5.1.Materials: 2 test tubes. Powdered chalk in one and 1 inch of water in the other, stirrer and stopper.

5.2.**Procedure:** same as Four. (shows matter in suspension). Place thumb on top of stopper when you shake the test tube

6.Mixture with Unmixed Substances

6.1.Materials: two dishes. Begin with substances mixed. Iron filings and fine salt or sand, handkerchief taken from the pocket to keep

magnet in handkerchief hidden. A magnet is stored with a bar on the side to keep it magnetized. Use handkerchief to easily separate iron filings off the magnet.

6.2.Procedure: with stirring rod, mix them. Place magnet through handkerchief close to the (when one is ready to go home) to separate them.

7.Mixture with Premixed Substances

7.1.Materials: similar to six but substances start off separately.

Again using iron filings and a fine sand or salt. (coarse iron filings work better than fine iron filings)

7.2.Procedure: dump one onto the other. Use stirring to mix them, use magnet covered by a handkerchief to separate them

8.Chemical Reaction with Copper Sulfate

8.1.Materials: test tubes (2), blue liquid = water and copper sulfate just blue enough so reaction can take place but you can see what is happening. Tube holder. Second tube has a drop or two of ammonia.

8.2.Procedure: slowly pour a couple drops of ammonia into the copper-sulfate solution. Slow and careful is only way we see precipitate falling.

9.Chemical Reaction with Potassium Dichromate

9.1.Materials: again two liquids, orange is solution of water and potassium dichromate. Don't make it too orange, use very little at first. Second test tube has several drops of a lead-nitrate solution. (it comes in dry form so add water to lead-nitrate)

9.2.Procedure: drop or two of lead-nitrate solution into the potassium dichromate solution. It forms a precipitate, something new.

10.Magic in a Cup

10.1.Materials: cup was not empty, it had a drop of hydrochloric acid in it. White cup and white saucer, hides a cotton ball with ammonia on it.

10.2.Procedure: Rub bottom of saucer with the cotton ball, just before you turn with the cup and saucer - before placing over the cup. Hold over cup for 30 seconds and lift to release the vapor.

11.Chemical Reaction Making Coal

11.1.Materials: larger beaker with liquid that merely covers the surface of the bottom of the beaker (sulfuric acid) with a Pyrex dish with 1/3 full of sugar on a heat-proof pad.

11.2.**Procedure:** dribble the acid around the outside of the sugar, working your way toward center. Just about cover the top – don't saturate the sugar (reaction will not occur). Dribble on carefully until top is almost covered. A Stirrer poking at it will sometimes get a good reaction. (Need good ventilation when the reaction works well.)

Date: September 23, 2004

Separation, Saturation, Super-saturation

Materials

Heat source on heat-proof pad, clear heat-proof container and a small amount of solution.

Prologue

The first thing we're going to look at is that solutions can be easily separated. You may have children working on making their own solutions. This might be what is going on one day when you just want to stop by with a group that perhaps has made a solution of water and copper sulfate – and pose the question “do you think those two can be separate again?”

Begin

You began with water and added copper sulfate. Can it be separated again?

(test tube with water and copper sulfate, pour it into Pyrex dish, turn on the hot plate.)

Let's see what happens when we apply heat. (pause)

We have activity now. It's boiling. We see steam going up. It is evaporating. I wonder if we'll be able to get that copper sulfate back?

(allow water to completely evaporate)

What's left in my container. There is blue copper sulfate. Can you see it? It's not exactly what it looks like in its crystal form. But if we scrape it all together we can see it is the same substance.

So, we were able to separate the solution using the heat. If we'd been able to capture the steam, we'd have the water also. Do you think we can separate a salt-water or sugar-water solution like this? Yes, in the same way.

With mixtures, we saw that mixtures can be separated, in our case attracting the iron filings using a magnet. What about matter in

suspension? Can they be separated?

We saw an example of water and chalk matter; they seem to have mixed but ultimately remained separate. They separated just by sitting. If we'd wanted to completely separate them, could we have used a filter paper to do so complete?

How about that piece of coal that we made? Can we separate that and get our sugar back. No – it has become another substance altogether. This is a key idea. This is separation.



Materials

½ inch water in a test tube. Copper sulfate in a dish.

Prologue

Several terms relate to using a **solution; one is said to have a solvent and a solute**. In this case, **water is the solvent**. The **copper sulfate is the solute**. The teacher should work these terms into the presentation.

With the children, one point of awareness to bring them to is that of when you've made a super-saturated solution. We need to remember that in the beginning **when the earth was forming, (we talked about how everything was very very hot), the liquids became super-saturated**. As they cooled, the different rocks and crystals and minerals formed. This is why we have all that variety around us on earth. **Each super-saturated solution forms a different crystal as it cools**.

The children have been working on their own solutions at this point. Suppose they are using water and copper sulfate. Stop by one day to challenge them further.

Begin

We're going to look at something else (today). We've been making solutions, right? Is there any limit to how much copper sulfate we'll be

able to dissolve in the water?

(pour some copper sulfate into the water and stir)

Once again, it all dissolved. Let's add some more and see what happens. Maybe I could go on adding all day and it would just continue. Let's see. I'll add more.

You have to check the very bottom to see if it has dissolved. We must need to stir more since there is something there.

Maybe if I put my stopper in and shake it will dissolve. This time, it all dissolved. Let's add more to see if shaking allows us to add and add and add. (add more and shake). No, there's still a little bit left.

It seems this time, no matter how much I shake, I can't get the rest of this to dissolve. We can still see at the bottom just a little bit there.

I wonder if it would make a difference if we applied heat to it.

(make sure to take the stopper out, place pan of water onto heat source and turn on heat source, use a test tube holder and hold it in the water in the pan. Use stirring rod and stir.)

Look how quickly it dissolved this time. Let's add a little more copper sulfate. We'll stir and stir while in the heated water. Let's see if we can get it all to dissolve.

(raise up test tube and observe)

Wow, look again, it all dissolved.

(add a bit more copper sulfate and put it back into the hot water bath and stir).

What do we see? It all dissolved again. Maybe now we will be able to go on adding all day. Let's add and stir some more.

(observe) I think it all dissolved again. (keep adding, slowly solution becomes darker).

I think this time, we might have reached the point where no matter how much I shake and stir and no matter how much we add, there is still a little bit of copper sulfate at the bottom of the test tube. I

cannot get that last bit of copper sulfate to dissolve.

The **point reached before I applied heat is called the point of saturation**. We had a saturated solution. This is the point where a solvent can no longer dissolve any more solute.

The **point reached after applying heat and where no more can be dissolved is called super-saturation**. We have a super-saturated solution.

End

Epilogue

If you make a super-saturated solution of salt, the cooled form would be quite different than that produced by copper sulfate. Obviously, this takes more solute than is used here. It's difficult to let it cool in the test tube because its hard to get out without breaking the test tube.,

It's best to make a super-saturated solution, pour it into a flat-bottom container (beaker or jar) and suspend a seed-crystal (just a small crystal of the same type) into it. You would want to tie this to a very thin thread such as a long hair. Suspend the seed-crystal into the solution; be careful not to disturb it. Place a sign "please don't move." If it's suspended into the super-saturated solution, after a time you'll be able to see the new crystal and just pull it up and out.

There are other substances in the training cabinet, potassium-permanganate, for example. It makes a beautiful solution and beautiful crystal if you make a super-saturated solution. The difficulty is it gets so dark.

From this work, the children often become interested in rocks and crystals; so you might make them aware of books on the shelf having to do with minerals or rocks. They might want to go out and visit a geologist. If there are a lot of children interested, this might be an appropriate point to bring a geologist in as a guest lecturer.

Date: September 23, 2004

Attraction and Gravity

Materials and Demos

1. tray of random solids, some attracted to magnet and not. Include several paper clips or nails. Something noisy is fun. An empty stamp pad box since they are noisy when magnet comes near it. Something heavy so one really feels the pull. Something like the index card, a broken piece of balsa will work like the index card - something not attracted to the magnet but will pull something through it.
2. Balance scale
3. Three test tubes: water, oil and molasses. The oil goes in the water first and then the molasses follows - different from the first time this demo was done in former presentation. Also needed is a stopper, since a stopper is put in and tube is turned over.
4. Solids demo shows how accommodating solids are (not): felt mat to put two bricks on with a balsa strip on top connecting them. Use kg weight to show stress but not going to bottom. Don't break through the balsa with the first weight. Then take off the weight and place a brick or sometimes two or three - to show pressure is so great that it eventually breaks through. (Balsa sets are distinguished by how they are banded together; there are two groups, one for practicals and one for the Further States of Matter, which have matched sets of balsa strips - sometimes called bass wood).
5. Clear plastic container, large ice bucket for example, of about two gallon capacity. Glass works but it is so much heavier. Fill about half full with sand on top of ping pong balls (hide them near the edge) and iron objects are placed on top near the center. Felt mat must be under the container or else you ruin the table-top. Procedure is simply to twist it back and forth. Make sure sand is dry, damp sand takes a long time. More dramatic with dry sand.

Begin

Today we're going to explore these things here using a magnet.

Demo 1

(touch each with the magnet to see what attracts and what does not. Demonstrate how paper clips move through the index card to follow magnet. Show how to pick up two or three paper clips from the touch of a magnet.)

Isn't that interesting? Maybe you'd like to explore the environment using the magnet to see what things are attracted to it and what things are not. But if you explore the environment, there are several things off limits. Bells are off limits and classroom computers as well.

End

Epilogue

The exploration the teachers offers depends on whether they've done this in primary; it may have been done and they are ready to go on. Otherwise, they will take off exploring. Once they've had a change to explore, sum up with them like the following.

As the center of the earth is pulling everything that's above it like a great big magnet, whether it's glass or rocks or water, or even people, everything is pulled toward the center of the earth. This is why things don't go flying off the surface of earth into space. This attraction that the earth has for things is called the force of **gravity**.

We made this statement in the barysphere presentation. Either presentation, in fact, can come first. Whichever one comes second, the wording is modified to recall that we already talked about the force of gravity.



Begin

This is why things have weight. Because they are pulled by the force of gravity. Things that are pulled more are said to weigh more. Things

that are pulled less are said to weight less.

Demo 2

(bring scale to table)

You might want to explore which things are pulled more and which things are pulled less. You can do this using our balance scale. I'll show you how to do this

What we want to see is not how exactly to weigh something, that's something for math. We want to see which things are pulled more and which are pulled less.

We can take two substances such as sugar and sand, measure an equal amount of each and put one on one side and the other on the other side of the balance scale. We can then see which one is pulled more.

End

Epilogue

This gives the young child an opportunity to see more and less without complicating it. We do have to make sure we put an equal amount of each substance to make a fair comparison. Bring test tubes to table.



Begin

Remember we said in our creation story the heavier things are always trying to get to the bottom and center of the earth.

Demo 3

(Raise test tube with water. Mix oil from second tube into the water

tube, then molasses from the third tube into the water tube as well. They settle according to weight.)

Here I have my three test tubes again. We can see that three liquids put together settle according to their weight. I wonder what would happen if I turn this upside down.

(put in stopper securely and turn test tube over)

Look at that, they rearranged themselves, with the heaviest at the bottom and the lightest at the top.

(make sure everyone can see the layers in the test tube)

These are liquids and we talked before how accommodating the liquids are. What about solids. Do you think solids are this accommodating? Let's see.

(bring felt layer and two bricks with wood strip as before in "Further States of Matter." Add weight on top of balsa strip.)

The heaviest hasn't gotten to the bottom here has it.

Demo 4

(add 3rd brick to break the wood)

But if the pressure is great enough, solids will rearrange themselves with the heaviest at the bottom.; The heaviest one wins once again. Now its time for another magic trick.

Demo 5

(on felt, put large bucket of sand with ping-pong balls buried and metal weights on top. twist bowl side to side. Metal submerges as you twist the container and the ping pong balls pop up to top.)

With movement, heaviest solids manage to find the bottom.
They rearrange themselves because there is so much movement.

End

Epilogue

This experiment leads to bridge building, pulleys and levers.

Date: September 23, 2004

The Sun and the Earth

Rotation of the Earth and Its Consequences

Material: charts 7a and 8a. Globe and faux-sun lamp. Plastic nut at end of string.

Prologue

Winter position is from teacher's vantage point: Hold the tilt toward your right from bottom to top, and hold the earth to the right of the light bulb (i.e., sun). Earth spins counter-clockwise (top view) and orbits the sun counter-clockwise (top view).

Begin

We consider earth and the sun as interrelated. The earth goes around the sun as if attached to a string (swing plastic nut on a string overhead). It takes one year for the earth to go around the sun.

The earth also spins around itself, like this (spin in place several times), never getting dizzy and never tiring. And we are glad. Because if not we would have a big problem.

Remember how hot it gets on certain days in the summer? It can feel like all the heat from the sun is falling on you alone. It can be terribly hot when the sun is beating down on us. But all the heat is not falling on us alone. There is another reason.

We already talked about how small the earth is compared to the sun and how far away it is. In reality, only a small amount of the sun's heat reaches us on the earth.

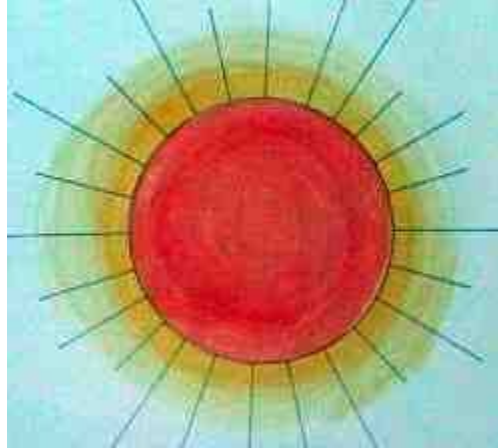


Chart 7a - sun

Most of the heat from the sun goes off in all directions around the sun. Only a 2 billionth part of its heat falls on us on earth. Yet it feels so hot. Can you imagine how hot the sun must be if only such a small amount of the sun's heat makes us so hot.

In fact, if the earth did not spin, this is what would happen:



Chart 8a - earth not rotating

The side facing the sun would burn up and dry up. All water would be gone. None of us would be able to live on earth. On the side away

from the sun, everything would freeze. Remember how we said that space was very cold? Since the sun's rays do not reach the side facing space, away from the sun, it is as cold as space. So again, nothing would be able to live there.



*The sun (lamp) and earth (globe) in winter position
(only do rotation - no orbit)*

Let's see what happens normally. **As the earth turns, there is a part facing the sun that warms up. And there is also always a part away from the sun that's cooling off.** As earth turns, the part in the light that receives the heat goes into the shadow and cools.

The part in the shadow then has its turn, facing the sun, and receiving heat from it. So what we see on this chart can never happen because the earth turns on and on and has since its beginning.

And something else happens that is in our lives each and every day. **As the earth turns on and on, we have day and night.** Here we are in the United States, just entering into the light. The sun is just rising in the United States.

The sun gradually rises over the horizon and into the sky. Little by little it rises until it appears almost right above us. This happens about the time many of us eat lunch, around noon.

The earth continues to turn on and on. It becomes 1 p.m., 2 p.m. As slowly as it rose, it now descends and eventually touches the distant horizon. This is about the time many of us eat dinner.

Gradually, the sun disappears. For our part of the world, it becomes dark. And for many hours, we are in this dark in the shadow of the earth. During this time when the sun does not shine on the surface of

this part of the earth, the earth cools off. This cycle happens every day for us, right?

As earth moves on and on, there's always some place where the sun is rising and somewhere else where it is setting. In between, is what we call midday or noon.



Materials: charts 9a and 10a

In reality, we don't see the earth moving. Why is that? To us it seems the sun is moving. Why is that?

Have you ever traveled down the street in a car and watched the streetlights and parked cars go by? It seems they run by us even though they are not moving. We are the one moving and yet it does not seem so as we sit in the seat of our car.

It is the same as the earth. We live on the earth. Therefore, we don't see it moving. It looks to us like the sun is moving, just as it seems to us the streetlights run by our car.

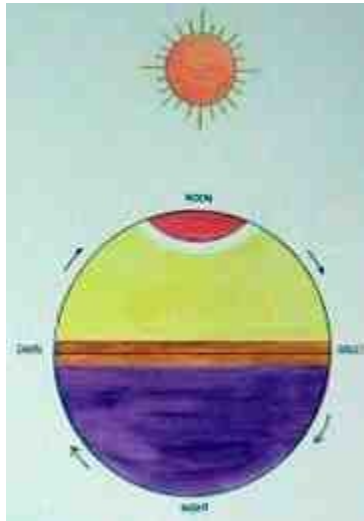


Chart 9a - dawn, noon, sunset and night

When we see the sun come up we say that it is **dawn. Or sunrise.** Then we see the sun get higher and higher as we go through the morning - until the sun seems to be directly overhead. This is when we have **noon or lunchtime.** The sun then begins to lower, gradually, toward the horizon and **over the edge of the earth; we say the sun sets.**

Then we go into the nighttime hours. We have all of this time in the dark for our part of the earth to cool off.

It takes 24 hours to make one trip around as the world spins. We see that here.

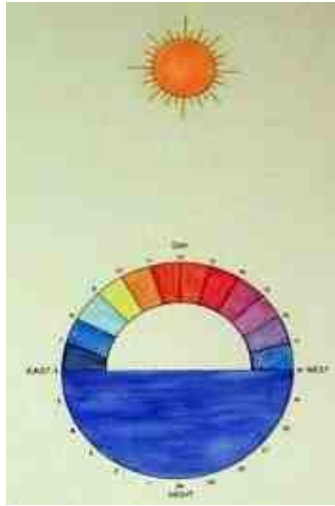


Chart 10a - day and night

If we divide those 24 hours in half, we have 12 hours of daylight and 12 hours of nighttime (count through the hours on the chart). At dawn the sun is just coming up; the sun rises around 6 a.m.

Maybe you don't get up until later and so you don't see it. You get up around 7 maybe, eat breakfast and take a bath, arrive at school around 9 a.m. (*narrate noon, sunset, sunrise*)

All of the nighttime hours allow the earth to cool off. Remember we talked about how cold it was in space, and is as well today. So the earth is able to cool since it faces the cold, dark space. The very coldest time is just before or just after sunrise.

When do you think the hottest time is? In general, it is between 1 and 2 p.m. that it becomes the hottest time of the day. On the chart, you see the red part that shows the most heat is present. Red means the hottest. Why is it not noontime, when the sun is directly overhead. It takes time to build up the greatest amount of heat on the earth and this is about 1 or 2 p.m.



The highest point reached by the sun in the sky happens at noon or midday. In Latin, the word midday is called 'meri' meaning mid and 'diem' meaning day.

All hours before midday are referred to as 'ante-meridiem.' Imagine if we always had to write this out each time. So people have come to abbreviate it. We say 'a.m.' instead.

The hours after midday are called post-meridiem. Again, this is a lot to write so we write p.m. Instead.

So, you can use these abbreviations also in your stories and other writing.



There is another aspect of the sun and earth I want to share. Longitude and latitude. When the earth is like this, we see that here in this part of Africa, the sun is directly overhead. It is midday. It's also midday in Europe and down here as well. If we could connect all these places where the sun is directly overhead, we would have an imaginary line like this - called a line of longitude. Also called a meridian.

(set up light bulb lamp shining on the globe over Africa. Hold string across the line of longitude at that longitude - point up and down the globe)

We have other imaginary lines on the earth. You may have noticed this one here.

(point to equator)

That's right, it is called the **equator**. And if you look on our globe, north and south, you can see another imaginary line like the equator. For example, can you see this line here. These are called "lines of latitude." Each line of latitude runs around the globe either north or south of the equator, parallel to the equator.

Knowing how to use these lines is important. These lines allow us to precisely pinpoint any place on earth. What if you were lost on a

desert island? How could you describe to someone on over a radio where you are? All you see is ocean all around. Actually, using longitude and latitude together allows us to find an exact location on our globe, no matter what surrounds that location – even if it is endless miles of ocean.

Longitude and latitude are measured in degrees. Degrees are presented and frequently used in geometry. A line of latitude, for example, is identified by degrees either north or south of the equator.

A line of longitude, is identified by degrees either east or west of the prime meridian, which was assigned as a starting point long ago, and is used much as we use the equator for latitude. The prime meridian runs between the north pole and the south pole and passes through Greenwich, England.

If I say 15° north latitude and 121° east longitude, where would this be in the globe? This would be in the city of Manila in the Philippines. We could take an atlas or a globe and with the information of longitude and latitude, we can find an exact location on the globe.

When is this important? If we are hiking and we find ourselves in the wilderness and lost, we can use longitude and latitude to find our way back to camp. Since all the trees look the same after a while, this is the sure way of finding your way home.

NOTE: The book “Longitude” by Dava Sobel is the story of John Harrison. This is a readable book about the origination and provides an explanation of longitude. Another such book is called “Galileo's Daughter.”



Date: September 23, 2004

The Time Zones

Prologue

This can be introduced after “day and night.” It could also come before the a.m./p.m. Presentation or the longitude/latitude presentation.

Materials

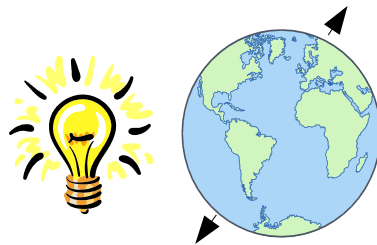
Chart gw1 ('gw' stands for geography work, meaning it provides the child manipulation of the chart using 12 black (night) and 12 white strips (day) and small clocks (12 red-bordered day and 12 black-bordered night) – all of which fit into the time zone spaces on the chart).

Begin

We should recall that in our study of the sun and earth we saw they were interrelated.

The earth goes around the sun once per year. The earth turns as it goes around the sun and it takes one whole day for the earth to spin around one time.

We can see here that the United States is just starting to come into daylight.



globe and light bulb lamp - in winter position

It is just sunrise where we live. But if one looks at South America, in Brazil, for example, one sees the sun is already overhead. It is midday

for them.

In Africa, the sun is just setting for them. We say their clock is ahead of ours since nighttime has arrived for them but not yet for us.

What about Alaska? This is part of the United States, isn't it. But it's way back here on our globe away from the rest of the states. It is still dark in Alaska because the sun has not yet begun to rise yet.

So some places are ahead of us and some are behind. Humankind has decided it should be dark at night and light during the day. So we created different time zones to help us adjust to the way in which light is present in some places on earth but not others.



Prologue

We introduce the time zone work chart by working from the left to the right, placing clocks and/or strips as we go. When the children are comfortable with this, introduce working from the right to the left. This assumes we are working in the Western Hemisphere. If working in the Eastern Hemisphere, begin working to the left and working up to working to the right.

As we place the clocks and strips, there are two alternative approaches. The first is to place all the clocks and then all of the strips. The second is to place a clock and then a strip, another clock and another strip, and so on. Either way can be used and both should be used in due time.

Material

Time Zone work chart , gw1. 12 Black and 12 white strips representing night and day. Clocks (12 red-bordered for day and 12 black-bordered for night) for 24 hours of a day.

Begin

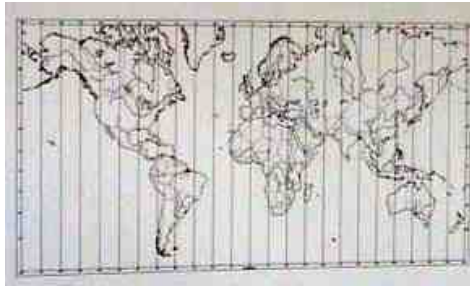


Chart gw1 - time zones

I have something new to show you (today) called "The Time Zone Chart." Along with this chart, we'll use strips to show if a place is light or dark at a specific point in time. We also have 24 little clocks which show how we divide the day into 24 hours.

I'll set them out across the chart as I read them and you can gather around me to see how they form one whole day. This is 6am, 7am ... 5am, and then we arrive back at 6am, where we started.

There are all the clocks set out for us to see how 24 hours spread across an entire day, in the same way that the sun goes around the earth in one entire day.

Now let's find where we are and use the clock that matches the time now.

(place clock for current time on local latitudinal area. place strip for day (white) over that region. Progressively place subsequent clocks on the chart. Count each region up one-by-one. Then fill in the strips for day and night.

Note known cities in regions as you go (e.g., Paris, France; Mumbai, India; Tokyo, Japan). Go right first saying, "They are ahead of us." Go left saying, "They are behind us."

This shows earth divided into 24 equal time zones. Though all time zones are not exactly the same size, more or less this is how time zones work to help us all make use of sunlight during the day and to rest at nighttime.

All of us eat lunch around midday or noon, in between sunrise and

sunset – no matter where we live on the earth – even though the earth continuously spins in and out of the rays of the sun.

Since we know that the date changes each new day, we have to have a change of date somewhere on the time zones chart, since the time zones represent a whole day.

We learned about the prime meridian when we studied the sun and earth. This was a line connecting the north and south poles that passes through Greenwich, England. It is used to mark zero degrees longitude.

On the opposite side of the globe at 180 degrees east or west of the prime meridian, we have a special line of longitude we call the international date line.

When we travel across this line, the date changes for us, either to the previous date or the next date, depending on which direction we are traveling.

Date: September 24, 2004

The Earth as a Sphere and its Result

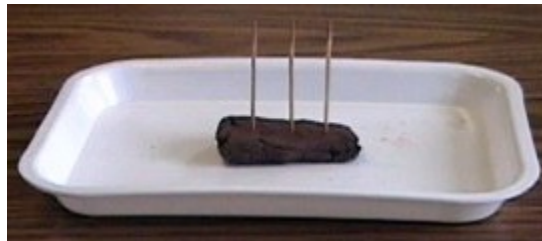
Material

Clay figures: one flat rock-like shape and one sphere and toothpicks. Black paper, light colored pencil, flash light, geography charts 11a, 12a and 14a (there is no 13a – as it became anachronistic)

Begin

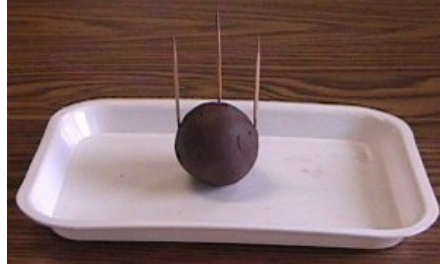
We looked at what would happen if the earth did not spin. We saw how day and night occurs. Now we'll look at another characteristic of the earth.

We saw yesterday how part of the day the sun seems to be directly overhead. When that is the case, the rays of the sun fall straight down on us (place toothpicks perpendicularly into clay). Or we say they fall perpendicularly. But there are however places on the earth where the sun's rays never fall perpendicularly. Why is that?



Perpendicular sun rays arrive at a 90° angle around the equator

It's because the earth is a sphere. If the earth were flat like the figure above, and the rays of the sun came down and heated the earth ... wherever they touched, it would be perpendicular to the earth. We can see a 90 degree angle at each of these points at which the rays arrive. Each ray would strike the earth perpendicularly.



But because the earth is more like a spherical shape, rays arrive obliquely (place toothpicks parallel into clay). As the rays of the sun come down and strike the earth at various indirect angles, they only make a right angle right here. So the rays only fall perpendicularly here. (where the surface is directly approached)

Where the rays of the sun fall perpendicularly, they are much more concentrated and therefore they get much hotter there.

(black paper, flash light shines on it, trace light on paper with pencil. Put flashlight at slant and trace again. We see non-direct light covers more area.)

When you hold it straight down – perpendicularly, less area is touched. Obliquely, the rays cover more area. We can see that perpendicular rays are much more concentrated. When the rays are at an angle, oblique to the surface, they are spread out over a larger area and are less concentrated. We saw that on the black paper. We can also see it here.

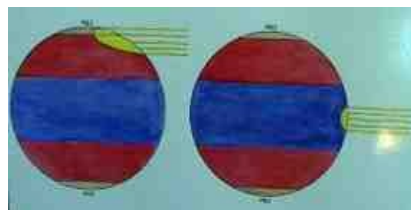


Chart 11a - oblique rays cover more area and are less concentrated as a result

We have the same number of rays falling here perpendicularly and here striking the earth obliquely. Look at the difference in the amount of area they cover when the rays strike earth obliquely. Maybe we should count the number of rays in each to make sure we have a fair comparison.

We can see the same idea but in a different way with *this* chart.

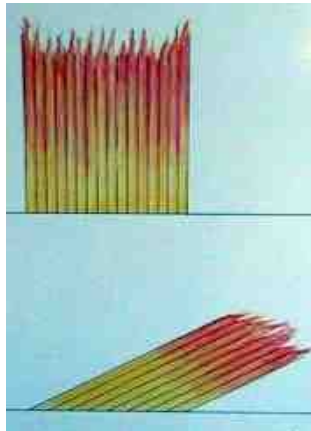


Chart 12a - same area is covered by less heat rays when arriving obliquely

We have the same amount of area covered here and here. But let's count the different number of rays up here (17) falling perpendicularly, hitting the earth here. We have the same amount of area here but how many rays. When the rays arrive obliquely, there are less rays (8). What is the result of this?

The result is that on the earth, in places where perpendicular rays are falling, it gets much hotter than in the places where the rays of the sun fall obliquely. Where are these places where the perpendicular rays fall? They are found around the equator. This is where the sun's rays arrive perpendicularly to the earth.

There's another reason why perpendicular rays provide so much more heat than the oblique rays do. Remember we talked about how the earth is covered with atmosphere? We talked about the atmosphere being one of the layers of the earth. Well, all of the rays of the sun have to pass through the atmosphere before they get to the surface of the earth.

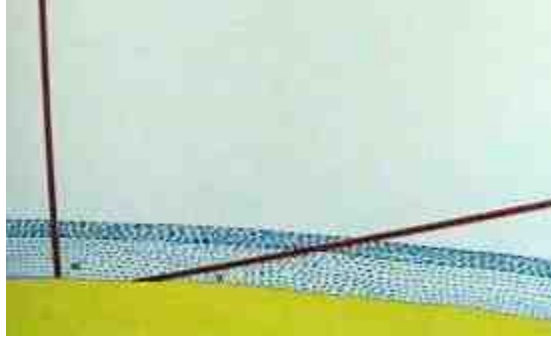


Chart 14a - Losing heat in the atmosphere

Here we see both a perpendicular ray and an oblique ray. Look at how much atmosphere the perpendicular ray passes through compared to how much the oblique ray has to pass through. The amount the oblique ray has to pass through is greater. It loses much more of its heat to the atmosphere before it gets to the surface of the earth. The perpendicular rays hold on to more of their heat. So they are hotter when they get to the surface of the earth.

End

Date: September 24, 2004

The Tilt of the Axis

Materials

A light bulb lamp, a ball of clay on a knitting needle with tape wrapped around it to prevent the sphere from falling down the needle, a thumbtack to mark our global position – all on a tray. The sandpaper globe.



Prologue

This presentation begins with a story. The earth spins in-place counter-clockwise, from the top view, and also moves around the sun counter-clockwise from the top view, on a plane. In both of these aspects of the earth's movement, the tilt always points the same direction – to the presenter's right, from bottom to top. With earth to the sun's right, we're in the winter position – where the north temperate zone is deep in the shadows.

Begin

In olden times, when many people used to spend much more time outdoors than many of us do today, and there weren't so many tall buildings blocking our view, people could see very clearly where the sun rose each morning. **They noticed the sun rose at different places on the horizon.** From a certain day in winter, the sun began

to rise a little more to the north and a little more to the north and a little more to the north, until there came a certain time in the summer when the sun began to rise a little more to the south and then a little more to the south. It started to go back where it had risen originally.

It is to this turning point in the sun's journey that they gave the name "solstice." This word comes from two Latin words: 'sol' which means sun and 'sistere' which means to stand. It was as if the sun stood still before it began to go back to where it had come from.

The people also noticed something else. They noticed there were two solstices; one in the winter and one in the summer. And so they called them the 'Winter solstice' and the 'Summer solstice.' They noticed that from the winter solstice, the days started to get a little longer each day - while the nights got a little shorter each night. This happened until summer solstice arrived, when they had their longest day and shortest night. It is still like this today. **We still have the longest day and shortest night on the summer solstice and the shortest day and longest night at winter solstice.**

Can you recall how long the days are in the summer. There is daylight for so long, we finish eating dinner with family and it is still light outside. Maybe we go out with friends. But in the wintertime, you may recall how coming to school in the morning sometimes, it's still dark. And when you finish dinner, it's already dark.

We wonder why it's like this. **Why do we have longer days in summer and shorter days in winter? (key question)**

(hold up sphere on needle and spin it)

It's because the earth does not turn around on itself like this (spin without tilt), but rather like this (add the tilt). If the earth went around like this (no tilt), the length of day and night would be the same. But **the earth has a tilt. It moves like this (add the tilt, mark a spot where we imagine we live, stick in thumbtack).** We'll imagine this is where we live and we will watch this spot on our sphere. When the earth is over here, watch that spot.

What do you notice about how long it's in the light and how long it's in the shadow? (spin it to right and left of the bulb, one shows light a longer time and one shows a shorter time as the sphere rotates). When the earth tilts away from the sun, the night is long. When it tilts toward the sun, the day is long. The tilt does not change

as it spins and rotates.

So, **it is because of this tilt that the days are sometimes shorter and sometimes longer.**

(make sure children see the phenomenon before going on)

It takes the earth one year to make one trip around the sun. Let's see that with our sandpaper globe. **When the earth is here** (between bulb and teacher, tilt is neither toward nor away from the sun) **the length of day and night is just about the same. (fall position)**

But as the earth moves on, the days get a little bit shorter and shorter until the **earth tilts directly away from the sun (winter solstice), when the day is the shortest of the year** and night is the longest.

As it continues around the sun to the opposite side from where we began, the days and nights again gradually become even in duration. **Half way around from where we began days and nights are again equal.**

As it continues toward the position in which the **earth tilts directly toward the sun (summer solstice), the days become longer** and longer and night becomes shorter and shorter.

As it passes summer solstice, the point where the tilt is directly toward the sun, again the days and night begin to become just about equal again - until we arrive back where we began.

End

September 24, 2004

The Seasons and the Two Tropics

Material

Light bulb lamp, sandpaper globe, charts 15a and 16a, 17a.

Prologue

The movement with the globe is the same as that in the Tilt of the Axis presentation. This time we are adding seasons. We present this related to the hemisphere we are in. Both have language dependent on the location the teacher presents from.

Begin

Here we are once again (fall position) in one of the positions where day and night are just about equal. As the earth moves on, the days where we live keep getting a little bit shorter and shorter, while the nights get a little bit longer and longer – until we get to here where we have our shortest day and longest night (traveled 90 degrees).

This is why this is when we have our winter. As we know the earth keeps moving on. Now the days start getting longer and longer, the nights a little shorter and shorter, until we get to here where day and night are just about equal (traveled 180 degrees). This is our spring.

As the earth moves on, the days get a little longer and the nights a little shorter, until we get to here (traveled 270 degrees) where we have our longest day and shortest night. This is our summer.

Now the days start to get a little bit shorter and the nights a little longer, until we get back to here, where day and night are just about equal (traveled 360 degrees). This is our Autumn or fall season.

(repeat two or three times as needed to get the picture across to the children)



There is something else we can see when we consider the movement of the earth around the sun (hold earth in **summer solstice position**, have children watch the **North Pole** as earth rotates, asking what they see: **light all day and night**).

When we focus our attention on the North Pole at summer solstice and we move the earth around the sun, we see that the **North Pole gets light all day and night**. It has extremely long days during this part of the year.

When we focus on the South Pole with the earth at winter solstice, we see a similar phenomenon - light all day and night. (hold earth in winter position, have children watch as earth rotates, asking what they see: light day and night).

The area at the North Pole that is light day and night at Summer Solstice is called the **Arctic circle**. The area at the South Pole that is light day and night at Winter Solstice is called the **Antarctic circle**. There are imaginary lines on our maps showing these areas.

All of this is because the earth travels around the sun while on a tilt.

Now let's look at something else. (using sandpaper globe and light for the sun)

This time, the movements of the earth and sun are the same, but watch to see where the light is falling relative to the equator - on, above or below the equator.

(Zero degrees = on equator, 90 degrees = below the equator, 180 degrees is on the equator, 270 degrees = above the equator)

What we saw with the globe and light, we can also see here, on chart 15a.

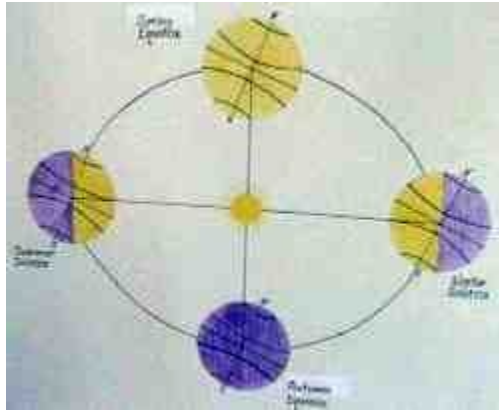


Chart 15a - points of solstice and equinox

If we start here (bottom totally purple sphere from chart 15a above), day and night are just about equal. As the earth moves on around the sun, the days get shorter and nights get longer until we get to here (90 degrees) where we have our shortest day and our longest night. Remember we saw that here, the sun's perpendicular rays are falling below the equator.

As the earth moves on to here (180 degrees), days and nights are again of equal duration; the sun's perpendicular rays again are directed to the equator.

The earth continues on to here (270 degrees) where we have our longest day and shortest night and the perpendicular rays of the sun fall above the equator. And when we arrive back where we started, again the perpendicular rays are falling on the equator.

In between our two solstices, we have what is called the spring equinox and the autumn equinox. These are the times when day and night are just about equal. Equinox, the word, comes from two Latin words: 'equi' meaning equal and 'nox', meaning night.

(subsequent presentations add, "Of course, we are looking at this from where we live. The description here is relative to the northern hemisphere. A summer solstice here would be a winter solstice in the southern hemisphere.")

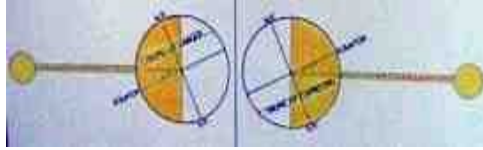


Chart 16a - perpendicular rays and the equator

If we were to draw an imaginary line connecting all of these places, as far north of the equator as the perpendicular rays ever fall, and as far south of the equator as the perpendicular rays ever fall, we would get two more imaginary lines, like the arctic circles. That is what these lines on the chart represent.

In the northern hemisphere, that imaginary line is called the **Tropic of Cancer**. In the southern hemisphere, that line is called the **Tropic of Capricorn**. Those lines are situated $23\frac{1}{2}$ degrees north and south of the equator.

What that means is that if we could extend one of the perpendicular rays of the sun from that spot on the line of the Tropic of Cancer, for example, into the center of the earth, it would make an angle of $23\frac{1}{2}$ degrees with the equator. The same can be said of the Tropic of Capricorn as well. This is why on our map, we see $23\frac{1}{2}$ degrees marked.

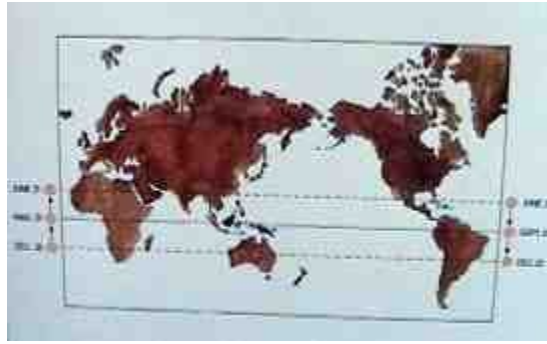


Chart 17a - beginning right side moving down (facing audience) and then moving to the left side moving upward

1. June 21 (Tropic of Cancer-on left-summer solstice)
2. September 22 (Equator-on left-fall equinox)
3. December 22nd (Tropic of Capricorn-on left and right-winter solstice)
4. March 21st, (Equator-on right-spring solstice)
5. June 21st (Tropic of Cancer-on right-summer solstice)

So the perpendicular rays of the sun seem to always be moving up and down in between the two tropic lines. If we start here, at the Tropic of Cancer, we know that when perpendicular rays are falling on the Tropic of Cancer, it is June 21st.

As the earth moves on, the perpendicular rays are falling down on the equator, it's September 22nd.

As the earth moves on, when the perpendicular rays are down here on the Tropic of Capricorn. It's December 22nd.

After December 22nd, the perpendicular rays come back here to the equator and its March 21st.

The perpendicular rays keep falling back to the north where again it arrives at June 21st.

End

Epilogue

Certain stories of cultures in north or south locations that are affected by the tilt are good stories to bring into the story at this point. Scientists in these areas as well might be useful.

Date: September 28, 2004

The Zones

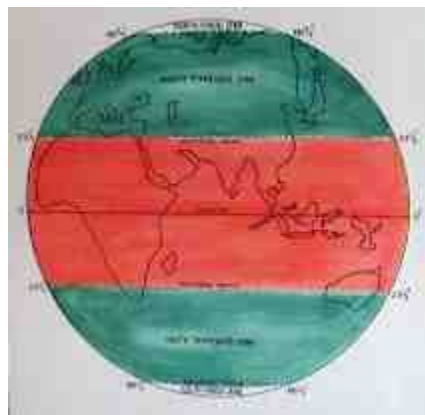
Material

Chart 18a

Begin

Remembering the work from the Tropics, the last chart we saw was a map of the world (17a), we could see how the perpendicular rays of the sun always fall between the two imaginary Tropic lines. What we will do now is a conclusion of that lesson.

Remember we talked about how hot the perpendicular rays of the sun are? We talked about how those rays are always falling between those two imaginary lines, always falling over this (refer to the middle area of red heat) - the torrid zones.



Char 18a - arctic, torrid and temperate zones

Therefore this part of the earth gets very very hot. That's why it's called the torrid zone. Torrid comes from the Latin word *Torridus*, meaning parched or dried up.

We also talk about this place - the arctic zone here and the Antarctic zone here, places where during part of the year, the sun never rises. Imagine how cold it gets when the sun does not rise at all. This is why these places are called frigid zones. Frigid comes from the Latin word

frigidus and it means cold.

There is a north frigid zone here and a south frigid zone here. (point to chart)

In between the frigid zones and torrid zones, we have these areas here and here – where the rays of the sun never fall perpendicularly. They always fall on the earth obliquely.

These are called the temperate zones. Temperate comes from the Latin word temperatus and it means to blend or to mix. It is hot and cold mixed together. So we have a north temperate zone and a south temperate zone.

These are other details about the earth that you might want to be aware of.

End

Date: September 28, 2004

Work Charts

Zones Work Chart

Materials

Zones work chart (gw2), small paper sun and a set of dates for solstices and equinoxes.

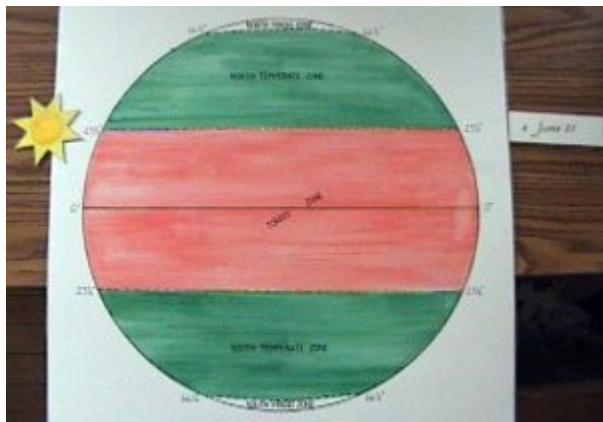


Chart gw2 - Zones work chart

Begin

The perpendicular rays of the sun are falling right here (point to chart gw2 and place a date card for each new location) - on the Tropic of Cancer. It is June 21st.

(place sun on left side of chart and place date at right side, both at Tropic of Cancer)

But as we know the rays don't stay there. The rays begin moving a little more to south until they fall right here on the equator. Now it is September 22nd.

(place sun on left side of chart and place date at right side, both at the equator)

But those perpendicular rays of the sun keep falling further and further south until they get down here to the Tropic of Capricorn. Now, it's December 22nd.

(place sun on left side of chart and place date at right side, both at

Tropic of Capricorn)

After December 22nd, the perpendicular rays start to fall to the north again, little by little, until once again the rays fall perpendicularly on the equator. Now its March 21st.

(place sun on left side of chart and place date at right side, both at the equator again)

The perpendicular rays keep falling further and further to the north until they get back here to the Tropic of Cancer and once again, it is June 21st.

(place sun on left side of chart and place date at right side, both at Tropic of Cancer)

Now would you like a turn to talk your way through a whole year?

(Give children a chance to repeat what was just done)

End

Epilogue

Prepare by having dates in the order you will use them. Notice we moved down the chart on the left side and the dates change on the right side. The focus is on the dates.

There are two variations possible. The first variation is to pick a date such as December 22nd, show it to a child and ask the child to place the sun. "Where would the perpendicular rays be falling on December 22nd?" (Tropic of Capricorn). Place date in Tropic of Capricorn.

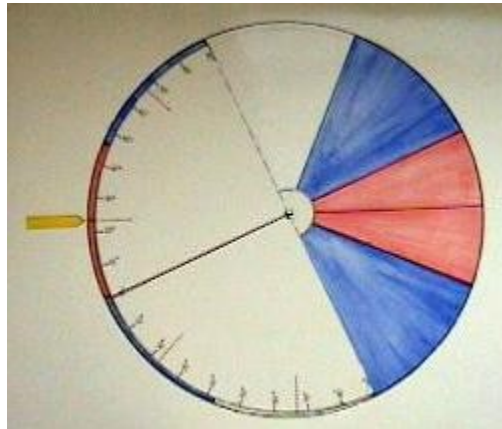
Second variation is to place the sun, and have the child choose the date. Point out to children at some point that Summer solstice in the northern hemisphere is the winter solstice in the southern hemisphere.

Date: September 28, 2004

The Protractor Chart

Materials

The protractor work chart (gw3).



Sun's perpendicular rays point to the Tropic of Cancer

Prologue

This chart is fundamentally important because up to this point, we talk about how the earth moves but with our charts, we always show the sun moving; with this chart, **the earth moves**. No need to go into much explanation however. The children only need to know what is represented initially.

Begin

This yellow arrow is the perpendicular rays of the sun. The piece that moves represents our earth as it moves. So, if the sun's rays go north, the protractor rotates downward. If the rays go south, the protractor rotates upward. This may seem counter-intuitive at first.

The red line on the 'earth' marks off the equator. We also see dotted red lines on the 'earth' surrounding the red line; they mark off the tropics.

Let's choose a date to begin on with perpendicular rays on equator (we choose March 21st - not September 22nd).

Up to June 21st, the rays move upward or northward. (rotate earth downward)

Then rays start south up through September 22nd. (rotate earth upward, pause at equator)

Rays continue south to the Tropic of Capricorn December 22nd. (rotate earth upward)

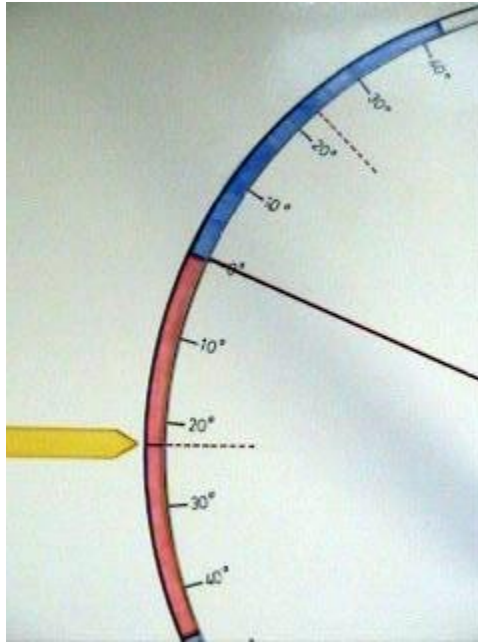
Then, they begin north again arriving at March 21st again. (arrive at equator)
(rotate earth downward again up to where red line and yellow arrow match)

There is nothing new here except the movement which now more accurately reflects what it represents – the sun's rays arriving at a constant point and the earth rotating up and down, as well as around - as it revolves around the sun.

Colors along the outside edge represent where it is hot. Blue is warm to cool and white is cold. The colors represent all the temperatures. We also notice numbers on the edge of 'earth' that represents latitude. ('earth' is the protractor.)

Now we can pick a city. We choose Milwaukee, latitude 43. 43 degrees north latitude. If we look at the protractor, find 43 degrees north, and point it out. We can walk through the year, watching what happens temperature-wise in our city.

We line up with perpendicular rays on the equator (spring or fall, March or September). Finger on Milwaukee, perpendicular rays are falling on the equator. Milwaukee in on the blue (warm to cool) area. After September 22nd (we chose fall this time). After September 22nd, rays fall more to the south. Milwaukee becomes cooler and cooler as the rays go south (protractor rotates upward).



When rays are on the Tropic of Capricorn, rays don't go any further south. So the rays head back north. On March 21st, the rays are again on the equator. The rays continue to the north and Milwaukee gets into the hot area. On June 21st, Milwaukee is in the red, hot zone. This is as far north as the perpendicular rays fall.

After June 21st, the rays begin falling south and Milwaukee reenters the blue area – up through September 22nd again.

This chart is from the viewpoint of the rays. The perpendicular rays don't move; the protractor does.

Now we choose Helsinki, which is 62 degrees north latitude (something really far north). Let's find it on the chart, and point to it.

Again let's choose between March or September. Let's choose March.

The perpendicular rays now fall on the equator. After March 21st, rays move north. Helsinki moves into the blue where it's warm-cool. Rays continue north until they arrive at the Tropic of Cancer. Helsinki only barely makes it into the blue area. Then earth moves on southward as it approaches September 22nd, when Helsinki is in the cold. Rays keep falling southward and Helsinki goes into a white - cold area – until perpendicular rays reach the Tropic of Capricorn. After December 21st, rays go northward. Helsinki gets back into the blue area. We arrive back at March 21st.

End

Epilogue

Assuming you've done some work with the southern hemisphere, pick a city in the southern hemisphere (child must have already done the light and globe work for the southern hemisphere). Choose Melbourne Australia at $37\frac{1}{2}$ degrees south latitude.

Remember, if rays go north, the protractor rotates downward. Also, notice that after March 21st, rays go north. After September 22nd, rays go south.

Date: September 28, 2004

The Seasons Work Chart

Materials

The seasons work chart (gw4), color strips (one green for spring, one brown for fall, two white strips for winter, one blue strip for arctic winter, one pink for hot, two red for very hot, and two dark red for very very hot). A second set of sun / dates for equinox and solstices is useful in the classroom.

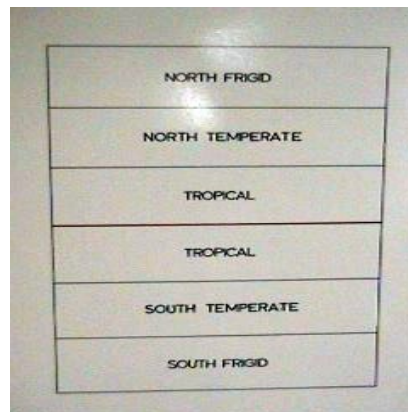


Chart gw4 - seasons work chart

Prologue

Organize the dates by the order of use. When presenting it, introduce the strips and lay out to the left of the chart. Tell the children what each color represents. On the chart across middle is a red line that marks off the equator. The zones are marked on the charts - the torrid zones (sometimes called the tropical zone and is so marked on this chart), the north-south temperate zones and north-south frigid zones. The sun moves up and down on the left and dates are changing on the right. As you place the strips, always start with the zone you are teaching in, work your way down and then finish the top.

Move the sun, change the date, then change the strips - in that order. Temperate zones never get very very hot. Temperate zones are the only ones having spring and fall. The prerequisite for this lesson is early sun and earth work for the southern hemisphere and the work on the dates and zones. You might ask the children after time has passed which zones change least - or - which zones change the most.

Tropics are the only place very very hot. They are either very hot or very very hot. Very hot is used for summer in temperate zones. Hot is used for arctic summer in the frigid zone.

Begin



Summer in the Temperate Zone

Today, we'll pretend it's June 21st. Remember on June 21st, the perpendicular rays of the sun fall here on the Tropic of Cancer. We live in the north temperate zone. June 21st is the beginning of summer for us. So what is the temperature like? It's hot.

(move the sun and place the date first)
(place hot strip on the north temperate zone)

Here in the north tropical zone, those rays have just passed right over. What is the temperature like here? Very very hot; let's place the strip for that over the north tropical zone.

How about in the south tropical zone? Since perpendicular rays are here, it would be only very hot down here. (place strip).

What about down here in the south temperate zone? It is their winter. (place white strip).

Down here in south frigid zone, well those perpendicular rays are as far away from the south frigid zone as they ever get. It's really really cold. They are having what's called arctic winter. (place blue strip).

Up here in the north frigid zone, the perpendicular rays are as close

as they will ever get. It is as warm as it ever gets and what we call our arctic summer. (place pink strip).



September
22

Fall in the Temperate Zone

After June 21st, the perpendicular rays of the sun move down to the equator and now its September 22nd. What's it like for us now in the north temperate zone? It's beginning of fall and cooling off. (place brown on the temperate zone).

What about the north tropical zone? Perpendicular rays just passed over the north tropical zone and it is still very very hot (no change).

What about down here in the south tropical zone? Now perpendicular rays are right on equator, very close to the south tropical zone. It's becoming very very hot. (place dark red strip).

What about here at the southern temperate zone? Since it's September 22nd, for us its fall - so for them it must be spring. (place green). The seasons in the southern temperate zone are always ordered like a mirror reverse image from that of the northern temperate zone.

Down here in the south frigid zone, perpendicular rays have come down a little closer. So it's not arctic winter anymore but just winter (place white).

What about the north frigid zone? It's no longer arctic summer and it's cold (place white strip).



Winter in the Temperate Zone

After September 22nd, the perpendicular rays continue to go south to the Tropic of Capricorn and now its December 22nd. The north temperate zone is now winter (place white).

The north tropical zone, since perpendicular rays are far south, it isn't very very hot, just hot (place red).

The south tropical zone, must be very very hot since perpendicular rays just passed over.

What about down in the southern temperate zone on December 22nd ? It's their summer and very hot (place red strip).

What about at the southern frigid zone? Perpendicular rays are as far south as they go and as close as they get. It is their summer (place pink), arctic summer. At the north frigid zone, it's very very cold since perpendicular rays are as far away as they get. It is arctic winter. This is what it's like on December 22nd.



Spring in the Temperate Zone

After December 22nd, the rays fall north and they get back to the equator. Now, it's March 21st. In the northern temperate zone, it's spring (place green).

In the northern tropical zone, where perpendicular rays are back to the equator, it's very hot (place red strip).

At the southern tropical zone, perpendicular rays have just passed over, it must be very very hot. At the southern temperate zone, we have spring on March 21st, so they have fall (place brown). At the southern frigid zone, it's no longer arctic summer so they are only just cold (place white). At the north frigid zone, it's also cold there (place white). Now it's March 21st.

(repeat summer)

After this, perpendicular rays move north to Tropic of Cancer and we arrive at June 21st. What season is it for us? Since it's summer, we put a red strip. In the north tropic zone rays just passed by; it's very very hot.

In the southern tropic zone it's very hot. What about in the southern temperate zone? It's winter since we have summer. (place white).

In the southern frigid zone, it must be arctic winter since rays are far away. What about at the north frigid zone? It's as warm as it ever gets so it's arctic summer (place pink). This picture is now the same as our first.

End

November 1, 2004

Protection of the Atmosphere and the Rains

Material

Heat source on a heat-proof pad. A spoon or metal rod – spoon is faster to heat up but the iron rod stays hot longer. Long (fireplace) matches. Four charts (19a, 20a, 21a and 22a)

Prologue

We've been talking about the sun and the earth. Today, we're going to get into further specifics. This is the last section within the sun and the earth chapter. From this one, we can go into the work of air or the work of water - either one.

If you've read "To Educate the Human Potential", you've seen how Dr. Montessori addressed these subjects. Before we go into those aspects of the universe, there is a bit more about the earth to understand; we make the children aware of the fact that there are two kinds of heat.

We do this with a simple demonstration.

Protection of the Atmosphere

Begin

(turn on hot plate)

There are two kinds of heat.

One that comes from something that is burning like a fire (light a match and let it burn)

Something that is burning gives off heat – like we see here with this fire.

The other kind of heat, comes from something which is not itself burning, but which is heated up by a source of heat and then it gives off heat.

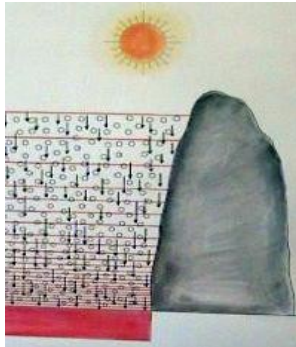
(place spoon directly over the heat source)

Can you feel the heat from this spoon now (let children feel heat from the spoon)

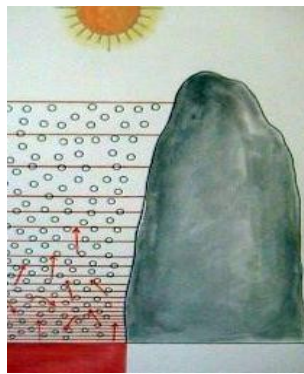
This kind of heat is called **radiant heat or dark heat**. We don't see

this kind of heat as we see heat from a fire burning. But we can only feel it as the heat is released.

The same thing happens with the earth. (hold up chart 19a)
Here, you can see the rays of heat coming down from the sun and heating up the earth.



Once the earth is heated up, the earth itself gives off heat. See how these rays of heat (show chart 20a) are released from the earth. These are the radiant heat rays from the earth – after it is heated up.



The rays of heat given off by the sun are shorter and stronger. The **rays given off by the earth are longer and weaker**. See how they are longer? (point to chart)

Well, **because they are longer and weaker, many of them are trapped by the atmosphere**. Remember we talked about the atmosphere as one of the layers around the earth. Many rays are trapped in the atmosphere – see some of them are turning back down toward earth.

So the reality is that this radiant heat – much of it – is trapped by the atmosphere. And it is that heat that keeps us warm here on earth – warmer than we become from the direct heat given off by the sun which ordinarily passes right through the atmosphere. The atmosphere wraps around us on earth and keeps us warm.

The layers of atmosphere are like blankets (show chart 21a) around us. We have the radiant heat coming from the earth, and that helps to keep us warm and cozy. Doesn't that man look cozy?



We can imagine being at the foot of a mountain, with all these layers of atmosphere wrapped around us. **The radiant heat - coming up from the earth - keeps heating our space**, keeping us warm and comfortable. But suppose you decide to go up that mountain.

Well **as you climb up the mountain, the layers of atmosphere get thinner and thinner**. (show chart 22a). The heat coming from underneath us becomes less and less. So even if it was a bright sunny warm day - at the foot of the mountain, when we get to the top of the mountain, it is very very cold. Look how thin the blankets are now!



There is very little radiant heat underneath us any longer and it is very very cold. Look at that poor blue face on that fellow.

This is why, when you get to the top of the mountain, even on a bright sunny day, you need to put a jacket on because it has gotten so cold.
End

November 1, 2004

The Rains and Atmospheric Temperature - Rains I

Material

Hair dryer for hot air. An upside-down metal lid or plate, sprinkled water inside. Turn on hair dryer and let them feel the hot air; blow it over the water on the plate to see what happens – the water disappears. Chart 23a.

Begin

Remember we said at ground level, the atmosphere layers are very very thick and the radiant heat is trapped in the atmosphere (show chart 23a).



So it's hotter at the foot of the mountain.

But as one goes up a mountain, its gets cooler and cooler.

Now, air, when it's hot, can hold a lot of moisture.

(turn on hair dryer, let them feel the heat and then blow over the sprinkles of water)

What's happening to the water? We can see that it has drawn the water from the lid!

So, hot air can hold a lot of moisture.

If this hot air is blowing over an ocean or a sea, it can gather up a lot of moisture. When it's at ground-level – it can hold all that moisture.

(hold up chart again)

But then, if it encounters something like this mountain and the air is pushed up the mountain, the air cools off. We talked about how it gets cooler as we go up the mountain. The air gets colder and colder and this cold air cannot hold as much moisture as hot air can.

So at a certain moment, it drops all that moisture – in the form of rainfall or if its cold enough, in the form of snow.

We can do another demonstration here to show this happen.

(heat a sauce pan with approximately 1/2 inch of water in it, a clear glass bowl with a diameter greater than the pan, preferably one with a lip to hold onto. Put pan on the heat source and turn on heat source. Wait for water to come to a boil.)

Now the water is just about boiling. So we've said hot moist air is forced upward by this mountain. This is like we see here. See the hot moist air going up?

(raise bowl from the pan, hold just above to catch condensation)

Do you see the water gathering on our bowl.

If we hold it here long enough, you'll see rain drops rolling down the side. Yes, now we can see little drops running down our bowl – just like little raindrops.

We can see the same thing here on this chart (23a). As the air is pushed up the mountain it cools off. And when it cools off enough, it just can't hold all that moisture any more – it drops it here on this side of the mountain (point to first side of contact)

As the air goes over the mountain, the air is much dryer – because it dropped all its moisture on the other side.

This is why if you have mountains by the sea, the side of the mountain closer to the water is green and lush – from all of this rain. On the other side of the mountain, the side away from the sea, it is much dryer – sometimes even like a desert.

Maybe you'd like to explore in the atlas and see if you can find any places like that, where on one side of the mountain, there is lots of vegetation and its very green – but on the other side of the mountain it is very dry.

End

Epilogue

From here, you're hoping to get the children interested in looking up more on this. They may run into the term orographic rain. It means coming from the mountains. We may hear of the orographic effect in California weather for example.

Other avenues for exploration would be other types and causes of rainfall. Different kinds of precipitation such as sleet and hail can be investigated; what causes these different kinds?

Some children become interested in different cloud formations and different types; at times, this information is required in the public curriculum. Know that in places like this you have opportunities to go into these areas – the lesson provides an opening.

Lastly, weather forecasting may be interesting. How did farmers and sailors predict the weather for the next day? They looked at cloud formations and so on. Weather forecasting is still not an exact science. Maybe we can predict almost as well with some investigation and study.

A barometer is a nice thing to have in upper elementary. This might be an opportunity to discuss again how particles become heavier when they get cooler.

The Work of the Air

November 1, 2004

Prelude to the Winds

Purposes

To bring to the children's awareness that air occupies space. To bring to the children's awareness that hot air rises. To bring to the children's awareness that when hot air rises, it creates an empty space which draws other air to take its place.

Air Takes Up Space

Material

On a tray: Glass flask, a funnel (cut straight across - the angle is there sometimes to let the air out.) with a narrow end. Seal the funnel and flask with clay. First make a doughnut with the clay, place on top of the flask and insert the funnel - then work the clay so its flush against the flask. Glass of water (if beaker, pour from side, not the spout).



Prologue

We need to make the children aware of some characteristics of air.

The first characteristic is that air occupies space. Note that a rubber stopper that fits into the flask also works but this is difficult to use. All three of these are done on one day, generally.

Begin

I have a funnel in my beaker. (pour water into beaker and it should stay on top)

Why? The water can't get in because the air can't get out of the beaker.

If I turn the top to break the clay, the air can get out and the water can then get in.

So we can see that **air takes up space**.

End

Hot Air Rises

Material

Metal platform, knitting needle held erect by Pleistocene (oil-based clay that never dries) – or if you have chemistry set you may have a platform with a needle-like rod that screws in. You need a spiral that is placed on the top of the needle. A candle (with Pleistocene to keep the candle upright, matches and water close-by if needed).

Prologue

Here we make the children aware that hot air rises.

Begin

(Apply heat under the spiral and see spiral turn since hot air is passing by).

The hot air goes up and pushes up the spiral. So we see here that hot air rises.

End

Heated Air Creates Air Displacement

Material

Cardboard tube – chimney, with an opening cut at the bottom. This tube should be wider than the average paper towel tube like empty gift wrap rolls – for example. They are sometimes made with metal and this is a hazard for all since it becomes so much hotter. A tea-candle (small), a stick of incense, a heat proof pad, matches and a bit of Pleistocene to hold the incense upright. A metal cup with a little bit of sand in it.

Prologue

Here we make the children aware that hot air rises and other air fills its former space.

Begin

(light the incense, blow it out.)

I want you to think of this smoke as air made visible. Can you see it?

Where is it going? It goes straight up. What if I put it here. Can you see it?

It still is going up for the most part.

Now watch. (light tea-candle next to it. Place hollow cylinder around the candle.)

Smoke is still going up right?

How about now?

(the smoke from the incense, which is now placed next to opening, is pulled into the tube and follows the tube upward. The original air inside the tube was heated and left the tube – drawing in the incense

smoke)

So we can see that when hot air rises, it creates an empty space which makes other air move in to take its place.

End

November 2, 2004

The Winds

Material

Chart 24a, 25a, 26a. For the demonstration, we need a paper circle and 2 pencils.

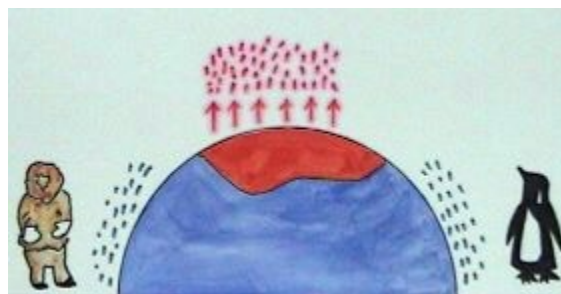


Begin

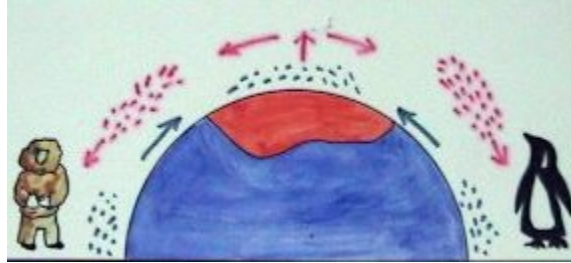
Remember how we talked about how when the sun's rays fall perpendicularly, they heat the surface of the earth at the equator.

(hold up chart 24a)

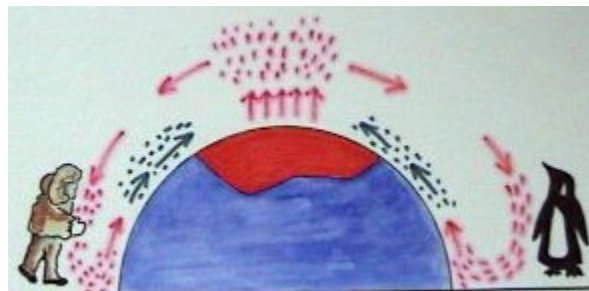
At the equator, since the perpendicular rays fall there, more or less, all year around, the land gets heated up. That in turn, heats the air above it and the air rises. (pic 1)



That creates an empty space. So the air toward surface of the earth, surrounding the area where air is rising, must also move, moving in to fill in the empty space. (point to 2nd mini-pic on chart 24a)



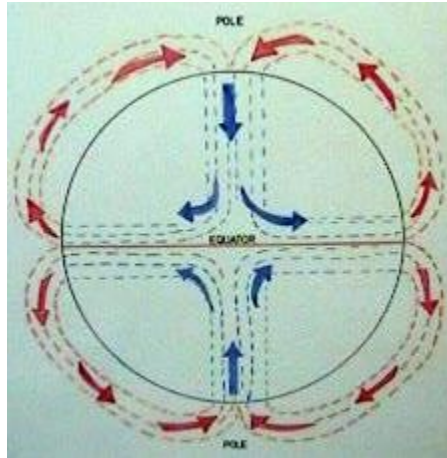
The hot air begins to cool off as it rises and it turns to the side after it has risen – toward the edges of where the air had risen. The cooler air that is drawn in, is then heated and it in turn rises (pic 3 chart 24a).



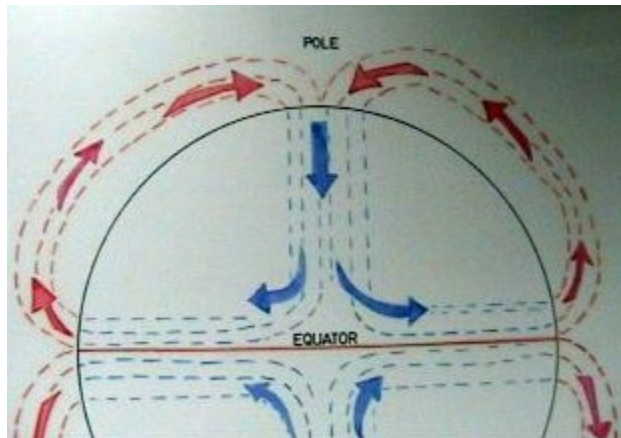
So, in effect, we get two streams of air: one along the atmosphere and a second along the surface of the earth. (pic 4 - chart chart 24a)



So there are two main streams of air. (hold up next chart 25a) One flows along the upper portion of the atmosphere and the other moves along the surface of the earth.



The stream of air that moves along the atmosphere goes from the equator to the poles. The stream of air that moves along the surface of the earth, goes from the poles to the equator.



(repeat from the equator to the poles or vice versa – each time you point to one of the arrows)

Air also has weight. So wherever air is rising, there is less weight – less pressure on whatever is beneath it. Wherever air is coming down to the earth, there is greater weight – the pressure from the air is increased.

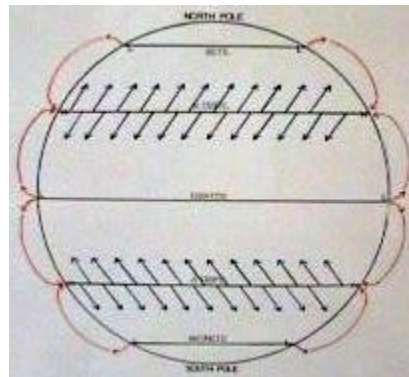
So we get **high pressure where there is greater weight** and **low**

pressure where there is less weight.



You might think, from what we have said so far - the air goes straight from the equator to the poles - and goes straight from the poles to the equator.

But actually, it looks more like this. (hold up chart 26a)



Movement Above Equator

The air rises from the equator and it turns in two directions. By the time it gets to the Tropic of Cancer or the Tropic of Capricorn, it has cooled off so much that it comes down again toward the earth's surface at those locations.

But what happens as it comes down? It splits in two directions. Some turns back toward the equator and some of it turns towards the poles.

Blue Arrows Pointing to Equator

So now we have air, again moving along the surface of the earth at the tropics. The air moving toward the equator gets heated and rises again.

So, in effect, we have a stream of air that moves along the atmosphere here (point to Tropic of Cancer - toward, upward and then away from earth). And the same occurs at the Tropic of Capricorn.

Blue Arrows Pointing to Poles

But what about this air that came from the Tropic line and fell before it reached the pole? Does it make it to the poles? No. Right **before the Arctic Circle or the Antarctic Circle, it meets a wall (front) of cold air and is forced upward.** (point to chart).

So we have many more streams of air than one first realizes. We see them all here (point to various points on chart), here, here and here.

The result of this is that we have even more high and low pressure areas. **Wherever the air rises, we have low pressure.** You can see that here (point to where air rises) We have low pressure areas here, here, here and here. (Look for L and H on the chart). **Wherever the air comes down we have high-pressure areas.**

This is how we get the winds. The winds on earth blow from high pressure areas to low pressures areas. And all this happens because hot air rises. If hot air did not rise, we would not have wind here on earth.

When the winds blow really hard, we get storms. If you watch or listen about the weather, you may have heard about high pressure and low pressure.

End

Epilogue

It might be interesting for the children to know that Eskimos are only found at the North Pole and penguins are only found at the South Pole. These two figures help to reinforce that we are looking at an area around the equator.

If a child asks why the blue lines are slanted, show them the demonstration.

When we show the movement of the air, be sure that the air rising from the tropics toward the poles turn toward the equator – after hitting the wall of cold air at the poles. Blue arrows provide the impression of the spinning of the earth – applied to the movement of the air.

Essential hand movement is that air rises at tropics and at the arctics.

Demonstration

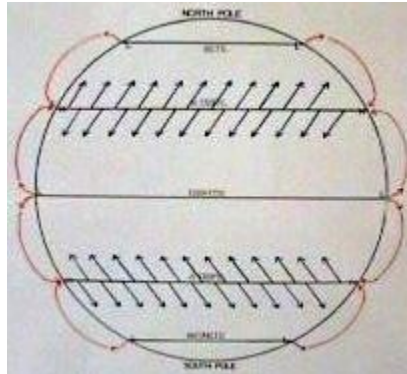
Have the child draw a straight line from the edge of the paper circle while you anchor the circle in the middle with the second pencil. As she begins to draw, turn the circle – so her line ends up being a slanted line instead of a line from the edge to the center.

November 2, 2004

The Names of the Winds

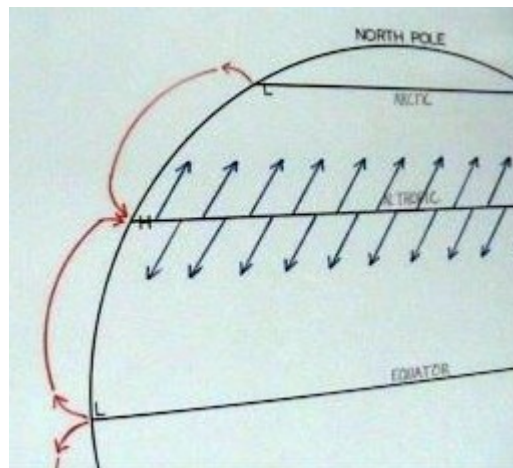
Material

Chart 26a.



Begin

The winds that blow towards the equator blow more or less steadily all year long. Therefore, they are called the **steady winds**. They are also called the **trade winds (blue arrows pointing to equator)**, because in olden times people used sailing ships and in their ships they traded goods; the sailors used the steady winds to go from one place to another. In this way these winds became known as the trade winds.

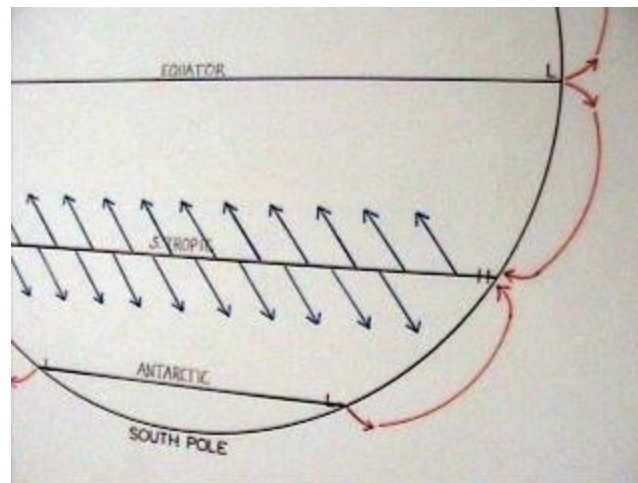


Steady winds blow toward the equator, variable winds blow toward the poles

The winds that blow towards the poles do not blow as regularly all year long. Sometimes they blow strongly and at other times they do not. So these have been named the **variable winds (outer blue arrows pointing toward poles).**

The winds take their names from the direction they come from. In the Northern Hemisphere we have the trade winds and they come out of the northeast. And so, they are called the **Northeast Trade Winds.**

In the Southern Hemisphere, these trade winds are called the **Southeast Trade Winds.** They come from out of the Southeast direction. So, just remember that winds take their names from the direction that they come from.



Southeast Trade Winds

There are other interesting names for the winds. In the Northern Hemisphere, the variable winds are called the **Stormy Westerlies.** Why?

At around 40 degrees south latitude, the winds travel at very great speed and are called the **Roaring Forties.** Why? We could look in a book and see if we can figure out why they blow like that.

There are places around the tropic lines, for example, where the air is neither going up nor coming down. Therefore, they are also not going

sideways. These we call The **Areas of Calms**.

Near the Tropic of Cancer, we have an **area of calms called the Horse Latitudes**. I wonder how it got its name. What do you think happens when a sailing vessel came into an area of calm. What happens? It stops! That would be a big problem, right?



You may want to tell some stories at this point. For example, sailors threw horses overboard; if they had run out of food, they ate the horses.

There is a term used for the area of little wind near the equator; that is called the **Doldrums**. We sometimes use that word to describe how we are feeling.

End

Epilogue

Once you've given these presentations, point the children to books and other resources for exploration. "Planet Earth" by Martyn Bramwell is a good choice. "The Visual Fact-Finder of Planet Earth", by Curtis and Allaby is another. A good teacher reference is also called "Planet Earth", by Jonathon Weiner.

November 2, 2004

Land and Sea Breezes

Material

Heat source, heat-proof pad. Lab thermometer and clock with a second hand. Two small metallic containers in a popcorn pan-container, one with water and one with sand. Chart 27a and 28a.

Begin

In here, I have some water and in here, some sand. We're going to apply the same amount of heat for the same amount of time to both the sand and the water in these two containers.

(ask for volunteer with a second hand on his watch - 3 minutes)

(ask for a second volunteer to read a thermometer)

We have two metal containers with equal volumes of sand and water, both at room temperature. We're going to heat the two for three minutes.

Afterward, we'll measure the temperature of the sand and then the temperature of the water.

(turn off hot plate and remove popcorn pan from heat.)

(push thermometer into sand halfway down into it and wait for the temperature to stop rising on the thermometer)

What temperature did we get? Oh, it's still going up! What did we end up with? 85C

Okay, let's measure temperature of the water. (thermometer mercury shoots downward)

What did we get in the water? 45C.

So, we can see that **land heats up faster than the water**. 85 degrees versus 45 degrees.

Then you can take the two containers to a cold place, like outside, for a longer period of time - 10 minutes - to see how they cool off. The result is that the sand will cool off more than the water as a percentage of their respective temperatures. (It does not become colder but it does cool off to a greater extent relative to its starting temperature.)

How does this relate to the earth?

In places where there is both land and water, the land is heated up more. Then the air by the land also is heated and it rises. (show chart 27a)



We know that creates an empty space that must be filled. And it is filled by cooler air off the water coming onto land. This is what we call a **sea breeze**.

But at night, what happens? At night, the land cools off faster than the water does. (show chart 28a which shows reverse flow of pressure - land cools faster and air above it replaces warmer air over the water).

So now the air over the water is warmer and it rises.



That also creates an empty space that must be filled by the cooler air - which comes in from over the land. This is called a **land breeze**.



This is why, when you live near a large body of water, it is so pleasant. During the day, we get a sea breeze (or lake breeze). During the night, we get a land breeze.

End

Epilogue

Lab thermometers go up to a higher amount and are longer for greater precision; they measure on celcius. Using the single thermometer provides the strongest impression because the children see the temperature drop quickly on the single thermometer.

November 2, 2004

Changes in the Wind Caused by the Seasons

Material-making

Be sure to use the same blues, yellows, etc. for charts 1 and 2.

Material

Chart 1 and 2.

Prologue

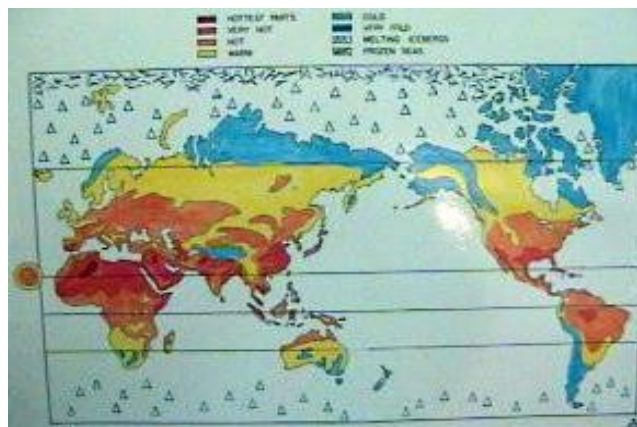
If its been a while since we did the seasons, review that first. We've finished charts 1a through 28a. From now to the end of Geography, charts are number 1-30 (no a).

Begin

Today, we're going to look at the effects of winds in various parts of the world at various times of the year.

(present chart 1)

Here we see the sun's rays are falling perpendicularly to the Tropic of Cancer. If you look up here you see a key; dark red areas are the hottest areas. Yellow is warm. This is our summer.



Lined areas are the frozen seas.

Triangles are icebergs.

Let's use this map to first find some of the hottest places on earth when the sun's perpendicular rays are at the Tropic of Cancer.

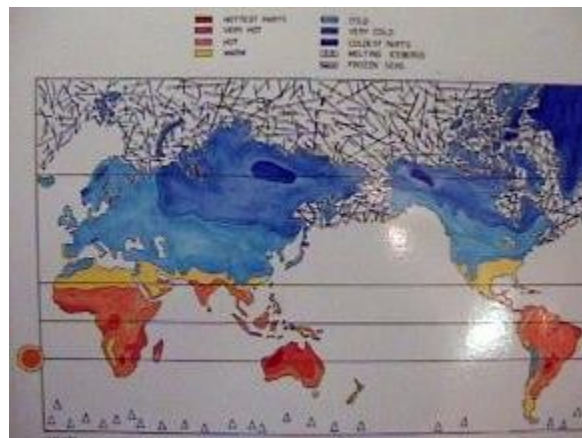
(walk with chart and let children find the hottest places - in red - on the map)

But here, near the Tropic of Cancer, we can also find places that are not nearly as hot. Right there, near the Tropic of Cancer in Africa, its only warm. Why is that?

What about here. This is near the Tropic of Cancer and yet its cold. Why?

So on this map, the perpendicular rays are as far north as it ever goes. And the northern hemisphere is as hot as it ever gets.

But after, the rays begin to move southward. We saw that already. Until finally, the perpendicular rays fall on the Tropic of Capricorn. In the northern hemisphere, it's really really cold now. (next chart 2)



And yet, there are places near the Tropic of Capricorn that are cold. Why?

End

Epilogue

Explore these areas with the children as well as the exceptions. Point out that places on land are the hottest places – because land heats up more readily than water. And since there is more water in the southern hemisphere, it stays warmer year-round as compared to the northern hemisphere.



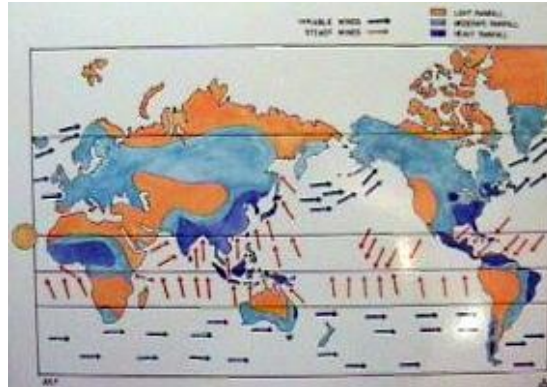
Begin

Now let's look at the winds. (present chart 3) Variable winds are blue and steady winds are red.



Here we see how the winds blow when the sun is at the equator. It gets very hot at the equator and we can see that the steady winds are flowing towards the equator. And the variable winds are blowing away from the tropics.

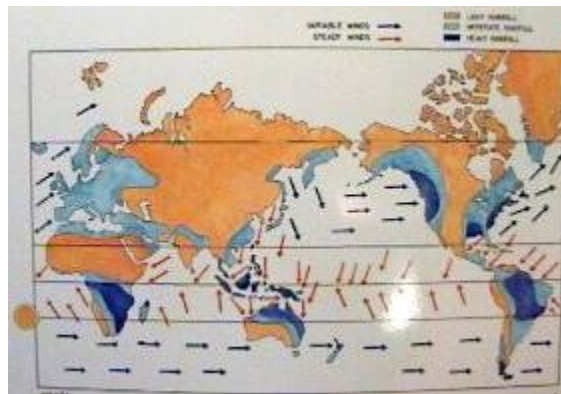
Now let's see what the winds look like when the sun is at the Tropic of Cancer. (show chart 4). Let's see if there are any changes in the winds. In the Pacific Ocean, the steady winds are more or less the same, aren't they?



But let's look here between Australia and Southeast Asia. What has happened to these steady winds? They have turned and are blowing towards Southeast Asia. This is an area where there is lots of water but also lots of land. The air is heated, heating air above it, creating an updraft of air. So the steady winds come to take their place.

(show chart 5)

But what happens when the perpendicular rays move further south, when we are now we're over the Pacific Ocean? What happens to our winds? The steady winds are the same.



But let's look here between Australia and Southeast Asia. They've turned and now they are blowing towards Australia (since perpendicular rays are now heating Australia acutely).

We can also see levels of rain fall. Here in the key, we can see that orange is light rainfall and dark blue is heavy rainfall (chart 4). When the perpendicular rays are again at the Tropic of Cancer, Southeast Asia, where there is lots of land, the winds are coming across the water blowing onto the land – and dropping their moisture. We can see here by the blue that there is very very great rainfall at this time of year.

Let's see the results of the winds (chart 5) when the sun's perpendicular rays are at the Tropic of Capricorn. Winds blow over the water towards Australia. We can see that it also get very heavy rainfall this time of year. Winds are drawn in because land heats up faster than water. And once the wind arrives, the moisture is released onto the land.

We also see on these charts (4 and 5) areas that get very little rainfall. These are the orange areas. Look at South America, for example. There is one area that gets very little rainfall during both times of the year. Mountains are producing an orographic effect on this coast due to their great size which forces the air to rise and release its moisture before arriving inland.

We can also find places that have heavy rainfall during both times of the year. Hot humid winds are always blowing over Indonesia, Malaysia and the Philippines.

End

Epilogue

There are lots of details the children can go into that we do not touch on. Remember the charts are still simplified representations. But they do allow the children to discuss and speculate the reasons or combinations of reasons why rainfall and wind patterns are as they are during different times of the year.

November 3, 2004

Local Rainfall, Equatorial and Coastal - Rains II

Material

Chart 6, chart 7.

Begin

We have been talking about rainfall caused by steady winds and variable winds. There is also rainfall caused by local conditions.

One cause of rainfall caused by local conditions is when we have warm air blowing over a body of water that arrives at a coastline which rises as cliffs. (chart 6)



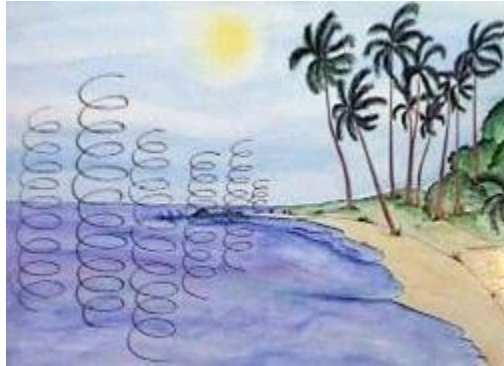
The air is forced upward by the raised land. And when air is forced upward, it cannot hold all of its moisture. So the moisture is dropped and we call it **coastal rainfall**.

There is also something else that can cause **local rainfall**.

We have talked about how hot it can get in the equatorial regions due to the perpendicular rays of the sun. It can get very hot and that hot air captures lots of moisture as the day goes on.

(hold up chart 7)

You can get a sense of that here. The hot air is drawing up the moisture.



(Chart 8)

But at about 3 or 4 o'clock, as it cools off, it cannot hold any more moisture. So, again, it drops this moisture as rain and we call this a torrential downpour - very heavy rainfall. These are called **equatorial rains** because they happen by the equator. If you've ever been in the Caribbean, you've seen this; it happens everyday in the rainy season.



November 3, 2004

Work Chart for the Winds

Material

Work Chart of the Winds (like a map), blue and red arrows and a sun cutout.

Note: there are masters for the blue and red arrows

Prologue

Blue arrows are steady winds and red arrows are variable winds.

Begin

Let's see if you remember where the winds blow. Let's start by building up that general picture (chart 3 with sun at the equator).

The steady winds (red) blow towards the equator (place arrows on chart pointing toward equator from both hemispheres).

What about those variable winds. Let's take the northern hemisphere first. They are blowing from the tropics out of the southwest towards the northeast and poles.

What about variable winds in the southern hemisphere? They blow from west to east. There is no land to deviate these winds so they travel directly from west to east.

(verbalization here is secondary to recalling chart 3 and being able to recreate it.)

(emphasize the picture - don't verbalize - or do as little as possible)

(chart is complete - use chart 3 as control of error)

Now let's add in the sun. (let children decide which tropic to begin with.)

At the Tropic of Cancer, our arrows will change. Arrows turn up around Southeast Asia due to the hot air that is rising over the Tropic of Cancer - between equator and the Tropic of Cancer. (chart 4 is the control of error for this)

(Tropic of Capricorn shows changes around land masses - use chart 5 for control of error.)

Epilogue

Start with the sun at the equator, have the children put the arrows as they appear in earlier charts. Then put the sun at the Tropics, one at a time, and have the children illustrate the changes in the arrows due to the lack of or presence of land. Avoid speaking generally, except for when you may want to explain the arrow adjustments.

November 3, 2004

Currents

Material

Large deep (2 inches) baking pan with water inside, filled up to half-point or so (half-sheet baking pan). Wood shavings, a flexible straw to blow through (blow through long end and bend short end toward water), a brick or some other obstacle placed in the water and centered, towel for spills.

Heat source and heat-proof pad. Flask about $\frac{3}{4}$ filled with water containing fine sawdust – ahead of time so that it thoroughly sinks to the bottom of the water. The sawdust allows us to see what is happening in the water.

Jar of hot water and a small beaker of cold water with a little food coloring (use an ice cube to keep it cold). Pour from spout down the side just a little bit. A white tray underneath also helps to see it.

Chart 9 and 10.

Prologue

Here we discuss ocean currents caused by the winds.

Begin

(place pan of water on table. Sprinkle wood shavings into the water)

When the winds are blowing over the ocean, they create currents.

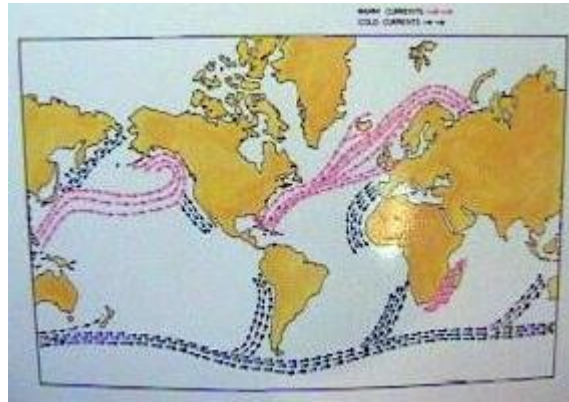
(straw to create the winds, see the wood shavings move across the water)

What happens if a current comes across an obstacle? Some are deviated around the obstacle but some are turned back against the wind.

Water blown out of cold regions creates cold currents. Water blown out of warm regions creates warm currents. If those currents come across an obstacle in their path, they can be deviated or turned.

(hold up chart 9)

Here we see some of the warm and cold currents in the oceans. Cold currents are in blue and warm currents are in pink.



Here we see a warm current that comes out of the Eastern United States, crosses the Atlantic Ocean and comes up here clear around the north side of Scandinavia.

This keeps this area much warmer than we would expect it to be.

At North 69° latitude, there is a port named Murmansk in Russia, which is open all year long. It never freezes even though it's so far north.

Over here in Asia, there's a warm current. It comes across the Pacific and turns around the south of Alaska. So, this area is warmer than you would expect it to be also.

Let's look at some cold currents. Down here at the bottom we have one. Does that remind you of anything? Yes, the winds. But it is deviated here at the coast of Africa and the coast of South America.

(set up heat source)

Do you remember we asked why there is so little rainfall here in South Africa? It's because of this cold current. The winds, because of the cold currents below, are not containing moisture as they arrive at the

coast – so it does not rain as much at that coast.

Remember we talked about hot air rising. Hot or warm water also rises.

We can see in our flask that the hot water is rising and bringing the sawdust along with it. As the water cools, it turns and begins to fall again, taking the sawdust with it again.

Let's look at our chart again. We see some cold current which seem to come out of nowhere. Let's look along the coast of North America. (off coast of California). This cold current seems to come from nowhere.

Well, what happens when a cold current meets a hot current.

(next demo)

In here I have warm water and in here I have some cold water. Watch what happens.

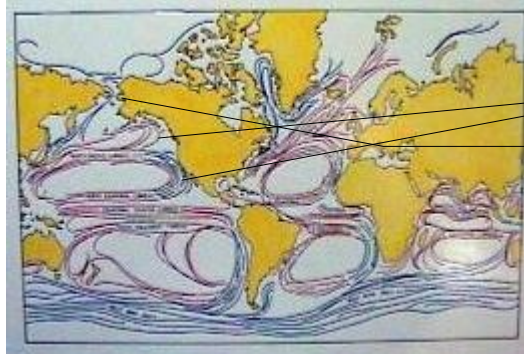
(cold water goes right to the bottom)

So this is why we have cold current popping up here. When a warm current and cold current meet, the cold current goes under the warm current. After the cold current passes under, it will rise to the surface again. This current actually travels through the Bearing Straits, goes under the warm current off the coast of Alaska – then pops back up further south along the coast of California. This makes the climate more temperate than one would expect because it cools them off



(hold up chart 10)

Human beings have named the major currents of the world. We like to name things don't we? Here we have the major currents of the world with their names.



This cold current travels through the Bearing Straits, under the warm current off the coast of Alaska – and then pops back up off the coast of California.

Remember the currents are caused by the winds. So if you remember the wind directions, it helps you remember how the currents flow. In the Northern Hemisphere, the steady winds come from the Northeast. So the Northeast trade winds come in this direction (toward Southwest).

This one is the North Pacific current. It goes clockwise. It is a bit deviated down around Asia and it turns and becomes the equatorial counter-current. (moving east from Asia)

Here in the Atlantic, it's a bit different than the Pacific. We have the Gulf Stream, originating in the gulf and it travels all the way up to Greenland. We also have the Northern Equatorial Current running clockwise here, that circles the Atlantic. Around the top of South America it is deviated and creates the Equatorial Counter Current.

And we have the Southern hemisphere. The trade winds come from the Southeast. So we have the Southern Equatorial Current here – counter-clockwise. Then this is this current at the bottom called the West Wind Drift. Part of this West Wind Drift is deviated along the coast of South America and becomes the Humbolt current which travels up the coast of South America. And part of West Wind Drift is deviated along the coast of Africa and this becomes the Benguela current. The Humbolt and Benguela currents keep these regions cooler and dryer than one would expect.

The cold current along the western coast of North America is called the California Coastal current. It keeps this area more temperate than one would expect.

This North Atlantic Drift is the current that keeps the port of Murmansk open all year long.

End

Epilogue

Allow the children time to investigate this on their own. At this point the children have everything they need to understand currents. With the children we want to pick up on a few of these names and then let them explore on their own.

We review the winds, named some of the currents, and given some idea of how currents affect the climate. We have told some of the stories of sailors and how currents, helped along by the winds, help boats to travel from one place to another. Help the children investigate the routes used by early explorers and traders.

Have books for the children to look at which gives more details about the currents – perhaps the same books as the winds. Help the children explore animal life and how currents affect migration and other patterns of whales and other wildlife. The children can make slips with the names of the currents and place them on the work charts in the correct locations.

November 3, 2004

Wind as a Sculpture

Material

Chart 11 and “Wonders of Nature” picture or other picture of wind erosion.

Prologue

This lessons discusses wind erosion.

Begin

We have seen a lot about what the wind does. It causes currents and rainfall. There is another thing that wind does. But it is much less obvious and very hard to see because it takes place over a long period of time.

This is the work of carving by the wind. Erosion, by the wind. Have you ever walked on a beach with bare legs on a very windy day, and felt the sand being blown against your legs?

The wind, as it blows, carries along tiny bits of sand and pebbles. They hit whatever is in their way.

(present chart 11)

It's a bit like this. They hit whatever is in their way. This happens especially in the desert where it is dry. Bits of sand hit and hit, over and over again, against the rocks - as they are blown by the wind. This is especially true in dry areas.



So over time, they wear parts of the earth away – especially softer parts. They cut away parts of the earth over time.

Here's a place where you can see what the wind did.
(hold up book)

We know it was wind because the sand and pebbles are carried only so far up. We can see only the lower areas were affected on this rock cliff.

End

Epilogue

Have books around for the children; “The Wonders of the Nature”, by Jacqueline Seymour is a great one to have. Most erosion is a combination of water and wind and this is something to be discovered.

The Work of the Water

November 15, 2004

The River

Nomenclature, Carving, Carrying and Depositing - I

Material

Wooden frame (see paper handout). Prepare the materials inside: start with wet newspaper and build up the base with the paper balled up. On top of that, put a thin layer of clay. The wet news paper helps keep the clay moist so you don't have to use so much. Put some larger sticks and rocks and cover the whole thing with damp sand. You can sprinkle the sand on and use a plant mister to make it moist. If you will not present right away, cover it with plastic or something to keep it from drying out. You can only leave this out for several days, as mold will begin after that time.

Charts are used extensively. Review charts 4 and 5 for rainfall for at two different times of year. We use chart 12; chart 15 is the rivers of the world. Chart 16 is the carrying and depositing. Chart 17 is the V-shaped valley. The canyon is chart 18. See Appendix A.



Prologue

When you start your flow of water, start it from one side or the

other so the children see how it meanders and winds. Start with a trickle to see how it will flow and then add more as needed for illustrational purposes. This lessons can be done outside.

Begin

(begin pouring a trickle)

Much of the water that flows on the earth goes underground. But some of it stays on the surface. It gathers in brooks and streams and eventually into a river. A river flows both downwards and sideways. When it comes across little particles in its path, it just carries them along.

(pour water down ramp slowly)

When the force is less, those particles slow down again. When the force is greater, the particles get carried along and are deposited into the sea. We can see that a river carves, carries and deposits.

(pour water down ramp rapidly)

Look how it has carved up here. The force of the water was so great. It carves and carries and deposits wherever it goes out. If it meets something very large, it goes around it.



Up here where the river starts is called the **source** (point to top)

Here (point to surface) we can see the **river bed**. The sides are called the **banks**.

Down here where the river empties into the sea is called the **mouth** of the river.

(for this part, catch the runoff in a clear glass loaf pan. Use colored sand contrasting with the bed – sprinkled on the river bed.) (see layers of sediment)

During one period of time on earth, there was this kind of sand and soil (pour water) covering the earth. It was washed down to the sea and deposited as sediment.

(sprinkle more sand of a new color)

During another period of time, there was another kind of soil or sand covering the earth. And when the water flowed into the river (pour another) and was deposited.

And during another period of time, there was still another kind of soil or sand which also washed into the seas. (pour third water to allow the third colored sand to be deposited)

(with luck you will begin to see layers of strata in the loaf pan, created from the three colored deposits.)

Epilogue

Sometimes you might have a triangular formation at the mouth – this would be called a delta. It is generally triangular and flattened and at the mouth. You might have the children explore rock polished by water – to compare it to a regular rock. You can show pictures of rivers. This allows one to see the winding course they take. Go out and visit the local river if possible. Another follow-up is the nomenclature material for the river. After this lessons is a logical time to do this work.



River Sources, Highlands and Lowlands - II

Begin

Remember how we saw the water of the river carries particles with it, forming its own path as it goes.

(hold up chart – orange is the highlands and yellow is the lowlands, light green is still lower and dark green is the lowest land.)



chart 12 – high and low lands of the earth

On this map we have illustrated some of the high lands and low lands on earth.

Look at this river (Yellow River). It starts here in the highlands and goes down to lower land. A little bit lower and finally out to the sea. Let's look at another.

Again, it starts in the highlands, goes toward lower lands and finally ends up out in the sea. (repeat with several rivers to get the idea of flowing from highland to lowlands).

We also have rivers like this one. (do one or two like this) This one starts in the highest land, it flows through the lower land and still lower land – heading north – now it's flowing over the lowest land. (children tend to think the rivers flow south, so pick several of different directions).

(use charts 4 and 5 along with 12)

We have chart 12 showing us major rivers of the world and 4 and 5 which show places on earth that get much rain fall during the entire year. Where do we have a place that gets a lot of rain fall both times of the year? (Around the Nile for example) Low and behold, there's a major river in that area. (let the children find other examples where rainfall is heavy year-round and a river is present).

I wonder if there are any rivers in places that don't get a lot of rainfall during these times of the year (these times per chart 4 and 5). Well, are all rivers caused only by rainfall? What about snow that is melting? A lake can also be a source of a river. What else can cause a river do you think? (only give them enough to get them discovering on their own)

Major and Temporary Rivers, Cities and Canyons - III

We can look at the main rivers of North America. This map gives an impression of the rivers that flow all year long in North America – the continent we live on (hold up chart 13).



But not all rivers flow all year long. Some that are caused by melting snow, only flow in the spring and summer; once the snow has melted, the river dries up. If we were to try to show all of the rivers that flow for at least part of the year, (hold up chart 14) we would have a map that looks more like this.



We can look at a transparency map of North America. Here we have just the outline of the continent. Next, we can overlay the first transparency, the highlands and lowlands. On top of this, we're going to add the waterways. Now, we can look at where some of the major cities have been developed. The top transparency has red dots to show where they are located. (align all together). We can now look at

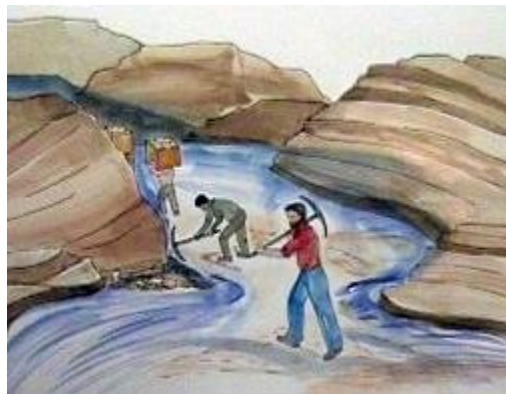
this and see that many major cities grew up along the waterways.

Do we see some that are not next to these waterways? First, what cities are in this category? Why did they develop there? (natural resources, transportation like the railroads, natural defensive positions).

Here are some of the major rivers of the world (new chart 15). (identify several we know). Can you imagine what this map would look like if we showed the rivers that also flow during only part of the year? There are many here but if we showed those flowing only part of the year, there would be many many more. Where do they come from and where are they?



The river does its work constantly day after day. And yet much of the work done by the river is done silently. The river is always carving out, carrying and depositing. It digs down to make the surrounding land lower and lower. (chart 16)



If we could cut across a river, we could see the shape of the valley that is carved by the river. We would see that it tends to be V-shaped. Since it rains a lot in many of these areas, the land along the tops get washed down - so it becomes what is called a V-shaped valley. (hold up second new chart, chart 17)



Sometimes a river flows through very very dry areas. Then it cuts through the land more like a saw that cuts straight through - like you see here (new chart showing Grand Canyon - like image, chart 18).



There's not a lot of rain here so the sides are not washed down into the water. So what we end up with is called a canyon; a canyon is a river cutting through otherwise dry land. Sometimes the sides are very steep and deep. (hold up picture of the grand canyon).

November 15, 2004

Rain Carves Earth - Rains III

Prologue

We again start with a demonstration to show that rains also carve the earth.

Material

Two balls / mounds of wet newspaper, thin layer of clay covering them, some damp sand on top of the clay. On top of one of them, some grass or moss. All of this built up on a tray or shallow pan. A watering can with a sprinkler head to simulate rain.

Begin

When it rains on earth, let's watch and see what happens. (pour aggressively onto each mound separately)

Where there are no plants, with their network of roots that hold on to the soil, the rains can wash away the land – just like we see here. But in places where there are a lot of plants and trees with their big roots, the soil is held onto and not nearly as much is washed away by rain.

This washing away of the earth is called **erosion**. As the rains fall onto the earth, year after year, some of the softer land is washed away – just like we saw with our demonstration. This happens very very gradually, so that we don't see it happening but only notice the results of it.

Sometimes a rock of harder substance is resting on top. And that harder rock protects what is underneath it – just like an umbrella protects us from the rain and wind. The softer rock underneath slowly are washed away and we can end up with a formation like this.

(hold up chart 19) The harder rock is on top and the softer rock underneath it has slowly been washed away. This is called an **earth pillar**.



(hold up pictures in a book)

Here are more examples of earth pillars.

Epilogue

As the children look into earth pillars, they find that there is a combination of causes which cause such formation. We present them separately for the sake of clarity. If possible, go out and see such formations for the full impact.

November 15, 2004

Ocean Waves

Material

Long mound of earth (sand) atop clay that forms an earth pillar, a land arch or whatever you want – covered by a layer of damp sand, on a tray with edge about 1 inch high. Some balsa strip or something else to create waves with and a pitcher of water.



Ocean beach with sand and rock shoreline

Begin

What happens when I create waves? (pour water into tray and began making waves)

We see that the shoreline sand is washed away, day after day. Just as the river and rain carves, we also see that the ocean waves also carve the land.



over time, softer earth is increasingly carved away from the rock by the water

They come in and bounce off the coast line.



The rock erodes more slowly than softer earth and becomes exposed

Over and over again, year after year, they carve beautiful formations like land bridges, earth pillars and caves.



As shoreline erosion occurs, caves and land formations sometimes (re)appear

They wear away the softer rock and leave behind the harder rock.

So the ocean waves also cause erosion. That's why we find coastlines go in and out - not just a straight coastline. They tend to flatten things out right at the water's edge - in the form of a beach at the water's edge.

Epilogue

You can go in different directions: various kinds of geographical features, land and water forms, regular and irregular coastlines. The details of a coastline – whether there's a promontory or a spit. Discuss varieties of highlands or lowlands, the foothills, mountains. Show pictures of these, show them through the nomenclature material and go out to see such formation and features if they are present in your area.

November 16, 2004

Ice

Material

Charts 20-24. Small bottle with a long neck, filled to the very top with water. Put the cap on as tight as possible. Put it in the freezer – on a tray and cover it with a plastic bag in case it explodes.



Begin

This is what happens in the rocky areas where the weather is very very cold. Not all of the water that falls on the earth stays on the surface of the earth. Much of the water sinks into the ground. Some of the water seeps into the cracks and crevices, going deep down as if it wants to find out what is down there.

And that water remains in hiding where we cannot see it. But then what happens? If it's in an area where the winters are very very cold, when it gets cold the water freezes – like the water in my frozen bottle. It becomes ice.

As it becomes ice, it expands and the force is so great that it can even break the rock, just like it broke the glass. The rocks explode gradually over a long period of time. If its a big rock the cracks will just get bigger and next season, more water can seep into the crack. That winter then, when it freezes, the crack gets a little bigger - and so on, year after year.

This continues until the time comes when the broken pieces of rock fall off as the ice melts. This happens very often along the edges of mountains. Have you ever driven through the mountains and seen the signs that say, "Caution, falling rocks!" This is why and what is happening there.

(hold up chart 20)

It's like a little boy who goes into hiding. Maybe he takes a book with him and falls asleep reading.



But then, as he wakes up, (new chart 21) he stretches out his legs and breaks off a piece of the rock. So this is another way that water works on the earth - in the form of ice breaking up the rocks.



Glaciers

High up in the mountains, where it's very very cold, it's so cold that all of the snow which falls during the winter cannot always melt the next summer. So it stays there on the ground. Then as it rains – or snows more – that rain and snow coming in contact with snow on the ground, causes it to turn to ice. It never warms up enough for it to melt.

So this snow and ice goes on building up. Over the years, quite a lot of ice can accumulate. As the weight of that ice becomes too much, it starts to slowly move down the mountain. It becomes like a river of ice. This is what is known as a **glacier**. A glacier moves very slowly – so slowly that we can't see it move with our eyes; but we can mark it off and measure it later to see it is in fact moving.

Just like the river, a glacier flows along scraping the bottom and sides where it's flowing – making a valley just like the river carves out a valley. Unlike a river though, if a glacier comes to a big obstacle in its path, well - it's not a liquid so it can't go around. It is pushed up and over the obstacle. The pressure of that ice is so great that it pushes it up and over. If it comes across smaller rocks or branches and so on, it pushes those along in front of it.

(present chart 22)

In fact it can gather up much debris as it goes along. Sometimes you will see rocks being pushed along at the front of the glacier. That's called a **frontal or terminal moraine**.



Sometimes, the rocks or debris are gathered up along the edges (point to chart at edges of glacier). These are called **lateral moraines**.

And sometimes, just as two rivers can join together, two glaciers can join together, like we see happening up here (point to chart). Then we get a moraine going right down the middle of the glacier - this is called a **medial moraine**. It is made of what the two smaller glaciers had gathered on their sides but they have now become one medial moraine. (hold up picture of a medial moraine)

Icebergs

Sometimes, in the arctic where it's really really cold, great big portions of a glacier flow right into the sea. There, they break off. These are the **icebergs**. Any rocks that they carry are washed right into the sea along with them. (hold up picture of an iceberg).

The climate of the earth has not always been the same. There have been periods of time when for thousands of years, the earth has been very very cold. Many places on earth were covered with large sheets of ice, where today there is no ice at all. The glaciers extended much further south than they do today. Even though we don't see any ice in these places today, we can tell that glaciers have been there by some of the marks they left behind on the earth. In some places, you'll find

large **rocks laying around in the middle of nowhere**. There aren't any mountains around, so they couldn't have fallen off a mountain. They were left there by a great glacier.

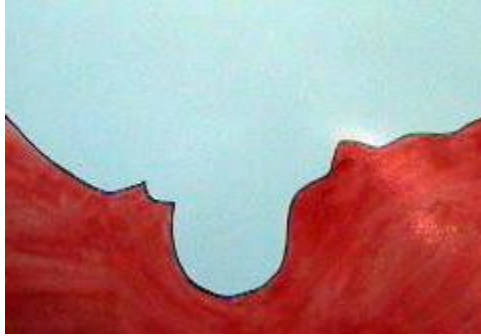
The Great Lakes - the shores of Lake Michigan, were carved out by great glaciers. The Wisconsin Dells, an area not covered but right next to a glacier, through the melting of the glacier and its great flow of water, carved beautiful rock formations. Kettle-Moraine National Park is one great location to see the work of water on the earth.

There's another way in which we can tell if a glacier has visited a place or not. And that is by the shape of the valley that is left behind. We already said, a glaciers scrapes out a valley just like a river. (hold up chart 23 glacier melted).



But a valley made by a glacier is much broader at the bottom and have sides that are steep. It is more like a U-shape than a V-shape. We can see some of the moraines left behind (point to the chart) as the glacier melted.

And sometimes, if there have been smaller glaciers that had joined with the larger glacier, they left behind a little valley like we see here (point to chart), just kind of hanging on the edge with water dripping down. It's called a **hanging valley**. (present new chart 24 U-shaped valley). Here we can see a valley created by a glacier and a hanging valley here (point to chart) on the edge.



Epilogue

“On Icebergs and Glaciers”, by Seymour Simon, is a great book to have in the classroom.

November 16, 2004

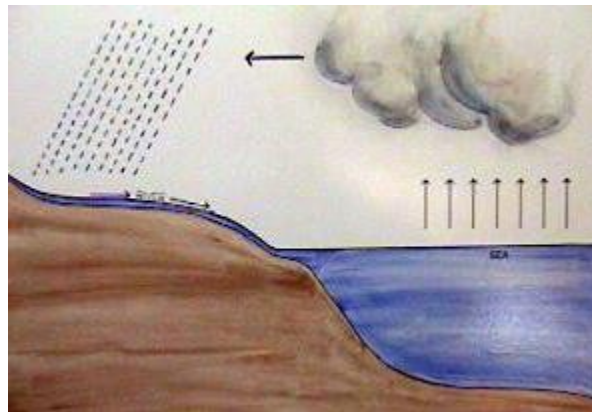
Summary of the Work of Water in Nature

Material

Charts 25-26.

Begin

Let's stop and think for a few minutes about the work of water.
(present chart 25).



From the ocean and sea, moisture is carried up along with that hot air which rises. Up in the atmosphere, it cools off and the moisture gathers together into clouds. Those clouds are carried far and wide by the winds which move on the earth.

Some of those clouds are carried over land and there that moisture is dropped in the form of rainfall or snow. On the earth over land, that water then collects together into brooks and streams, running through the forests and hills, gathering together into rivers. Eventually, it makes it's way back to the ocean or sea. And then the whole cycle starts over again.

The moisture is carried by the hot air that rises, gathering into clouds that are carried far and wide, some going over land where the

moisture is then dropped again - in turn gathering into brooks and rivers rushing back to the sea.

As it does all this, it carves, carries and deposits the earth. It is like a never-ending game played by children on the slide. (present chart 26). Up they go and down they come again - over and over. It goes on and on forever - in fact at this very moment, it is happening.



Here is the little one (point to person on the chart) who goes into hiding to do his work of breaking up the rock.

November 16, 2004

Spread of Vegetation

Prologue

This is after botany work with the children and hopefully The Needs of the Plant. It is well after the previous water lessons. It's also useful to have done The Leaf - varieties of leaves are discussed. If you haven't done this work, modify the wording here.

Material

Chart 27 with a population overlay (stored separately with its own container).

Begin

There is another work that's done by water which is work of a different kind that we've discussed. **If it wasn't for water, life on earth would not be possible.** Both plants and animals need water for their lives - we need water to stay alive.

Plants need water in order to make their food. And much of their bodies is made up of water. Where there is no water, there are no plants.

(present chart 27)

Let's look at the Torrid Zone, the area around the equator where the sun's rays fall perpendicularly all year long. In the Torrid Zone there are some areas that get almost no rainfall - like right here (point to yellow regions in the Torrid Zone), in areas where there is almost no rainfall.



We usually find deserts in these places. It rains very very seldom in these places – sometimes only after years have passed. We can see it is difficult for plants to live here. There is no continuous carpet of green like we have in other places during the summertime.

The few plants that can live in places like this generally have thorns in place of their leaves. Remember leaves give off moisture from the plant; these plants in the Torrid zone don't have any moisture to spare. They hold on to any water that they do have.

Some of these plants also have devised very clever ways of storing water. They will store it right in their bodies, in their stems – so that you'll find plants here like cacti and succulents. Those are the kinds of plants we tend to find in these desert regions in the Torrid Zone. Can you help me find other places like this on this map? (allow the children to look and discuss their findings).

But in the same Torrid Zone, we can also find places that get a lot of rainfall, like here (point to dark green in the Torrid Zone). These areas get a lot of rainfall and it's very very hot. So conditions are ideal for many plants to grow. Generally, the kind of plants we find in these areas have enormous leaves since these plants want to get rid of some of their moisture.

Look, we have a big area here in South America in the Torrid Zone of dark green. Imagine all of that green lush vegetation with enormous leaves. (point to chart). We have forests here and all kinds of plant life. In these areas are where we will find the equatorial rain forests.

There's often a lot of competition to get to the sunlight and so we'll also find the creepers - plants that climb up other plants - to get their leaves up toward the sunlight. As we know, they need sunlight to make their food. Would you like to help me find other places like this on our map? (let children help find more dark green areas in the Torrid Zone).

But then still in the Torrid Zone, there are areas that get much less rainfall than the equatorial rain forests get - though more than the deserts get. These are the light green areas on our map (point to chart). These are areas of grassland. For mile after mile, one sees different kinds of grasses growing. These areas don't get enough rainfall for big forests to develop. It is very hot here; so even though they get a good amount of rainfall, the water evaporates quickly. If there are any trees growing in these grasslands, they tend to be the kind of trees that don't have very many leaves - like the Acacia and the Baobab.

We've only talked about the Torrid Zone so far. What about the Temperate Zone?



We live in the Temperate Zone. In the Temperate Zone, we also have areas that get a fair amount of rainfall. Place like this (point to brown areas of Temperate Zone).

There are big sections like this. These are areas where we find trees like Maples and Oaks, a couple we're familiar with from around here. Remember, the Temperate Zone is cooler than the Torrid Zone. Just before winter, some of these trees lose their leaves. Smaller plants die altogether because it gets too cold and they can't survive. Here, in North America, we have a big section of brown like that (point Midwest and East NA).

Still in the Temperate Zone, we also have places that get very little

rainfall. We have these large sections of yellow (point Canada) where they get very little rainfall. In some of these areas you will find deserts since there is so little rainfall. If there are any plants growing in these areas, they would again be plants like succulents and cacti.

Then in the Temperate Zone, we have areas that get an in-between amount of rainfall – less than in the forested areas and more than in the deserts. This is once again where we find the grasslands – like here (point North Canada) and here (Northern Asia).

Remember we talked about the Steppes of Russia. That's these grasslands. These played a very important part in history because the grasslands were easy to travel through – easier than traveling through an equatorial rain forest or a desert.

As we go further north, we know it gets colder and colder. Here, we get to areas where we have the large trees (point to white areas of North America) with needles for their leaves. Because they get lots of sunlight, they get quite tall. But they have needle-like leaves to help them hold on to all of the water that they need. They don't want to give any up. Look at what vast areas they cover across Northern Europe, Asia and North America (point to chart).

We can go even further north and we know it gets even colder as we go further. The ground up here is almost constantly frozen (point to most northern regions above Asia). During summer, perhaps only the top couple inches thaws out. In the winter time, the sun almost never comes out so it's very very cold here. Few plants can survive up here. No trees but sometimes shrubs that grow close to the ground can survive. Not only is the ground almost permanently frozen, but there's almost always a lot of wind here too. If you see these shrubs, you can tell which way the winds come from because they cause the plants to grow in the direction the winds are blowing.

The kind of plants that can survive in these very cold regions are plants like Lichen and Moss. During the winter, they sort of close up shop and wait for the warmth of summer to return. Just think of how the plants have adjusted to these different kinds of climates and conditions. It's amazing how they've done it. Maybe you want to look

into some of these more interesting plants that survive in these difficult regions. How long does it take for a Lichen to grow one inch in diameter in these regions?

Where have the most human beings settled on this map? (use population overlay for chart 27). We can see here where most people have settled. They settled mostly in the brown regions. Why? Temperatures are suitable for people. It's easier to travel. Food can be grown more readily. The land was easier to develop - for shelter and so on.

If we look at an area like this in the middle of the desert (point to the Nile), we can see that rivers also influence where people have settled. There are people in other desert areas but life is perhaps more difficult. How about these cold regions. I don't see red here but we know there are some people living there. I wonder what their lives are like. We can look into that too.

Epilogue

Don't give too much; give enough to get them interested enough to begin exploration on their own.

Note: Make a copy of the overlay before leaving

November 16, 2004

People in Different Zones

Material

Charts 28-30.

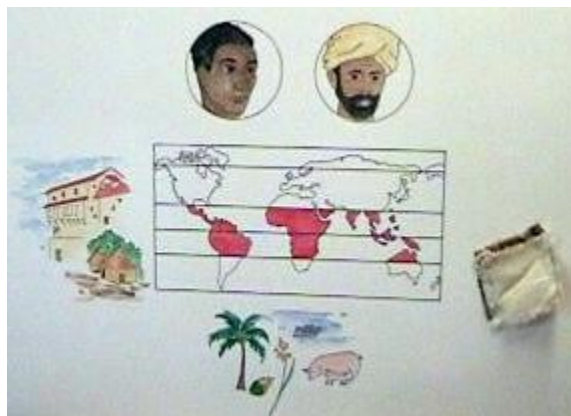
Prologue

This presentation draws from both the past and present; it draws on what is familiar past and present to relate the child's experience to the basic environmental aspects of different regions and their impact on how people have survived in those regions.

Begin

Just as we have a variety of plant life on earth, we have a variety of people on earth. We find that there are different kinds of people who originally lived in different places in earth. We're going to first look at the Torrid Zone.

Here are several original types of people who lived in the Torrid Zone.



Remember we talked about how very hot it is in this part of the world. So usually these people have a darker color of their skin to protect

them from the sun – since the rays are always perpendicular.

We talked about what enormous forests there are in some of these places. Where the forests have not been cut down, the people living there use whatever material from the trees that they can, to make their houses. Often times they'll use branches and leaves to make the roofs of their houses. They'll use the wood from the trees to make the walls of their houses. Whatever parts they can use, they will do so.

It's so hot in these places that their houses have very few windows in order to keep the heat out. Often times the people only go into their houses to get out of the rain or to sleep. For food, they can grow many kinds of different fruits and vegetables that we can't grow here; bananas, dates, coconuts are grown in these areas. When we go to the grocery store to buy these things, they often come from these parts of the world.

Rice also grows in these climates (point to the chart) because rice needs a hot moist climate. There's also quite a lot of meat and fish that these people can eat to meet their need for food. Especially in places near water, they tend to eat lots of fish.

Remember we talked about how hot it is in these places. Often times, these people don't wear much clothing. What they do wear, they want to be light-weight (point to chart) like cotton or even like grass. These materials help them stay cooler. For special occasions, they might wear something made of silk. So that's the people of the Torrid Zone.

We also have people in the Temperate Zone.



Here are three types of people from the Temperate Zone - a Native American wearing a ceremonial headdress that is only worn on special occasions, a Mongol, a person from this part of the world (point to Asia) and a Caucasian, a person from Europe. Notice that he has lighter skin. Since they lived further north, they needed lighter skin to get the benefits from the sun.

We have a wide variety of housing used in the Temperate Zone (point to chart). Here, we have a moveable type of housing - a tent or a teepee. Many of the people using these shelters followed the animals from one place to another. So they had to have a kind of housing that they could take with them. They'd fold it up and carry it with them, setting it up somewhere else later on. There are still people in parts of the world that do that.

Here, (point to chart again) we have a kind of housing with tile roofs. In these places, there's a lot of clay available for forming it into tiles for roofs and buildings. We find that in China, Japan and many other places. Of course, there were also big cities that grew up and there are large building here too.

For food, there's a wide variety that can grow in these areas - like apples, wheat, grapes and so on. Wheat needs a different climate than rice; so there is a lot of wheat grown in the Temperate Zone. People in this zone also have lots of meat and fish available. They often eat beef, chicken and so.

In terms of clothing, it gets a bit colder in the Temperate Zone than in the Torrid Zone. So people often wear clothing made of wool or thicker cotton. During the winter we're going to need heavy coats and boots. Do you think children in the Torrid Zone ever need heavy coats and boots? No. But those in the Temperate Zone will need such things.

(present new chart 30)

Then, we have the Arctic Zone and here at the top we have an Inuit person.



One of the interesting things about the Inuit is their eyes. If you've ever met an Inuit, they have very distinctive eyes. Their eyes open only a slit and they have heavy eye lids. That is to protect their eyes from the glare of the sun on the snow; if not for this, they could go blind. So their eyes have changed to protect them from snow blindness. We've talked about how cold it is here so what do they do for housing?

During the summer, those in Arctic regions sometimes make houses from blocks of sod – just the top portions of ground. They cut just the pieces that have thawed to use for their houses. During the winter, they used blocks of snow and ice to construct their housing (point to chart Igloo). It was shaped this way so that it can withstand the winds that blow so furiously there. Do you see any windows in this house? No, none at all. This time there are no windows in order to keep the cold out. Even the doorway is like a tunnel they had to crouch down to

pass through; again, this helped them keep the cold out. They were very clever.

For food, those in the arctic did not have many plants. So for the most part, these people eat meat and fish. Reindeer and the Caribou were some of the meat they typically eat. For clothing, these people need thick warm clothing. So they would use the fur from the animals in that region; they are thick enough to keep them warm, soft and comfortable and help keep the moisture in. With that cold wind blowing, they need to well protect their bodies. Maybe you'd like to look in some of the books we have to learn more about people from different parts of the world.

What we have to remember is that all people have the same basic needs. We all need food, clothing, shelter and transportation. We just met these needs in different ways around the world, depending on what was available in the environment to meet out needs. This is why we have different cultures and customs from different parts of the world.

Epilogue

Give enough information to get them interested. Pick up on a few of the differences to help the children realize why there are those differences. Today, our society is so mixed, the children wouldn't understand unless we went back to early origins of people. Ultimately, we hope to help the children realize that in spite of differences, there is unity among all human beings. Hopefully, we can help them arrive at that realization.

Interdependence and Economic Geography

February 6, 2004

Interdependencies

Age

6 years old

Prologue

In 1969 at a study conference in Italy, Mario Montessori tried to revive a set of exercises which for some reasons, was being neglected in the elementary classes. It is a set of exercises that brings to the awareness of the children the interdependencies of human beings in society.

One can wonder why teachers have ignored these exercises. Perhaps they are not sophisticated enough. On the surface, they seem quite simple. So, people may not realize their importance and their potential. This set of exercises is probably more important now than when these activities were first developed.

Roughly 29 percent of the crime committed in this country is committed by children under the age of 18. The nature of these crimes is getting more and more serious as we know. The problem of crimes committed by youth is not limited to the United States; it is becoming a world-wide problem. There is a certain alienation felt by young people. It is this alienation these exercises are addressing.

We need to help children reflect upon the fact that they are a part of society and that their well-being depends upon the contributions of other human beings with whom they often have no direct contact. Remember that part of cosmic education is to help the children feel gratitude to unknown human beings. This set of exercises is more directly aimed at bringing to the children's consciousness that gratitude toward unknown human beings.

Activities Prior to Presenting Interdependencies

Soon after a group enters the elementary class, introduce them to stories about how some of the products they use in their daily lives are made. These stories need not be exhaustive; present a couple items through our story. For example, tell the story of how bread is made or how milk is produced or how clothing is made. The items selected for presentations should be connected to the Fundamental Needs of Human Beings.

There are many books on this. In selecting books, make sure the focus is on the people involved and not machinery or resources. Show pictures in the book and tell the story – don't just read it. We may want to write stories; but remember, the **emphasis is on people**. We can streamline the process or story so as not obscure the interdependencies involved. If this story is original, include pictures and text for each of the important people at each stage. Begin with a small group.

Begin

(begin by asking what was eaten for breakfast – if no one says bread, work it in somehow)

Some of us had bread for breakfast. Let's see how it is made. I have this book ("From Grain to Bread"). I'm going to tell you how bread is made.

Here's our farmer. He's preparing to plant seeds to grow wheat (hold up picture)

Here's the wheat growing. (hold up picture)



After the wheat grows and when it's ripe, the farmer harvests the seed. He cuts it down and separates the seeds from the wheat. (show pic)

Once the farmer has the wheat grains, it is taken to a miller. The miller will grind the wheat berries into flour. (show pic)

Once the miller has made the flour. He takes it to a baker, who mixes it with other things to make bread and other kinds of things to eat. (show pic)

Once it's baked, the bread is taken to a bakery. We can go buy it and we can eat it for breakfast.

All of these people worked to make the bread that you and I had for breakfast today!

Another day, we'll talk about how clothes are made for us to wear! (begin early story themes with products that come from farms; later do cars, etc.)

Epilogue

Parallel to the story, help the children to do a going-out to a factory or bakery or dairy farm and so on. The children should see all the people working together to make one product that the children use day by day.



Materials

Three sets of cards, each with subsets. Each card is coded on the back with a letter and the number corresponding to its subset (e.g., a1, a2, a3, etc.)

The **first set** is a title card saying “where do we get out food from: bread.” Each set has pictures of human beings, a little symbol in corner – a store, for example. And then there is a title for that person's job – storekeeper, for example. This is set a1. We also have set a2. “Where do we get our food from: meat.” Set a3 is “Where do we get our clothing from: cotton.” There are other examples sets.

Second set: b1 is “Who does the farmer need”? b2 is “Who does the toolmaker need”?

Third set: c1 is “What does the farmer produce”? It has a picture of the farmer and then we have smaller cards with pictures of things he produces: dairy products, silk, cotton, etc.

Then there is a picture of a (or a real) cup that goes with later exercises.

Where Do We Get Our Food ? - 1

Age

Early lower elementary

Material

Set a1.

Begin

(make sure cards are in order before beginning)

We've talked about bread and where it comes from.

If you want a load of bread, one goes to the store and buys it from a shopkeeper.

(now place cards left to right a fair distance apart)

Here is my shopkeeper (place **shopkeeper card** on table)

But the shopkeeper doesn't make the bread. She gets it from the baker.

(place **baker card** on table – moving rightward)

Shop keepe r

Baker

It's the baker who makes the bread.

(we now have two cards on the table – box of cards at far right)

The baker needs flour to make the bread.

We call that person who makes flour the miller. Here is the miller
(place **miller card**)

Shop keepe r

Baker

Miller

The miller needs the wheat to grind into flour but he doesn't grow the wheat; she gets it from someone who grows the wheat. Here is the farmer who grows it (place **farmer card**)

Shop keepe r

Baker

Miller

Farme r

layout of cards in center of table

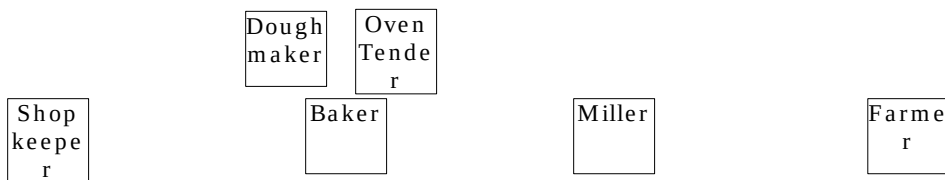
The farmer grows the wheat and then takes it to a miller who grinds the wheat and makes flour.

Then the miller takes it to a baker who uses the flour to make the bread. Then the bread is taken to the shopkeeper. He sells it to you and me.

Now ... the baker does not do all the work herself. She has someone to make the dough from the flour.

(now we put a **dough-maker card above the above and left of baker**)

We also have an oven-tender to watch the bread while it bakes. (place **oven-tender card**)

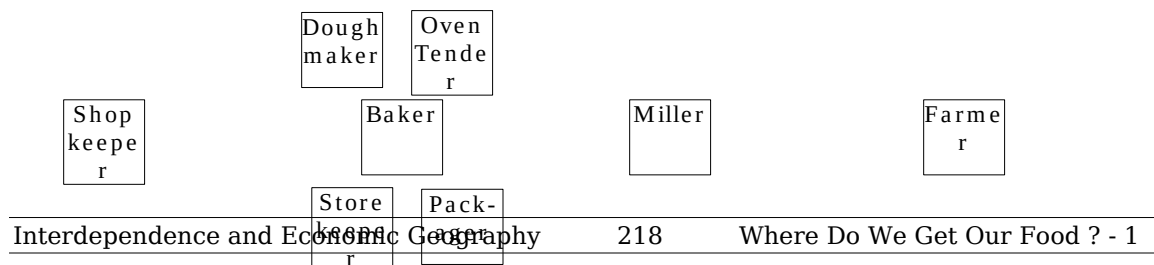


Once it's baked, we have to package it so here is the packager (**packager card**)

Then we need to place it where the bread can be tracked and purchased, so we have a store-keeper

(place **store-keeper card**).

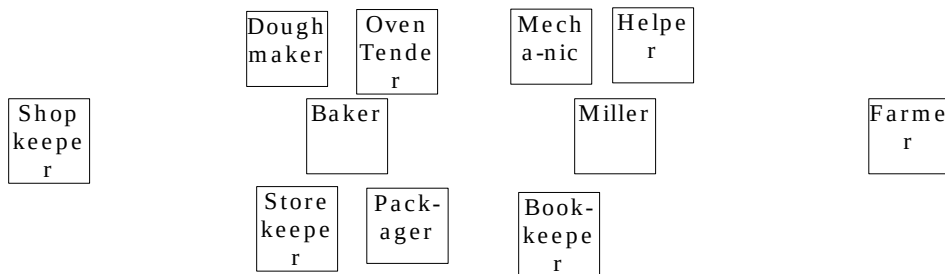
(the last four cards are placed above and below the baker)



The miller doesn't do all the work himself either. He uses lots of machines. So he needs a mechanic to care for the machines (place **mechanic card above miller**)

He also needs a helper to take the flour from the machines and to put it in bags.
(place **helper card**)

The miller needs to keep track of the amount of wheat she buys and sells and the money she makes. So she has a bookkeeper.
Here's the bookkeeper. (place **bookkeeper card**)



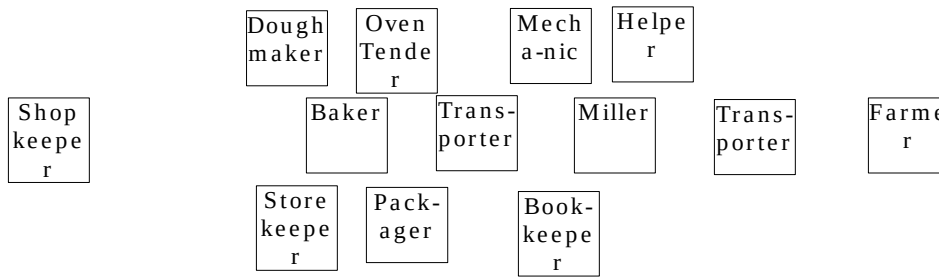
Farmer

The farmer grows the wheat but he needs to get it to the miller.

He needs a transporter. Here he is. (place **transporter card between the farmer and miller**)

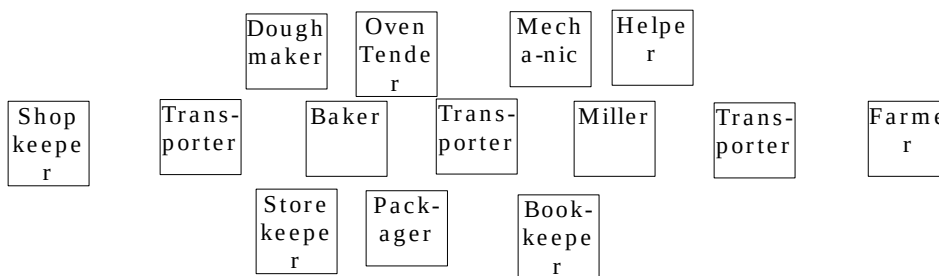
The miller grinds the wheat into flour so he needs a transporter too.
Here he is.

(place **transporter card between the miller and baker**)



Baker

The baker needs to get the bread to the shopkeeper so he needs a transporter also. Here is that transporter.
(place **transporter card between the baker and shopkeeper**)



Look at the number of people working in order to make the bread you and I eat for breakfast! And all we had to do was go to the store to buy it. Imagine if we had to do all these things ourselves. We would be too tired to do anything else.

Epilogue

If you pick them up in reverse order (left to right) they are in order for the next presentation. Title card goes on top. Let the children know we have sets a2 and a3 and they can put them together in order. Have books about how products are processed – how silk is produced for example. Don't change names for the main people but other names can be ad lib.

Children can make up their own cards for other products and add them to the sets. There are rubber stamps to help them make cards, featuring shapes of people and so forth. The important idea is to let

the children dwell on the people involved. The activity gives them more time to think about more than just our single story.

The important thing is to build the chain of people – to show the interconnectedness of what people do to produce certain things. Let the children work with this for a while.

How many Things Do we Get From the Farmer? - 2

Age

Early lower elementary

Material

From bread card, use shopkeeper, baker and miller. From a2 meat, use shopkeeper, butcher and slaughterer, From a3 cotton: shopkeeper, textile-maker, material maker and tailor. Set c.



Prologue

As the children work through set a cards, the last person each time is the farmer. So one day, take up the question:

In general, make cards into groups – first group in upper left corner – working cards into groups that go clockwise in a circle.

Begin

How many things do we get from the farmer?

Here's my farmer (place card - the only one on the table)

Farme r

Let's think about the things he produces.

He produces corn - like vegetables. (place card around farmer)

He produces cotton (place card around farmer)

He produces fruit (place card around farmer)

Silk (place card around farmer)

And dairy products too. (place card around farmer)

Corn	Cotto n	
Farme r		Fruit
	Silk	
Dairy		

Grains like wheat (place card around farmer)

Poultry like eggs (place card around farmer)

And wool (place card around farmer)

	Wool	Corn	Cotto n	
Poultr y		Farme r		Fruit
	Grains		Silk	
		Dairy		

(if children say something we don't have a card for, acknowledge it and go on)

(we now have placed all these cards around the farmer)

Look at all these things the farmer produces!

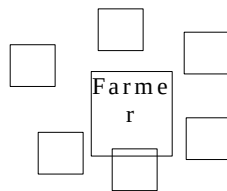
What happens to the wheat?

(pull wheat card into its own area, place workers around it as done with cotton)

The miller grinds it onto flour.

The miller takes it to the baker who makes it into bread.

The baker takes it to the shop so that we can buy the bread.



(take the cotton card from the farmer circle we've just built and make a new circle center at lower left on which to array all the roles involved with cotton)

What happens to the cotton the farmer grows?

(place cards around the cotton card as we did with the farmer card)

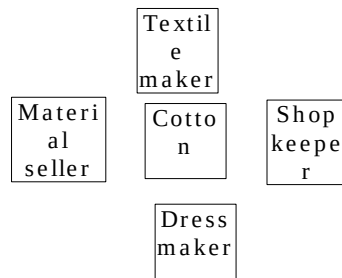
It's taken to the textile maker who makes it into fabric (place textile maker)

Textile maker takes it to the material seller.

The material seller takes it to the dress maker who makes it into pants and shirts and skirts. Here's the dressmaker (place dressmaker card)

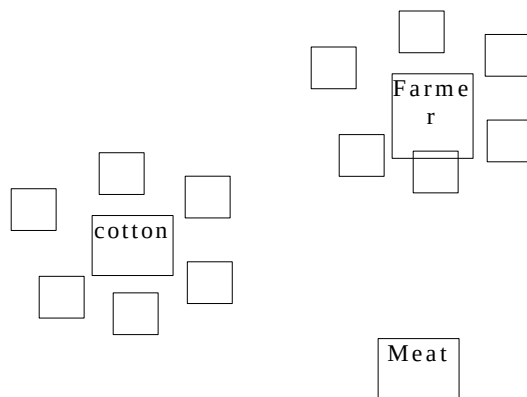
The dressmaker takes it to the shop so we can buy our cloths.

Here's the shopkeeper. (place shopkeeper card)



What happens to the meat the farmer grows?

(place meat card as new circle center and begin laying cards around it - 3rd circle now)



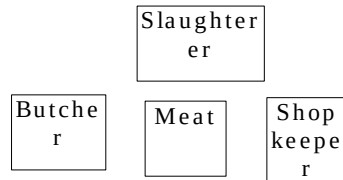
It's taken to the slaughterer who kills the animal and prepares the meat.

(place slaughterer - we working rightward, still below the first)

The slaughterer give it to the butcher who cuts the meat into pieces for cooking.

(place butcher card)

The butcher gives it to the shopkeeper so we can buy the meat. Here is the shopkeeper (place shopkeeper card)



(if children have made up more cards, keep going using theirs)
That's all the cards I have today.

Look at how many people depend on the farmer for their work!

Who Does the Farmer Need? - 3

Age

Later part of Lower elementary

Material

Set b1.

Prologue

When we did activity 2, we talked about all the people who depend on the farmer for their work. This leads us to ask the question, “who does the farmer need”?

Begin

Who does the farmer need?

Here is our farmer (place card at top of table)

Does he need to eat?

He might grow wheat but he doesn't make bread.

So here's the baker (place card)

Do you think our farmer wears clothes?

So he needs the dressmaker. Here's the dressmaker (place card round farmer)

The farmer lives in a house – so he needs a mason to build the house. (place **mason card** around farmer card)

Our farmer also eats meat so he needs the butcher. (place butcher card)

This farmer grows grain but he likes lettuce and apples. He needs the green grocer to get his lettuce and apples. (place green grocer card around farmer)

The farmer wears shoes – so he needs the shoemaker. (place shoemaker card)

The farmer uses lots of machines, so he needs the toolmaker to make the tools that makes the machines. (place toolmaker card)

(we have farmer card in center at upper portion of table with all cards surrounding farmer)

(we are moving cards from farmer to baker now)
Does the baker need anyone? (take baker from this farmer circle and begin new circle)
The baker wears clothes so he needs the dressmaker. (place dressmaker card around baker now)
The baker has ovens and other machines so he needs the toolmaker too. (place card)
The baker needs shoes so he needs the shoemaker (place shoemaker card around baker)
The baker like salad so he needs the green grocer.
He likes meat so he needs the butcher.
The baker lives in a house too so he needs a mason
The baker needs wheat, so he needs the farmer to grow the wheat.

Look at all the people the baker needs!

(now do the same for dressmaker as we did for the baker)
How about the dressmaker? Does he need shoes? (move cards around dressmaker now)
(continue as with baker – moving cards to a different circle each time)

Look at all the people the dressmaker needs!

(continue for all roles)

Look! Everyone here needs everyone else!

Epilogue

This lessons helps the child realize that everyone has the same needs but fills a different role. No one is self-sufficient. We depend on many different people to satisfy our needs and they depend on us. It is an interdependence of services and goods.

Exchange of Services - 4

Age

Later part of Lower elementary

Material

Farmer card and four others - here we choose mason, butcher, dressmaker, baker.

Large piece of paper and 5 colors of markers or crayons.

Prologue

This activity reiterates the interdependence of people.

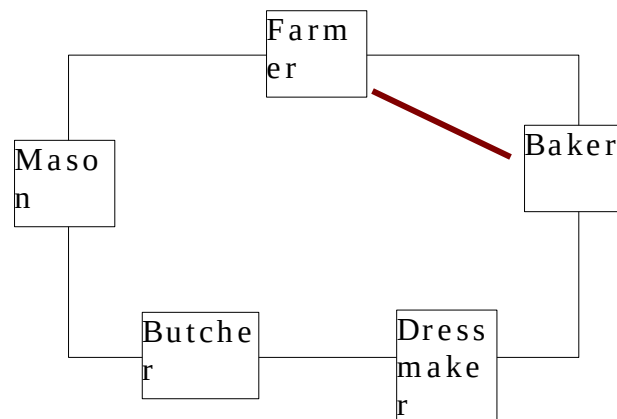
Begin

(lay cards around the paper - all in center of table)

Here we have the farmer.

He likes to eat bread so he needs the baker)

(draw line from farmer in red)



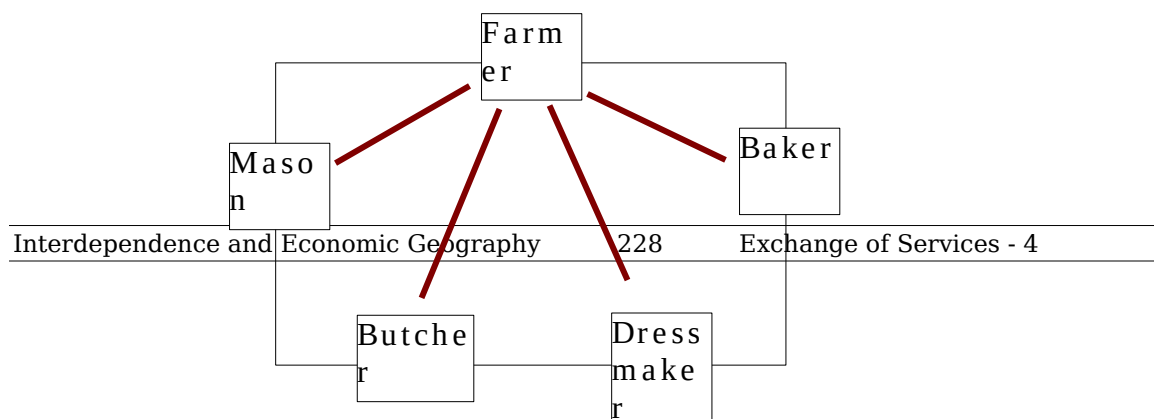
The farmer needs clothes so he needs the dressmaker

(draw line from farmer in red)

The farmer eats meat so he needs the butcher

(draw line from farmer in red)

(continue to describe and draw lines from this one to all among all others)

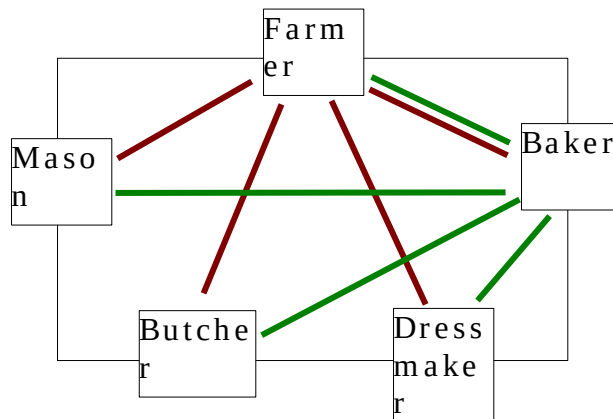


Here we have the baker. He wears clothing so he needs the dressmaker

(draw line in green from baker to dressmaker)

The baker lives in a house so he needs the mason to build the house
(draw green line from baker to mason)

(continue to describe and draw lines from this one to all among all others)



Here we have the mason.

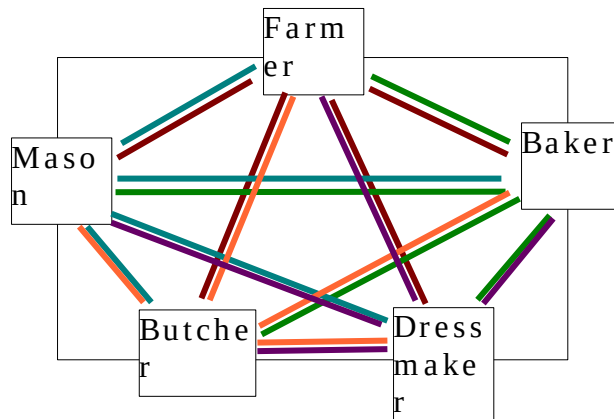
(continue to describe and draw lines in new color from this one to all among all others)

Here we have the dressmaker.

(continue to describe and draw lines in new color from this one to all among all others)

Here we have the butcher.

(continue to describe and draw lines in new color from this one to all among all others)



at the end, the lines are connected from each to all others - showing interdependence

Epilogue

Use specifics like lettuce - not food. Children can make up their own cards. They aren't restricted to five cards. They may be interested in the pattern they end up with.

Introduction to Taxes - Understanding Government - 5

Age

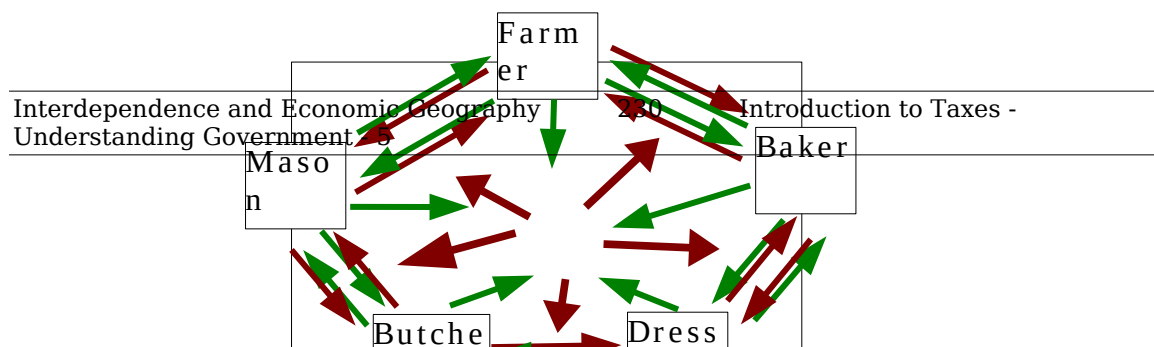
Upper elementary

Prologue

Introduce concept of taxes.

Material

Small bowl or pot. Two markers of different colors (green is good for money).





Begin

(begin by laying out cards around paper as in activity 4 – same 5 cards)

The farmer needs a house to live in. So he pays the mason some money.

The mason builds a house for the farmer
(draw arrows – 2 colors)

What about the mason? He likes to eat apples. So he gives the farmer some money.

The farmer grows apples and gives them to the mason.
(draw two more arrows, same colors as before but in opposite directions)

The mason likes to eat steak. He pays the butcher money to cut and clean steak he eats.

(draw more arrows). The butcher prepares the mason's meat.
The butcher lives in a house so he gives the mason money too (green arrow)

The mason builds the butcher a house.

The butcher wears cloths, so he pays the dress maker some money
(green arrow)

The dressmaker makes pants for the butcher (draw cloths-color arrow to butcher)

The dressmaker eats meat, so he pays the butcher some money.

The butcher cuts up steaks of meat for the dressmaker (draw 2

arrows.)

The baker also wears cloths, so she pays the dress maker some money. (green arrow)

The dressmaker makes dresses for the baker. (draw cloths-color arrow to baker)

The dressmaker eats bread too, so he pays the baker some money.
The baker makes bread for the dressmaker (draw 2 arrows.)

The baker needs apples from the farmer to bake his apple pie, so he pays him money.

The farmer provides the baker's apples.

The farmer also needs bread for his family and gives the baker money.
The baker provides the bread for the farmer.

(there are no arrows in the middle - all along the edges so far)

There are other things each one of these people needs but none of them have!

They need roads to drive on, bridges to cross over rivers, schools to educate their children and a fire department in case of fire.

Imagine if everyone had to build their own road, their own bridges. So everyone pays a little money into a collection.

(place collection in center of paper)

(draw green arrows from each role into center pot)

With this money, roads and bridges are built, streetlights are maintained, snow is plowed, schools are run and garbage is picked up.

This money, which everyone put into the pot, is the money with which the state or the nation or the city, carries out all of the work which helps all of us; we call it tax.

Who can name other services taxes might be used for?

Epilogue

Services and goods should be relevant to the locale in which the lessons is given. This is the basis for taxes. It needs to be given so the

children see the connection between paying taxes and fulfilling the needs of people in general. This exercise helps the children see why taxes might be necessary and to think about how these things might be done otherwise. They can then think on their own about how taxes should be used. For further work, they can explore what happens to the tax money – how it's spent.

We may not all agree. How do we settle these differences and who decides? This opens up the study of government at the city, state and federal levels in this country and others. Perhaps the easiest way is to begin at the local level with a brief explanation of each department of the government and then leading to the overall organization.

Again, we have people involved. So this is a part of interdependencies. Begin with brief explanations – leaving out details. Start with one or two departments and their functions. If teaching upper elementary, one may find this is part of a required curriculum. In that case, provide an activity for repetition concerning the departments and government structure for whatever information is required to be known. Card material or booklets can be made for each department. Include a picture and a brief text for each department. Then have a diagram showing the overall structure – including the people involved and the structure within each department. It could be similar to the card material used here. Most governments will provide organizational structures upon request. The children can visit these departments at city hall. This is an opportunity for a guest speaker.

Help the children realize, the person who leads this structure needs other people. The mayor or whoever it is, could not possibly do everything themselves. Emphasize that one person cannot do it all. So, some of the money paid in taxes is used to pay these people. Depending on how this is presented, hopefully the children will come to understand what government is and why it is needed. But the **most important point is an appreciation of all the work done by the anonymous people**. Hopefully, through that awareness, we can become better citizens. They can then understand the duties of elected officials, the role of elections and their own civic responsibilities related to that.

February 9, 2004

Economic Geography

Economic Geography Stamps:
Baumgarten
325 Washington Blvd.
Laurel, MD, 20707-4617
since1860@baumgarten.com
301-317-3933

Introduction to Activities

Prologue

This relates directly to the lives of human beings. One of the aspects of geography that we try to make the children aware of is **how the geography of a certain region affects the lives** of the people living there. This idea is first introduced in **geography folders in the primary** class, with pictures of people in each of the different continents, the animals and resources there, and reappears studying the **People in Different Zones** and in **early civilizations**. In history, the **History Question** charts bring in this idea of how people are affected by their geographical location.

Helping people understand the effects of geography on the lives of people is one of the main purposes for many of our presentations. What we'll look at today has an even more direct relationship.

We'll study natural resources and the industries of a given region. We can do this at a state or country-wide level. Then **this work gradually build up to a world-wide level**. We're going to look at several different activities with this work but **there does not need to be a fixed sequence to the activities**. So in this case, we will not number the sections.

We'll think of these activities by their titles only. We get an indication of how one could go back and forth between activities to help see how there is no sequence involved.

We want to keep in mind our goals in this process. Our **first goal is to help each child develop his own full potential**. This is why we do all the research we do in the elementary classroom. We're not concerned with how much they learn but how much they are able to

reach their potential. This also means we need to **be aware of the child's interests**. We need to be ready to be guided by those interests – changing directions when it appears fortuitous. The **second goal** is to help the children **realize the interdependencies of the whole world** community. This is the ultimate goal of this work with economic geography.

This **work begins later in the elementary** class – not at the beginning; the children are often 7-8 when this is first introduced. We will see there is a big difference between beginning and ending activities; the later is appropriate to the older children due to the research and math required. We want to think about how interest might be generated in the children for this work.

Interest can be generated in several ways. **Branch off from a study of the geographical features** of a region – and what the climate there is like. For example, wheat needs a climate that is temperate – cooler and not too wet. This is in contrast to rice – which needs a wet and warmer. Cotton also needs this warm weather – but it needs to be dry for cotton – a third type of climate. Wood is a product of the forest and we have talked about temperate forests – specifically in Spread of Vegetation – where we talk about rainfall, the climatic zones and what can grow in them.

Another way to generate interest is investigate a **geological era on the timeline** of life. In the middle of the first timeline of life, there are two rectangles in the center – iron and coal. Where is that coal? Where is that iron? How about oil and salt deposits? Where are they and why? How much do we have?

We could **generate interest through current events**. During the winter a deep freeze in the south happens at times; what effect does that have on the citrus crop? What about a drought? These things have an effect when we go to the grocery store. What happens when a river floods? How are we affected during an oil embargo.

What is going on **locally**? What industries are there and how does it affect those in your own area? What kind of people live there to fill the types of work that are available? How is Detroit affected by it's preponderance of auto manufacturing? Milwaukee has a brewery or two; what does it mean to those in Milwaukee?

We say that wheat is needed to make bread; there are **interdependencies** involved here too that relate to economic geography. This too can generate interest. You can start at the state level or the country level. The way one begins is determined by the interest of the children and what product one is considering.

Make sure that anything related to the public school curriculum is taken care of in this domain of economic geography. Don't limit it to

that – we don't teach to that mark. But be sure required subject-matter is covered.

What is produced and where?

Materials

A country and state map with product symbols indicating what is produced in various regions. Some empty outlined maps for the student's state and country. The economic geography rubber stamps – which include small figures representing certain commodities, as well as stamp pads with several different colors – and a pencil.

We also need some books such as an almanac – hopefully current. An atlas sometimes has this kind of information. World Book Encyclopedia works well as a source book too. Finally, National Geographic published a book called “Our 50 States” which lists for each state with their product symbols; the maps are a bit different in today's version but used book stores may have the older version that is more ideal.

Begin

We seem to eat lots of dairy products. I wonder where those products come from?

Do we produce dairy products in Wisconsin? I don't know, but let's look at our **map of Wisconsin** (hold up filled-in map, laminated with a key for the symbols).

Yes, we have dairy products here. And I see different areas of the state where dairy products are produced. Would you like to make your own map showing where dairy products are produced?

Let's take a blank map of Wisconsin and my stamp for dairy products – and let's make our own map. I see dairy products produced here – so I'll make a stamp there.

(look at prepared map, then stamp dairy product symbol in same places on the empty map)

We produce lots of dairy products here in Wisconsin!
Let's make a stamp for a key and write “Dairy Products” next to it.

We saw that we use cotton to make some of our clothing. Do we product cotton in Wisconsin? Nope – doesn't look like it. So we can't use our state map for this.

Let's look at our country map. Yes, I see cotton grown in Louisiana, Arizona and other places. So I can make a country map showing where cotton is grown and produced in the United States.

What else do we produce there?

Begin

I see we have the automobile industry in Wisconsin; let's put that on our map. We also produce fruit in Wisconsin – so let's add that to the map too.

(continue to stamp products specific to the chosen region)

Now we are producing a map that contains many products since we listing all of them for our state.

What if we want to do a different state? We can do that! I don't have a map so what can you do about that? (find a map of that different state)

If we do Kentucky – we need a horse stamp! What can we do about that?

We can make our own stamp or make a picture for that, right?

We always have the option of creating new maps and new stamps – and even new resources and commodities if we like. We just need to be sure to add it to our key.

Epilogue

These first two activities will lead to lots of work among the children. This is the 7-8 year old level for this area of study.

How much of a certain product do we produce?

Begin

We want to know how much our state or country produces. We'll use our maps and stamps and we also need an index card or piece of paper - and this time we'll use our resource books.

We looked into wheat in the past. We've seen where wheat grows around the country. Today we want to look into how much wheat is produced in various states.

I have my map of the United States here and I found my wheat symbol. But to find out how much is produced, I'll use my World Book Encyclopedia - and look up wheat.

(hold up book)

Here's a map that shows where, in the United States, wheat grows. Above it is also a table that gives us quantity. This is what we want to look at.

The first interesting thing about this table is it includes not only states of the U.S., but also Canadian provinces. I'll ignore the provinces.

I can see that Kansas is our main wheat producer. It says here that Kansas produces 366 million 300 thousands bushels of wheat per year. Bushels is our unit of measurement here.

Boy that's a lot of wheat - 366,300,000 bushels. I have my stamp here but I can't fit that many stamps, can I? So we need to **determine how much each stamp will be worth**. How will we decide to use this stamp to show what each state produces relative to each other.

Given 366 million, perhaps a single stamp should stand for 100 million bushels of wheat. Let's try that and see how it goes. So first thing I do is make my key here at the bottom of my map. (do so)

We're doing wheat production, so I'll title my map "Wheat." (write title at top)

The next thing I need to know is which state is Kansas. If each stamp is worth 100 million bushels, how many stamps do I put here in Kansas?

(use index card, put it down so part of the stamp is visible for the 66 million bushels)

So we'll put 3 stamps and then use our index card to make a half-mark with our stamp for that last 66 million bushels.

How about North Dakota. My book tells me it produces 269,120,000 bushels. So let's find North Dakota. And how many stamps do we need? (2 and a bit more than half) Yes, so here's my two stamps – and I'll use my index card to make a partial stamp for that extra 69 million bushels.

(continue for all other wheat-growing states)

(repetition includes investigating cotton, etc. in the same manner)

Epilogue

As you help the children with the resource books, be aware that sometimes different units of measurement are used; give them a quick lesson to understand the units they are working with. We need to use the same unit of measurement for an entire map – so this can lead to complications.

Once interest has begun, the children will want to look into other things like population. There is a stamp for this, a person, and other stamps to accommodate a free investigation of other quantities and resources pertaining to various regions.

How much of a certain product is consumed?

Material

Same as previous lessons but add: graph paper with varying sizes of squares, a ruler, pencil, colored pencils, glue, scissors and pieces of plain white paper.

Begin

(begin with a product the children can relate to – e.g., milk)
How much milk do you think a person drinks in a day?

Two to three glasses? What do you think?

Some people drink more than other people, right? Well, let's say the average is about a quart of milk a day.

If we drink a quart of milk a day, what do we drink in a month? (about 30 quarts)

Can we convert those quarts to gallons? So that would be $7\frac{1}{2}$ gallons per month per person.

If we drink that much in a month, what do we drink in a year? (90 gallons)
That's only one person!

How much do you think would be drunk by all the people in the United States? In order to know this, we need to know what? (population)

The figure I have is 267 million people in the United States. So now we have to multiply 267 million times 90 to find the amount of gallons drunk by everyone in the U.S.

If we do that we get 24,030,000,000 gallons. That's a lot of milk!

Epilogue

We can tarry here for quite a while, looking at consumption for various commodities. Sugar, soda – things the child can understand – are good candidates. If they want to, they can also make a representation of this saying, “We’ve been estimating how much of these products is consumed. Let’s put it down on paper.” Use the stamp of dairy products (for milk or whatever commodity is being investigated) as before – determine the reasonable amount that a stamp could be worth. In this activity, we’re using a piece of paper instead of a map. So it is acceptable to put 24 stamps, for example, for 24 billion gallons of milk. Another way to represent a number like this is to use graph paper. For example, use a square on graph paper to represent that 1 billion gallons. So, we would color in 24 (and a bit more) of them in a line to show the gallons of milk drunk nationwide. Outline it with the pencil using a ruler, color it in with a colored pencil, cut it out and paste it on paper to form a graph. Again, make

sure to include a key showing what 1 square represents – and give it a title like “Consumption of Milk in the United States.”

Realize that we have not looked up an exact number for this consumption. The realization for the child is that there are quantities produced and quantities consumed that co-balance each other. See Appendix A, page 2 for examples of charts.

As an extension to this activity, look at the same idea going state by state; this means we need state populations and allows us to compare states. For example, which state drinks more milk – Minnesota or Wisconsin. The same could be done on a world-wide basis, comparing countries around the world. Logically, one might work back to asking where the commodity comes from and that leads us back into our first activity.

Once we've done production in isolation and consumption in isolation, the next logical step is a comparison between production and consumption. In the earlier work in consumption, it did not matter what color the children used to create their graph bars; but they should now use color consistently. For example, consumption should always be red. Production could be represented as green – but should be green everywhere if so.

Production and Consumption

Begin

We worked on consumption and production of products. I wonder how many eggs are produced in the United States?

Let's see if we can find that answer in our resource books. (do so)
We find here that the United States produces 69 billion eggs per year.

We want to represent that in a bar graph. How many should we allow for each square? Since our squares are large on the graph paper, we'll choose 2 billion eggs per square. So we need 34 squares and a part of the 35th.

(Trace, color and cut it out. Paste it on the titled paper)

That's production of eggs. But how many eggs are consumed each year in the United States? Let's look that up too. (be sure unit of measurement is the same)

Here, I find the number 65 billion 455 million. We can represent that

the same way we represented production – but let's use a different color – one corresponding to consumption.

Let's paste it down next to our graph bar for production.
(we end up with a colored bar graph on a titled page)

Now we can do this for any product we are interested in. Let's find out which is higher – consumption or production. And we can find out where other product comes from when our consumption is higher than our production. If we produce more than we consume, what do we do with the extra that we've produced?

Epilogue

We can also do this sort of comparison between different parts of the country.

February 10, 2004

Imports and Exports

Prologue

We use more of some things than of others. Where does it come from? What do we do with extra? In the course of this investigation, the child will ask why some things are produced abundantly in some countries and not in others; climate, geographical terrain, type of soil, technology and capital available all contribute to this understanding. They'll find some countries produce a lot of a product and yet they still import material and things made from that product; this can happen if the labor needed to produce goods from raw materials is not available. They will find countries that don't produce any of a given product and yet export goods made from that product - in the case where labor is available as in the last example. Some countries earn their living carrying goods between locations.

Begin

We consume more meat than we produce. Let's look into where we get the extra from.

So let's make a graph - using the small-square graph paper. Consumption will be in one color and production in another.

(glue the tiny graph bars onto the map of the U.S. - then using production-color, draw an arrow from the country we get it from to the U.S.)

We import meat from Argentina. So I'll draw an arrow from there to the United States.

(we can look at any product in the same manner)

If we produce more than we consume of something, we would draw our arrow the other direction.

Epilogue

Begin with one item in isolation – like meat. Then, combine items, investigate what we import from a given country. There may be two or three maps made like this and kept in the classroom to get the children started.

For example, See Wheat chart handout, Appendix A. The size of circles represent amounts. This is an older map – about 1950. Looking at the key, there is production of wheat. The black circle means export, so France, for example imports and exports wheat – and produces it. A map or two like this in the classroom gets the children started. We aren't concerned with quantitative accuracy but with the impression of the extent of importing and exporting that goes on.

Another way this information can be represented besides the Wheat chart is the handout for sugar. The U.S. Produces lots of sugar but we see we still import it too.

The center of focus has now shifted from the child's own country to the world. If the children need to, make sure they look into aspects regionally. Some of this information might need to be included in state history, for example.

A natural conclusion of this is to consider the total amount of importing and exporting done by a country – its volume of trade. Logically, we'd begin with the child's own country; the internet is a good tool for this type of investigation. Typically, we use a unit of measure that is the currency of the country.

For the World Commerce handout, we connect with a lines of graded thickness, where trade occurs. Black represents exports and red is imports. The thickness of the line indicates the extent of trade. The circles are proportional to the value of the trade also – so there is a quantitative indication for both commodity and for dollars involved.

This is work for a group of children in the upper elementary class. Through these visual aids, this shows them how inter-connected the world is. Trade goes on between virtually all peoples today and this is the arrival point for all of our geography presentations. It is like blood-streams in the body. Wherever there is a circle on the map, it's like a heart beating. Veins connecting cells in the body might be the lines on our maps connecting the countries of the world – bringing in what is needed and taking away what is not needed. As time has gone on, international trade has expanded so that world inter-connectedness has intensified over recent decades. If one little market somewhere is cut off, the effect is felt all over the world – and the children can come to appreciate this. Just like with the body, if there is an infected toe, that pain is felt everywhere at some point.

Dr. Montessori said the world has become, in the **sense of inter-connectedness**, one nation. Through faster and faster means of communication, this is more and more true as time passes. Yet, there

are many ideologies among people that don't accept this. Through our intellects, we form organizations such as the United Nations, but they don't seem to be able to fulfill their missions. Member states are not always ready to be generous. Even if we have all the resources in the world, we could not live in isolation – and this is the important realization for the child. The child naturally accepts this view of interconnectedness. He still sees the world with his feelings as well as with his intellect. Montessori's hopes, and ours, are that children coming through Montessori classes will develop a different attitude toward the world – a more accepting and generous attitude. This is what is needed to make the difference in the world, that leads to peace.

Appendix A